

Object Detection as Regression Problem

- **YOLO: Single Regression Problem**

- Image $\rightarrow$ bounding box coordinate and class probability.

- Extremely Fast
- Global reasoning
- Generalizable representation

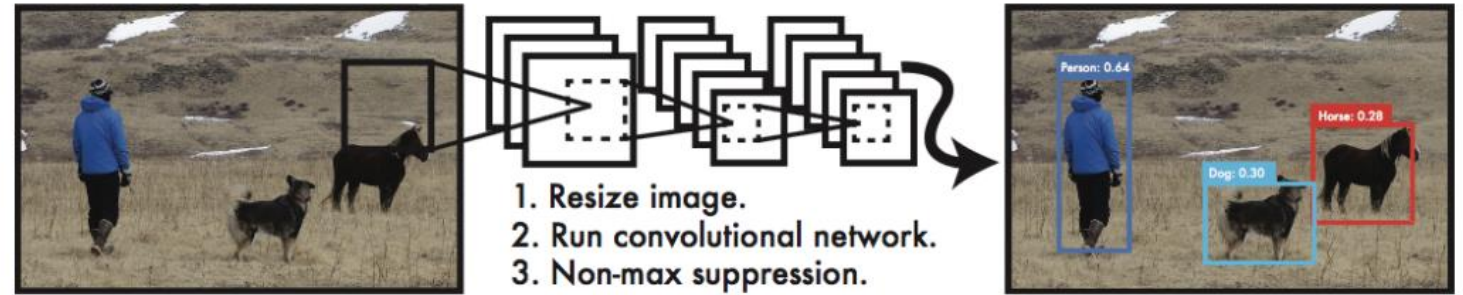


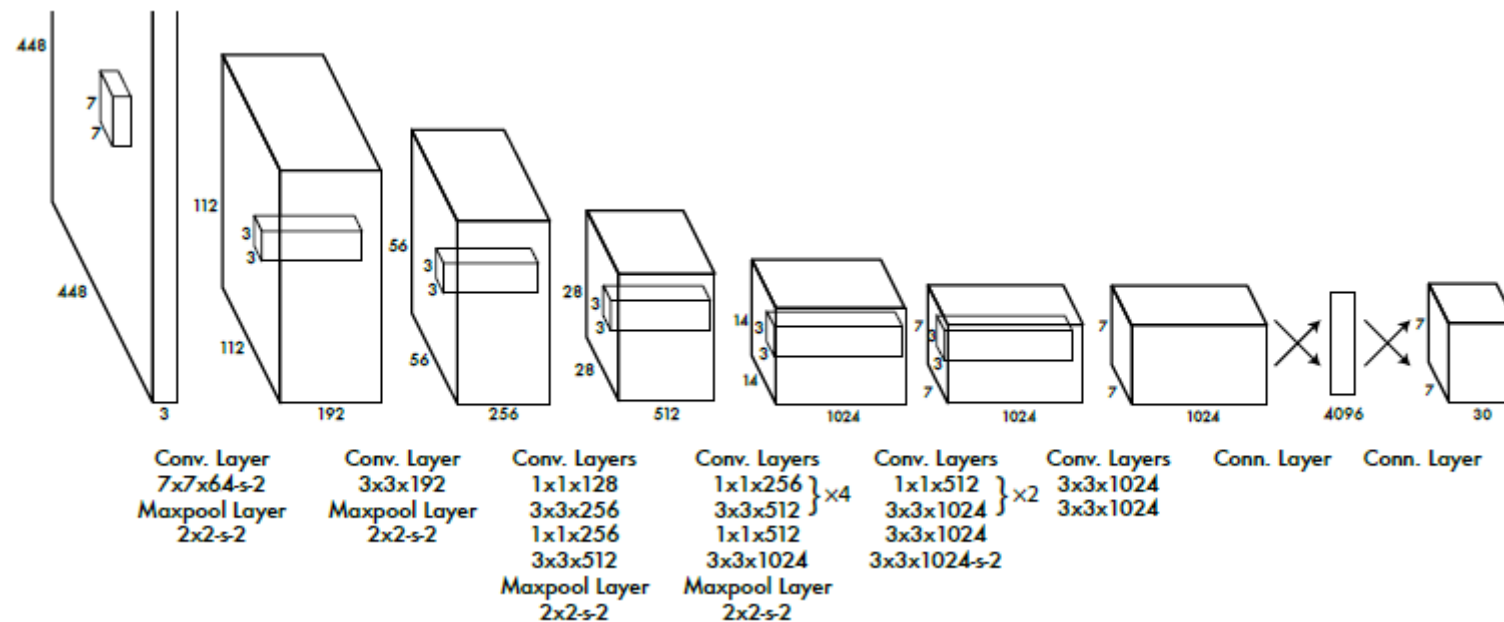
Figure 1: The YOLO Detection System. Processing images with YOLO is simple and straightforward. Our system (1) resizes the input image to 448×448 , (2) runs a single convolutional network on the image, and (3) thresholds the resulting detections by the model's confidence.

YOLO

You Only Look Once: Unified, Real-Time Object Detection

- ? Tool?-- A single neural network, unified architecture
- ? Framework?— Darknet
- ? Technology background?--related methods are slow , not real-time and devoid of generalization ability
- Solution and Advantages:
 - @Simpler structure of network
 - @much more faster, even with real-time property:
150fps: able to process streaming video in real-time with less than 25 milliseconds of latency
 - @Maintaining a proper accuracy range

How YOLO gets its goal?



- For speed: **Also! structure advantage!**
- No bounding box proposals and subsequent pixel or feature resampling stage
- ✓ A neural network predicts bounding boxes and class probabilities directly from full images in one evaluation

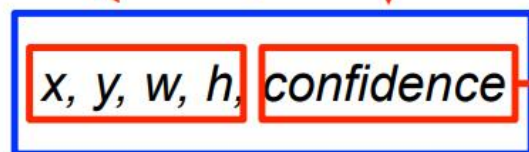
Unified Detection

- All BBox, All classes

1) Image $\rightarrow S \times S$ grids

2) grid cell

$\rightarrow$ **B**: BBoxes and Confidence score



$$\Pr(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

$\rightarrow$ **C**: class probabilities w.r.t #classes

$$\Pr(\text{Class}_i | \text{Object})$$

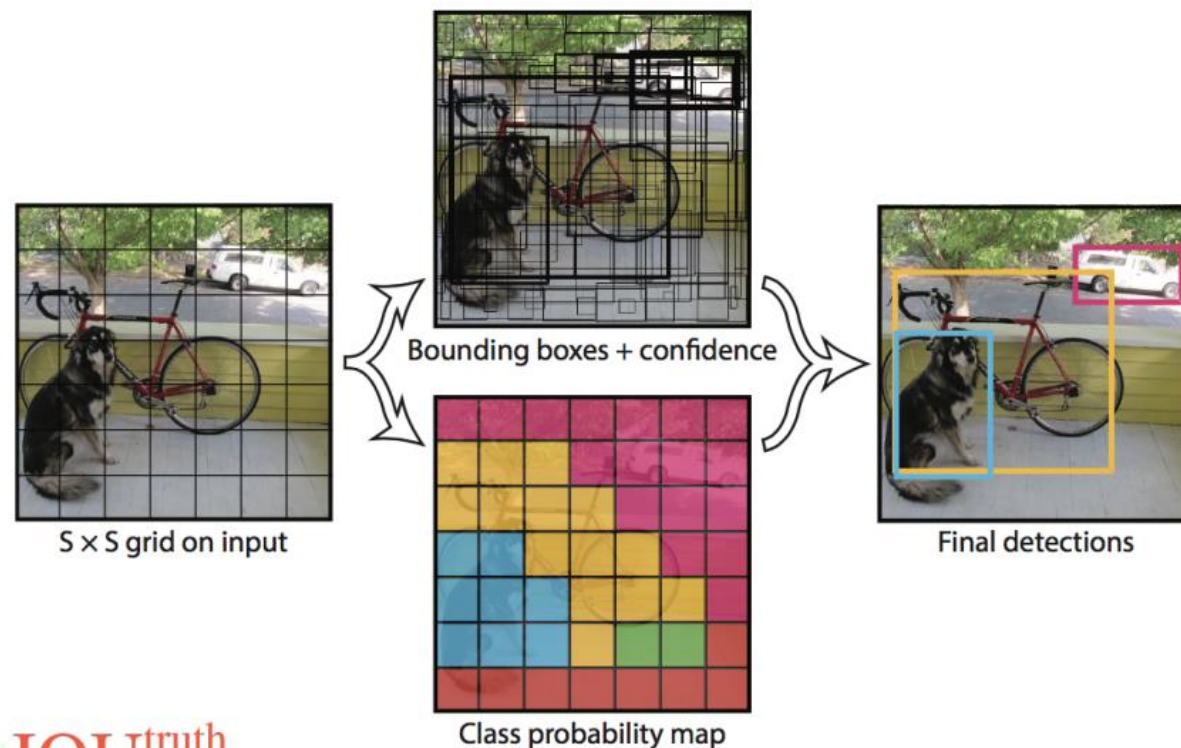


Figure 2: The Model. Our system models detection as a regression problem. It divides the image into an $S \times S$ grid and for each grid cell predicts B bounding boxes, confidence for those boxes, and C class probabilities. These predictions are encoded as an $S \times S \times (B * 5 + C)$ tensor.

Unified Detection

- Predict one set of class probabilities per grid cell, regardless of the number of boxes B .
- At test time, individual box confidence prediction

$$\Pr(\text{Class}_i | \text{Object}) * \Pr(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}}$$
$$= \Pr(\text{Class}_i) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

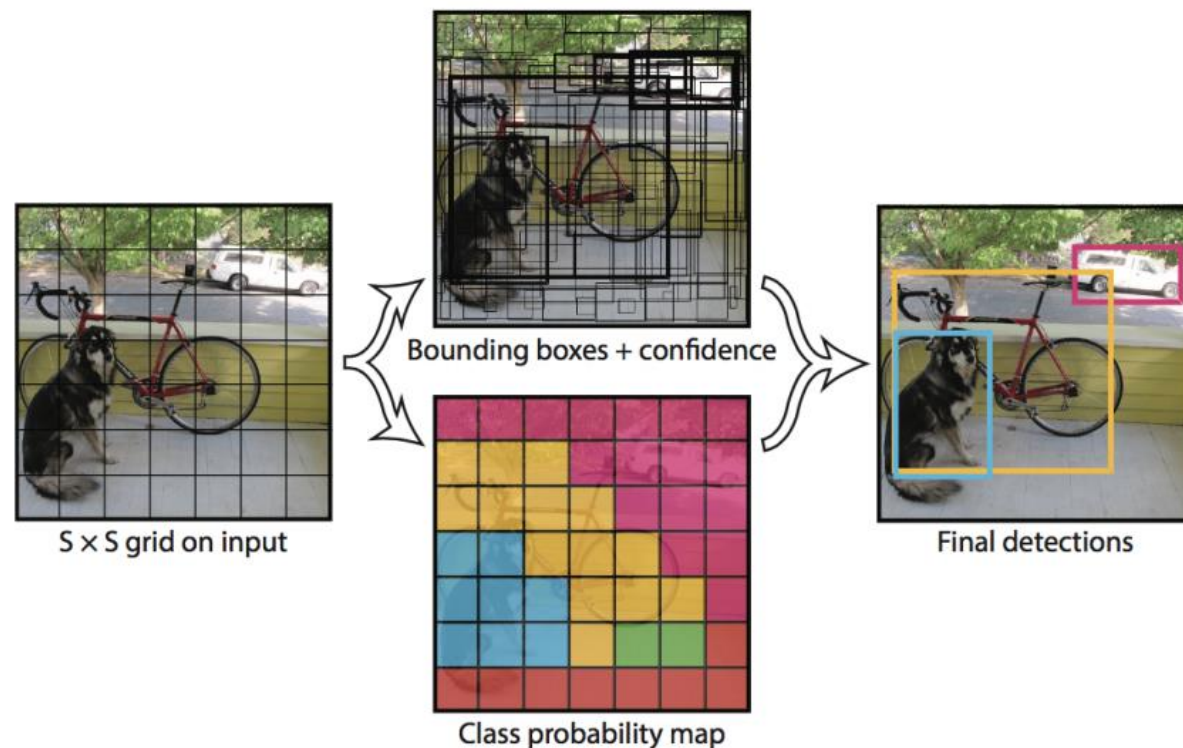
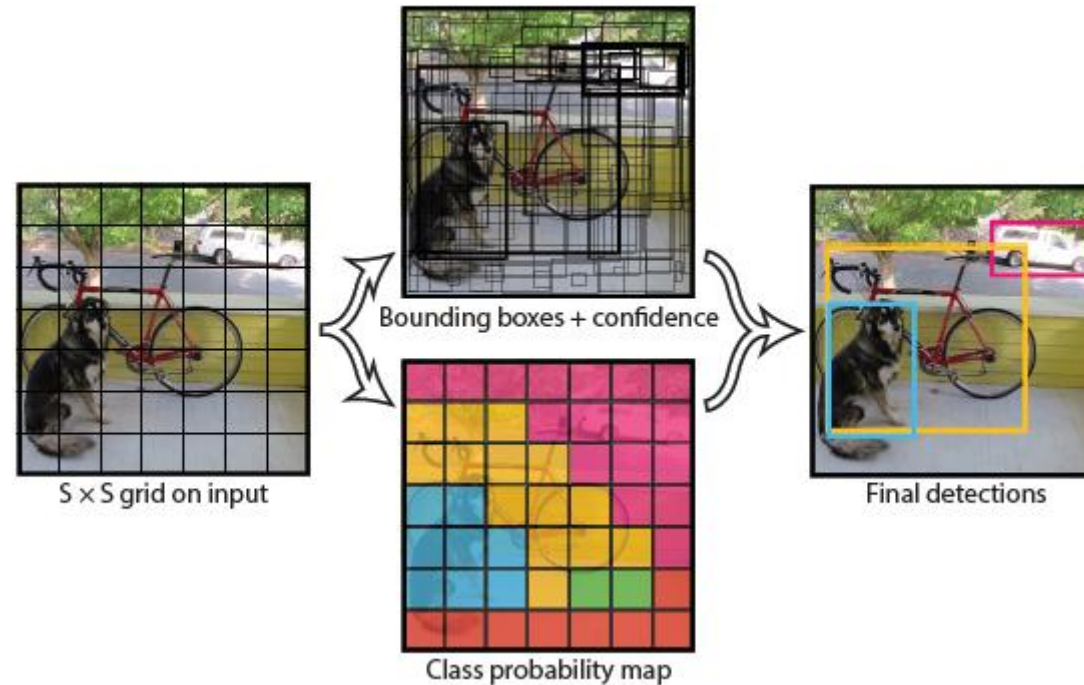


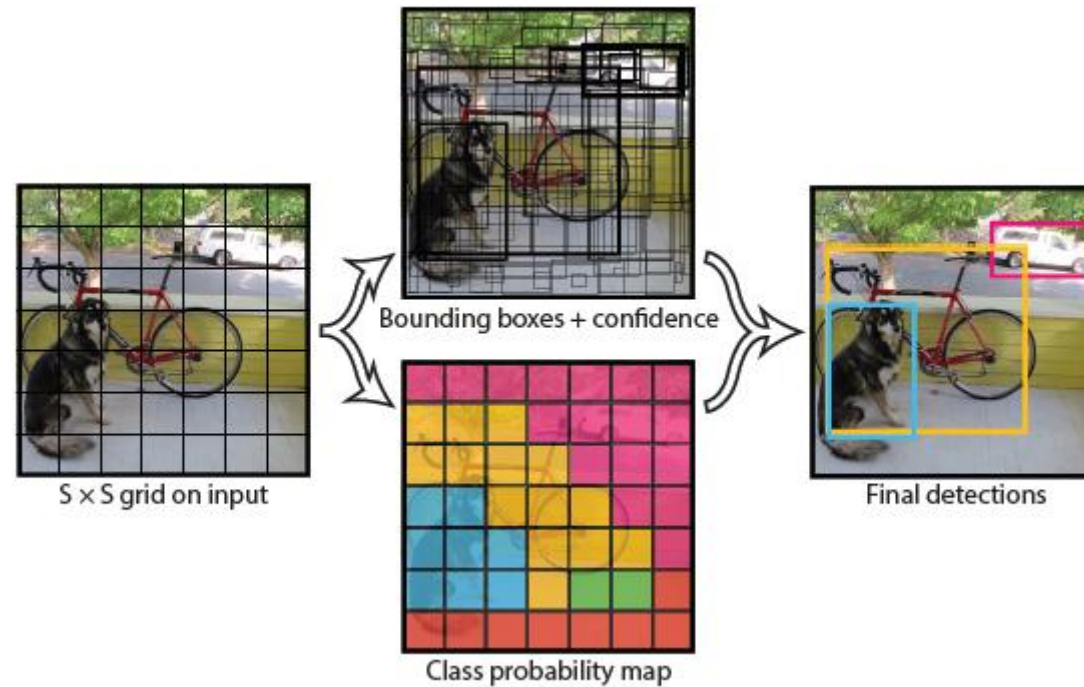
Figure 2: The Model. Our system models detection as a regression problem. It divides the image into an $S \times S$ grid and for each grid cell predicts B bounding boxes, confidence for those boxes, and C class probabilities. These predictions are encoded as an $S \times S \times (B * 5 + C)$ tensor.

How unified detection works?



- uses features from the entire image to predict each bounding box
- predicts all bounding boxes across all classes for an image simultaneously?
- divides the input image into an $s \times s$ grid. If the center of an object falls into a grid cell, the cell is responsible for detecting that object.
- each grid cell predicts B bounding boxes and confidence scores for those boxes
- Each grid also predicts C conditional(classified on the grid cell containing an object) class probabilities

How unified detection works?



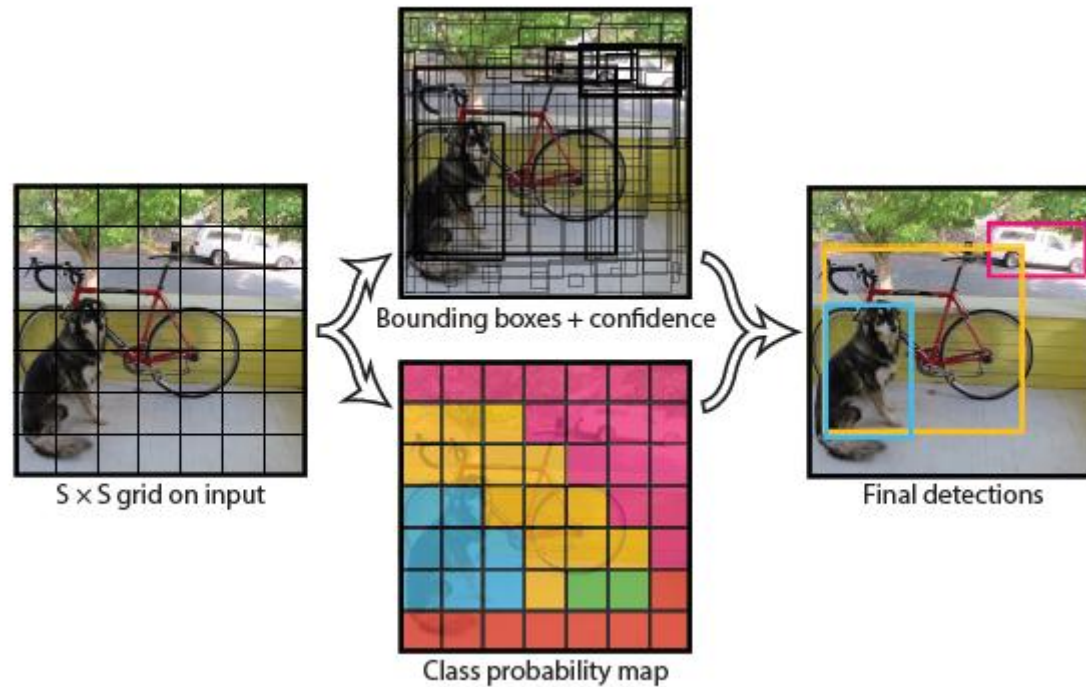
confidence scores: reflect how confident is that the box contains an object+how accurate the box is .

$$\Pr(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

conditional class probabilities: conditioned on the grid cell containing an object

$$\Pr(\text{Class}_i | \text{Object})$$

How unified detection works?



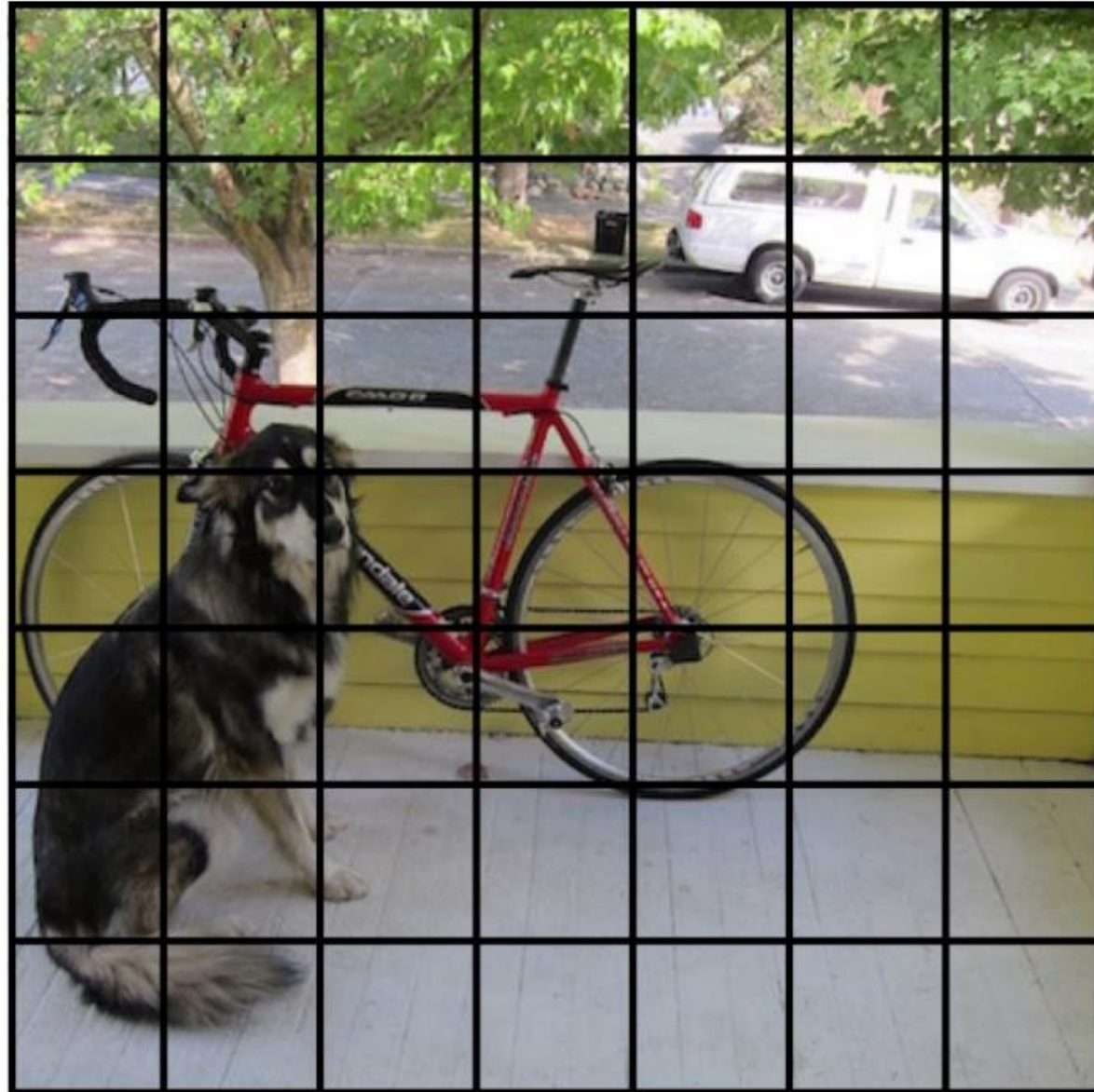
$$\Pr(\text{Class}_i | \text{Object}) * \Pr(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}} = \Pr(\text{Class}_i) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

- At test time, multiply the conditional class probabilities and the individual box confidence predictions
- giving class-specific confidence scores for each box
- Showing both the probability of that class appearing in the box and how well the predicted box fits the object

Automatic Object Detection from Image

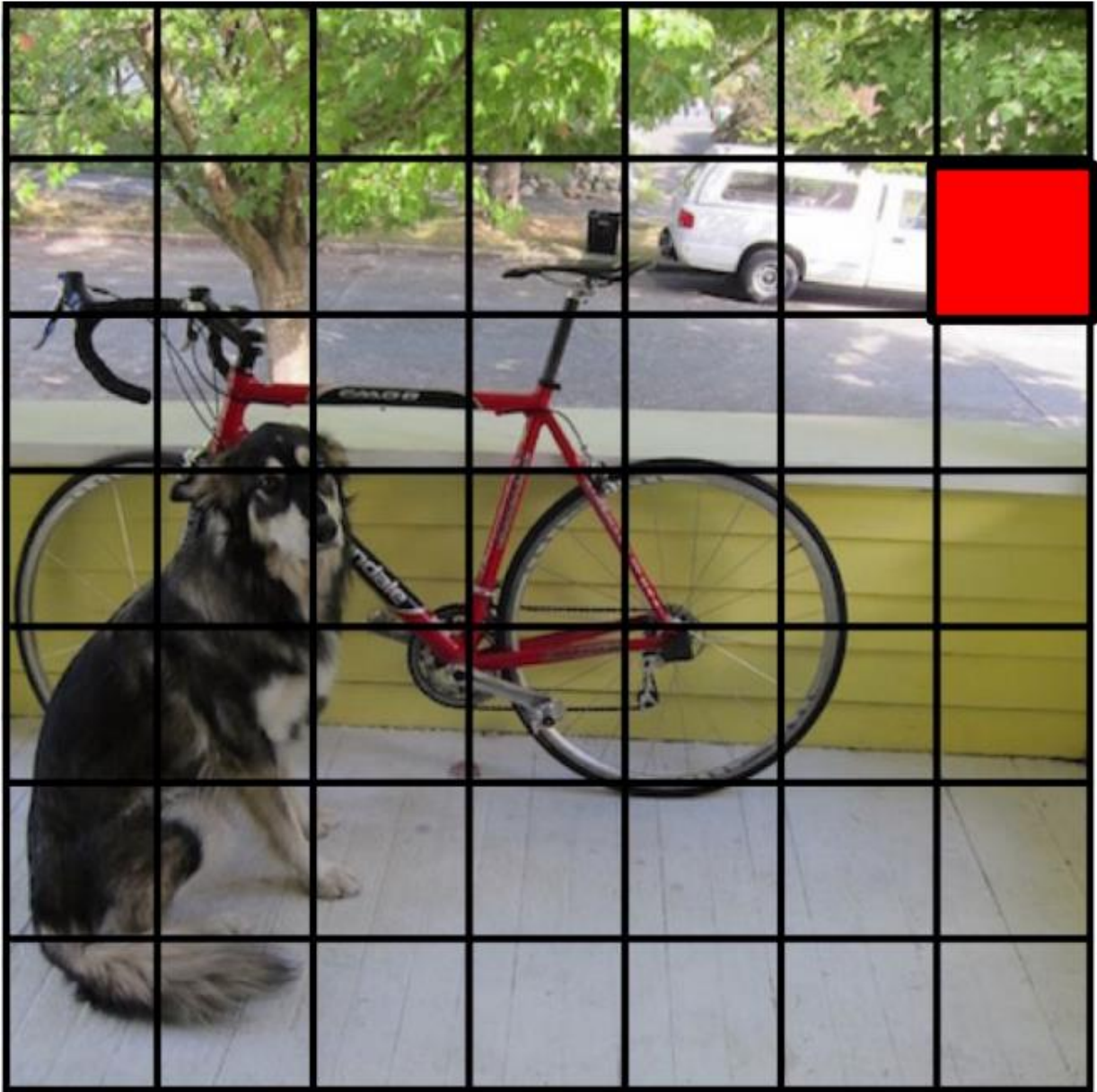


We split the image into an $S \times S$ grid



7*7 grids

Each cell predicts B boxes(x,y,w,h) and confidences of each box: P(Object)



Each cell predicts B boxes(x,y,w,h) and confidences of each box: $P(\text{Object})$

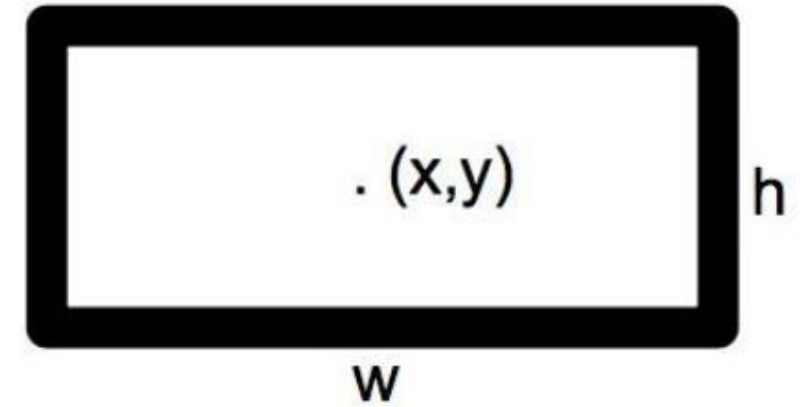


Each cell predicts B boxes (x,y,w,h) and confidences of each box: $P(\text{Object})$

$B = 2$

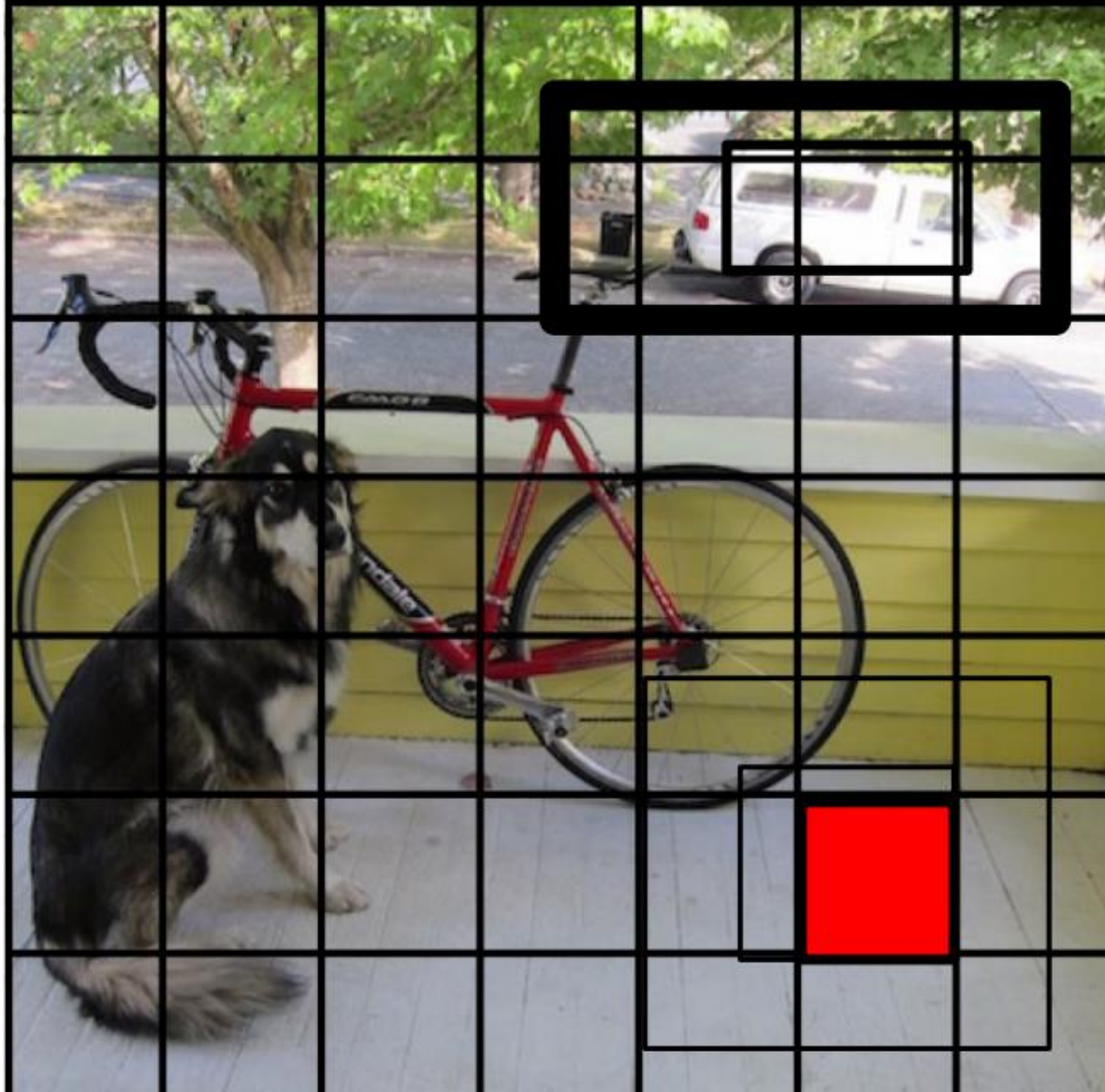


each box predict:



$P(\text{Object})$: probability that the box contains an object

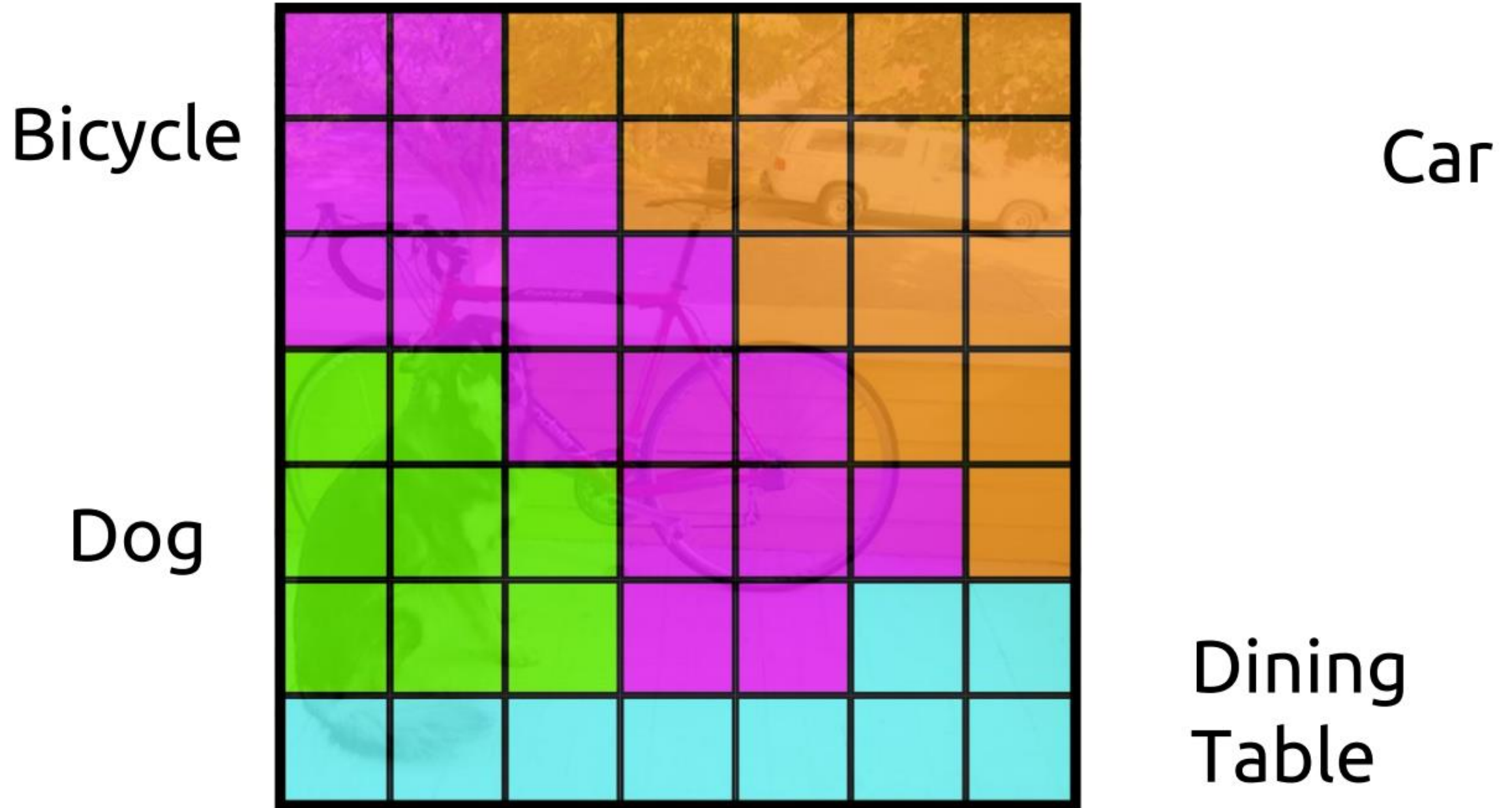
Each cell predicts B boxes(x,y,w,h) and confidences of each box: $P(\text{Object})$



Each cell predicts boxes and confidences: $P(\text{Object})$



Each cell also predicts a class probability.



Conditioned on object: $P(\text{Car} \mid \text{Object})$

Bicycle

Car

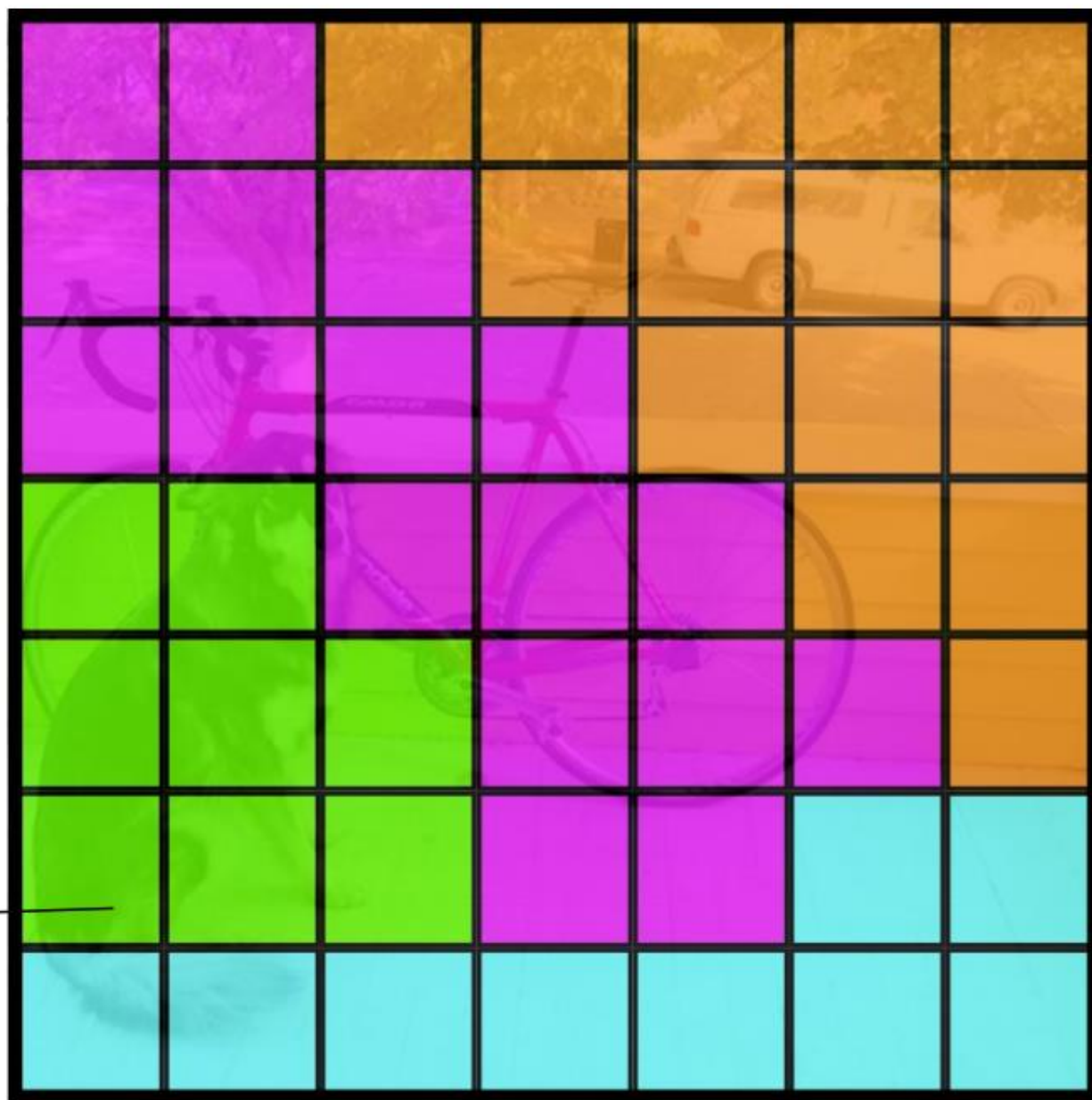
Dog

Eg.

Dog = 0.8

Cat = 0

Bike = 0



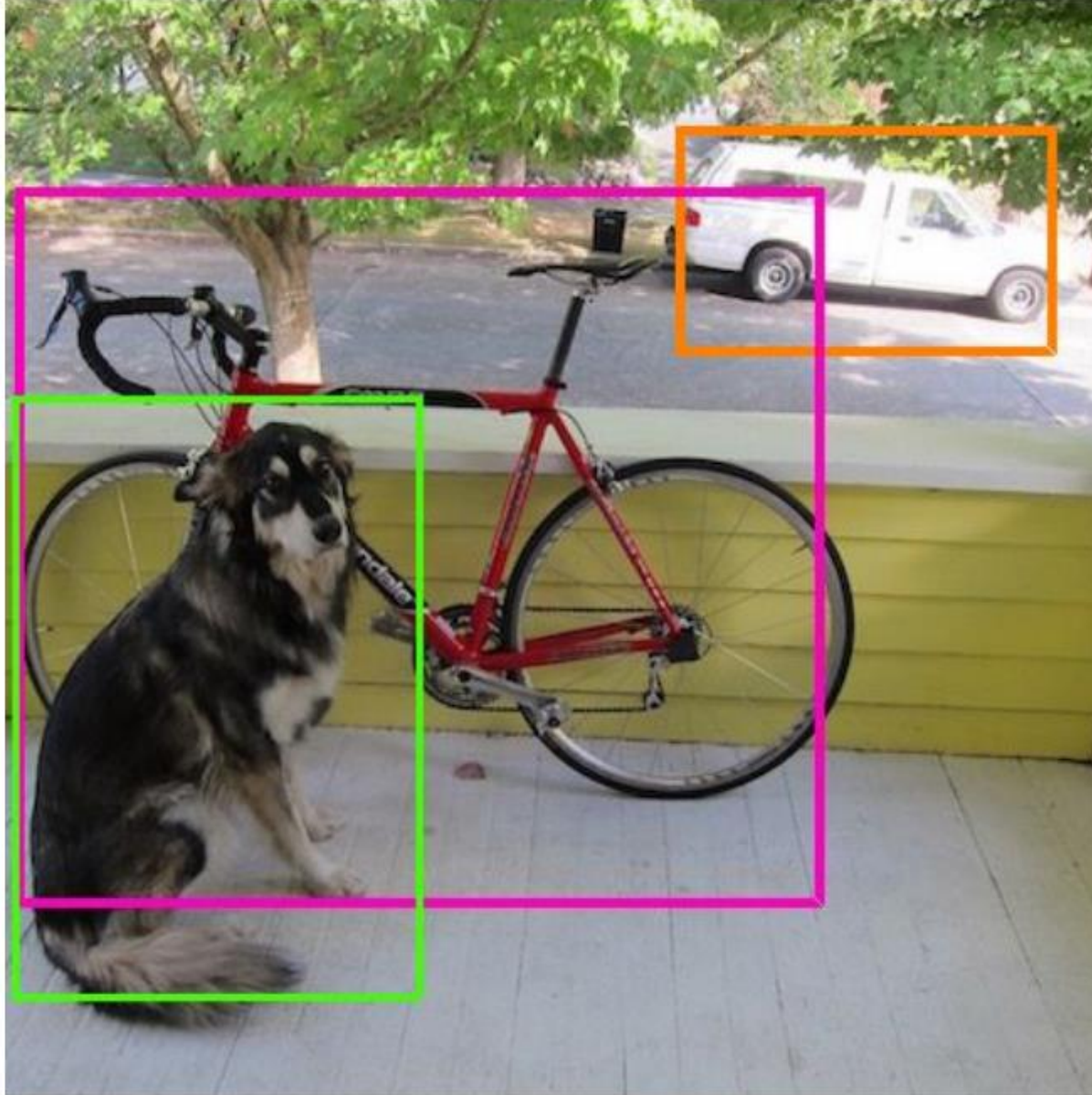
Dining
Table

Then we combine the box and class predictions.



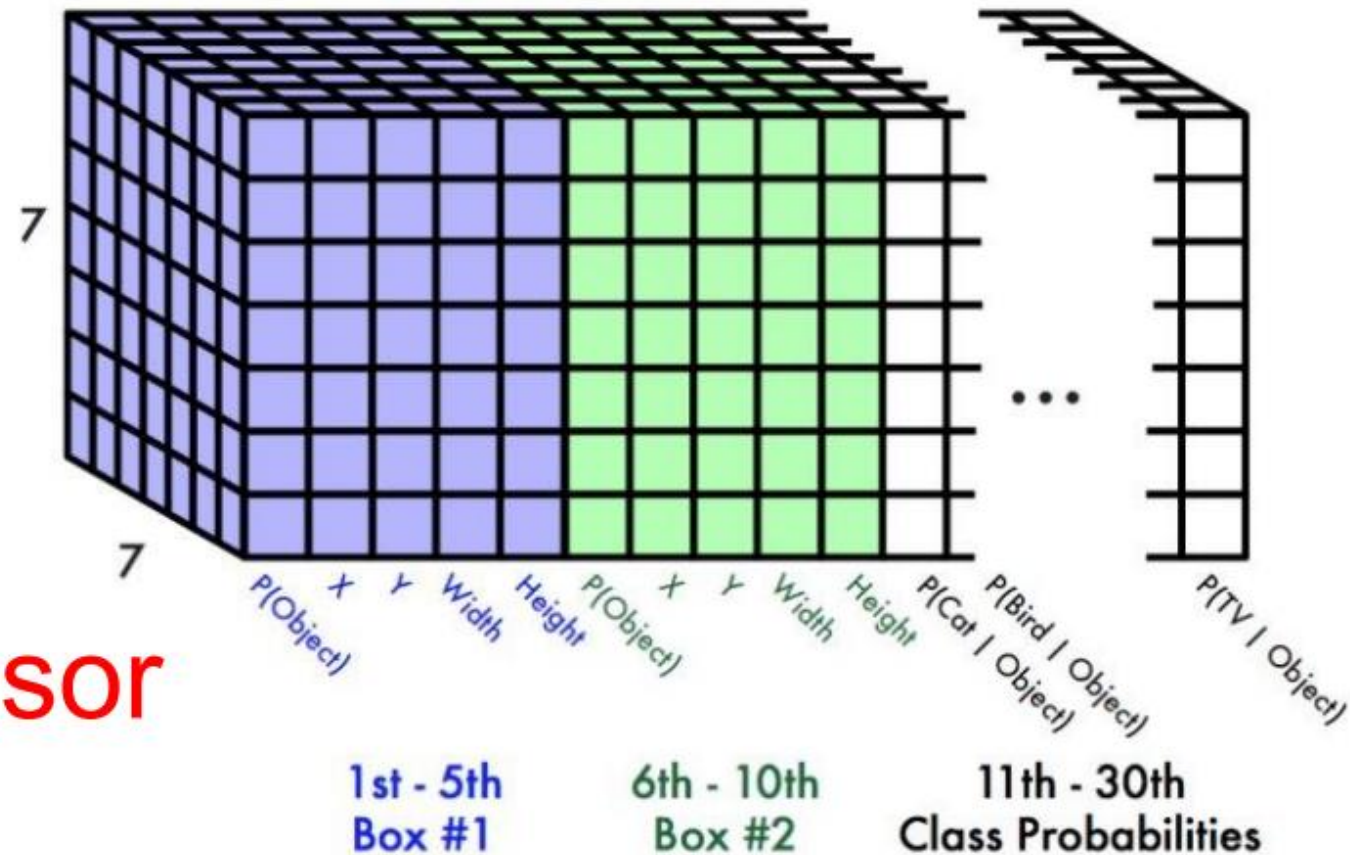
$$P(\text{class}|\text{Object}) * P(\text{Object}) \\ = P(\text{class})$$

Finally we do threshold detections and NMS



Each cell predicts:

- For each bounding box:
 - 4 coordinates (x, y, w, h)
 - 1 confidence value
- Some number of class probabilities

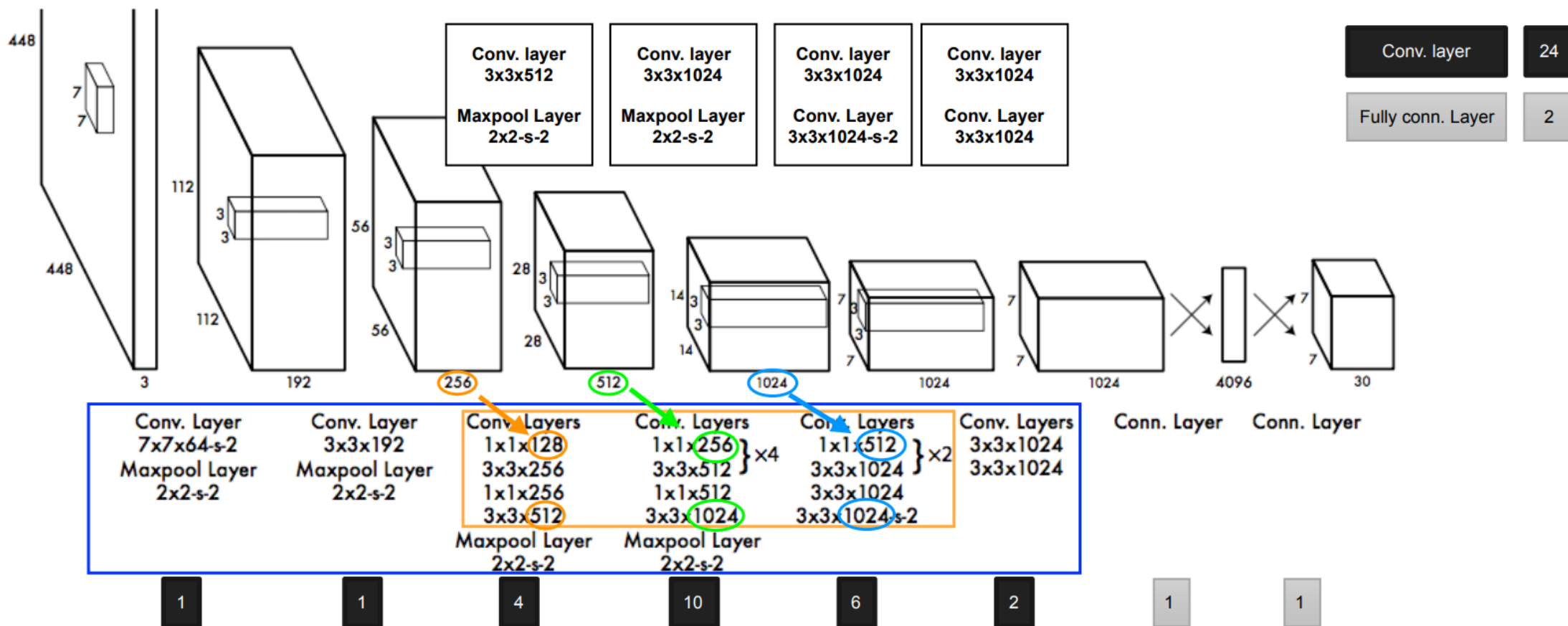


$S * S * (B * 5 + C)$ tensor

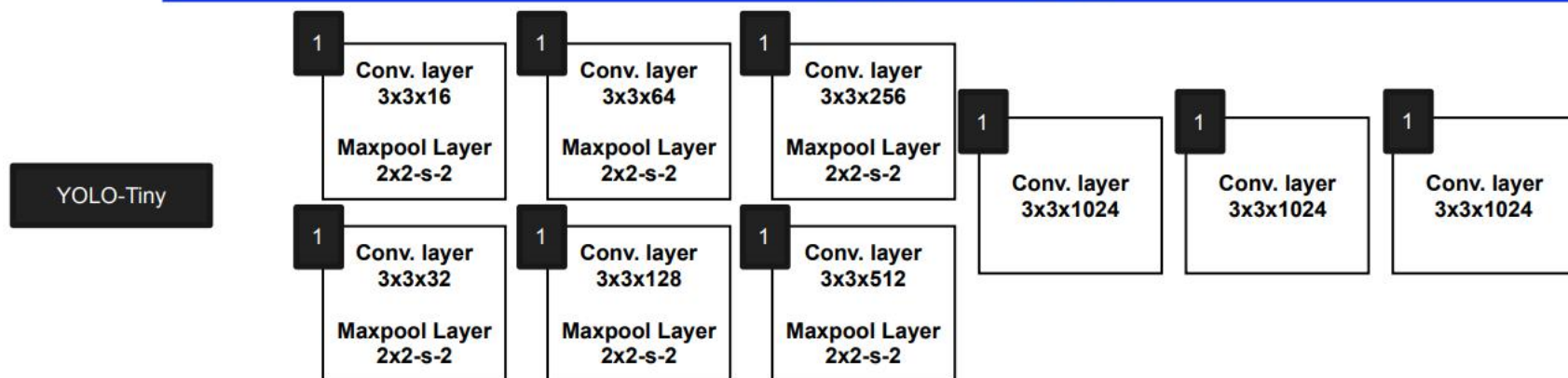
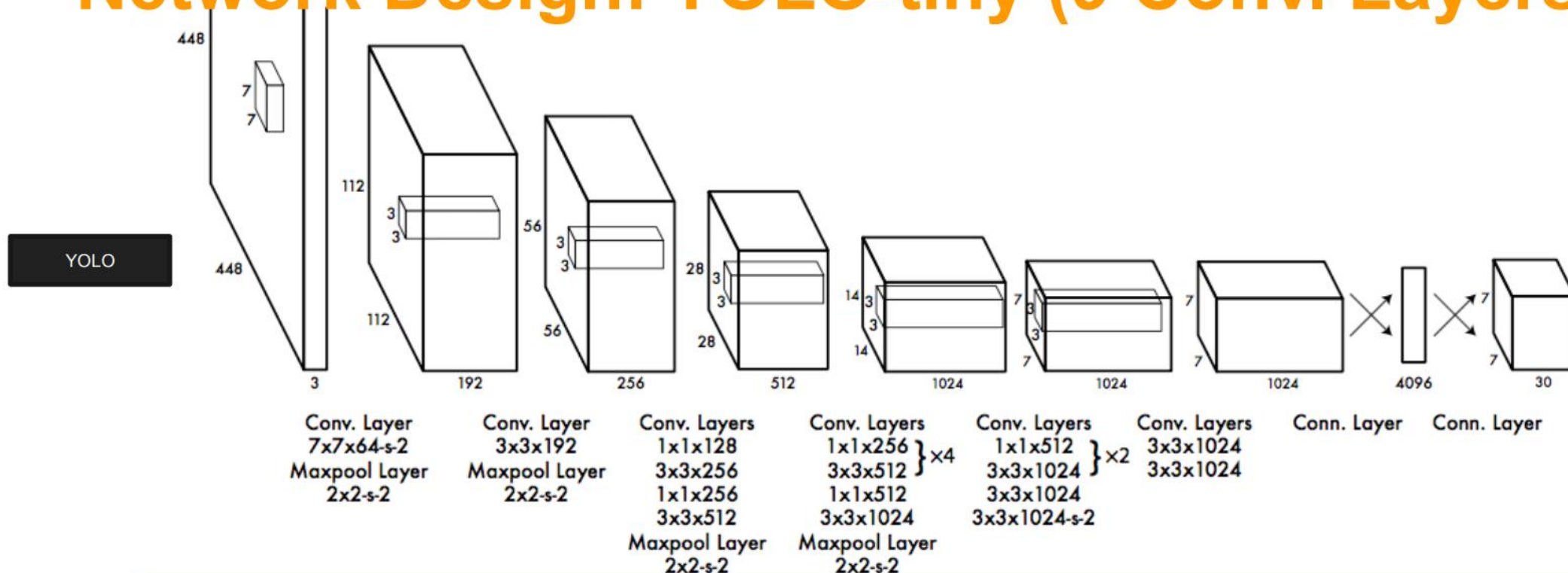
Network Design: YOLO

- Modified GoogLeNet
- 1x1 reduction layer (“Network in Network”)

Our network architecture is inspired by the GoogLeNet model for image classification [34]. Our network has 24 convolutional layers followed by 2 fully connected layers. Instead of the inception modules used by GoogLeNet, we simply use 1×1 reduction layers followed by 3×3 convolutional layers, similar to Lin et al [22]. The full network is shown in Figure 3.



Network Design: YOLO-tiny (9 Conv. Layers)

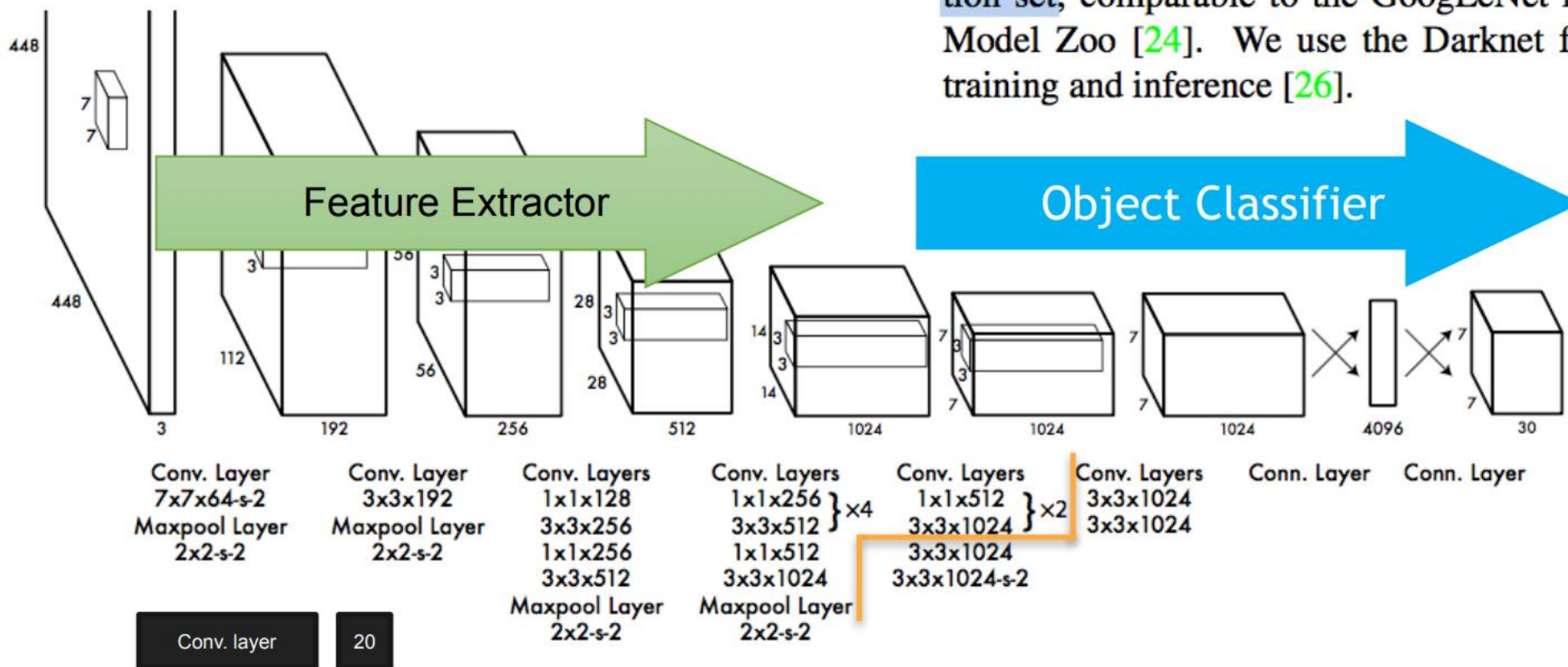


Conv. layer	9
Fully conn. Layer	2

Training

1) Pretrain with ImageNet 1000-class competition dataset

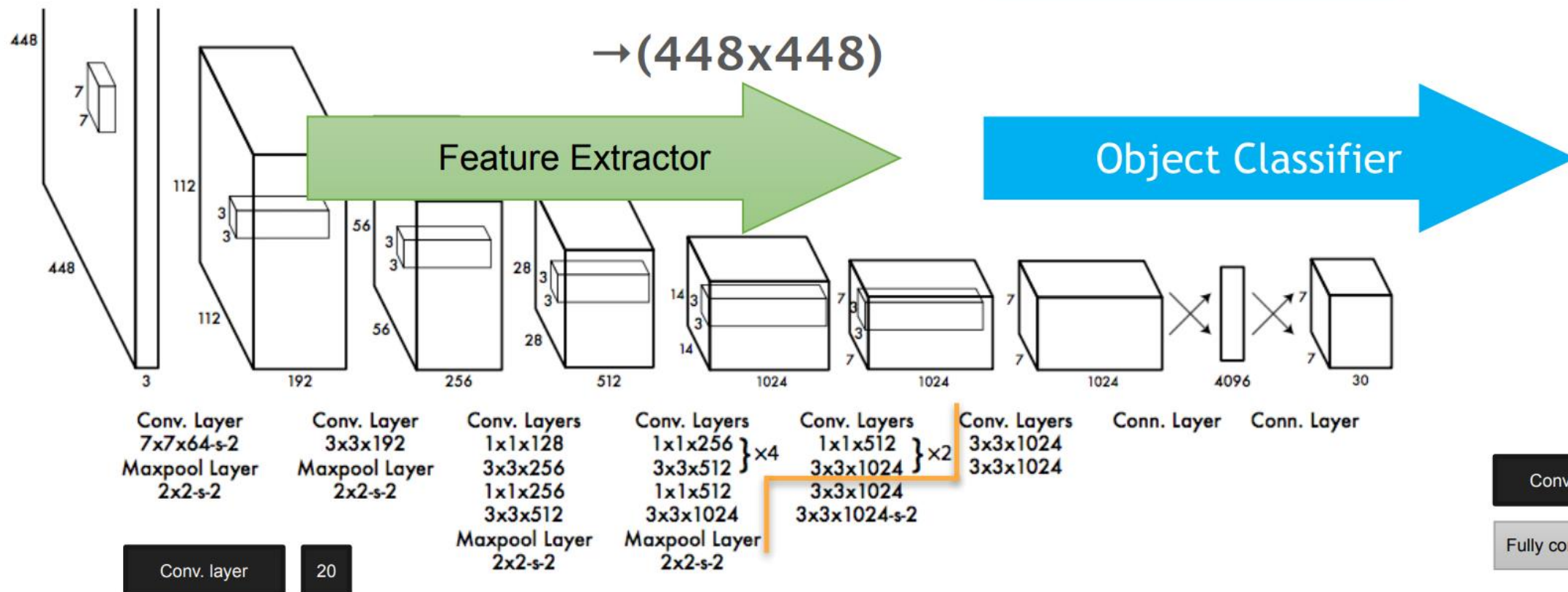
We pretrain our convolutional layers on the ImageNet 1000-class competition dataset [30]. For pretraining we use the first 20 convolutional layers from Figure 3 followed by an average-pooling layer and a fully connected layer. We train this network for approximately a week and achieve a single crop top-5 accuracy of 88% on the ImageNet 2012 validation set, comparable to the GoogLeNet models in Caffe's Model Zoo [24]. We use the Darknet framework for all training and inference [26].



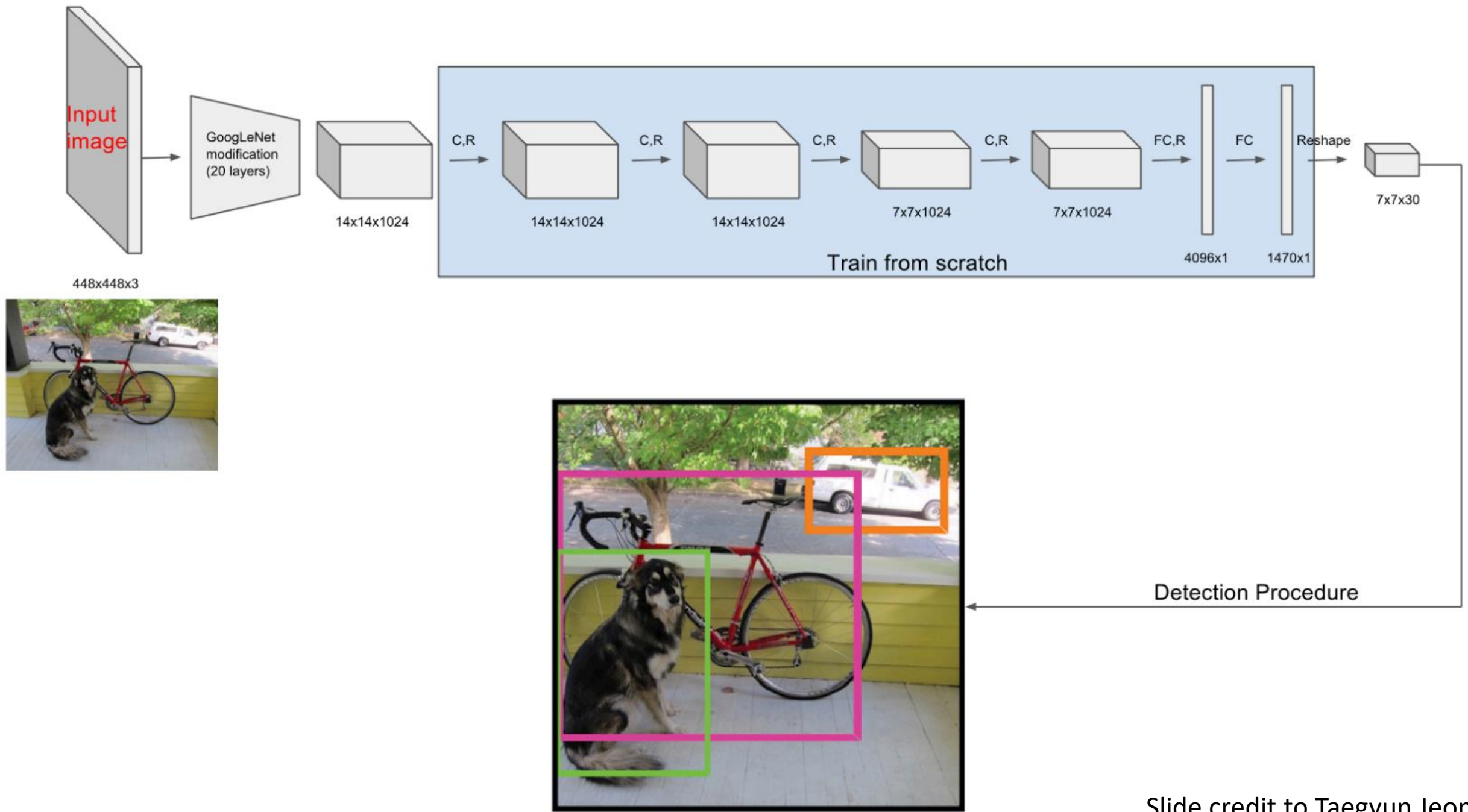
Training

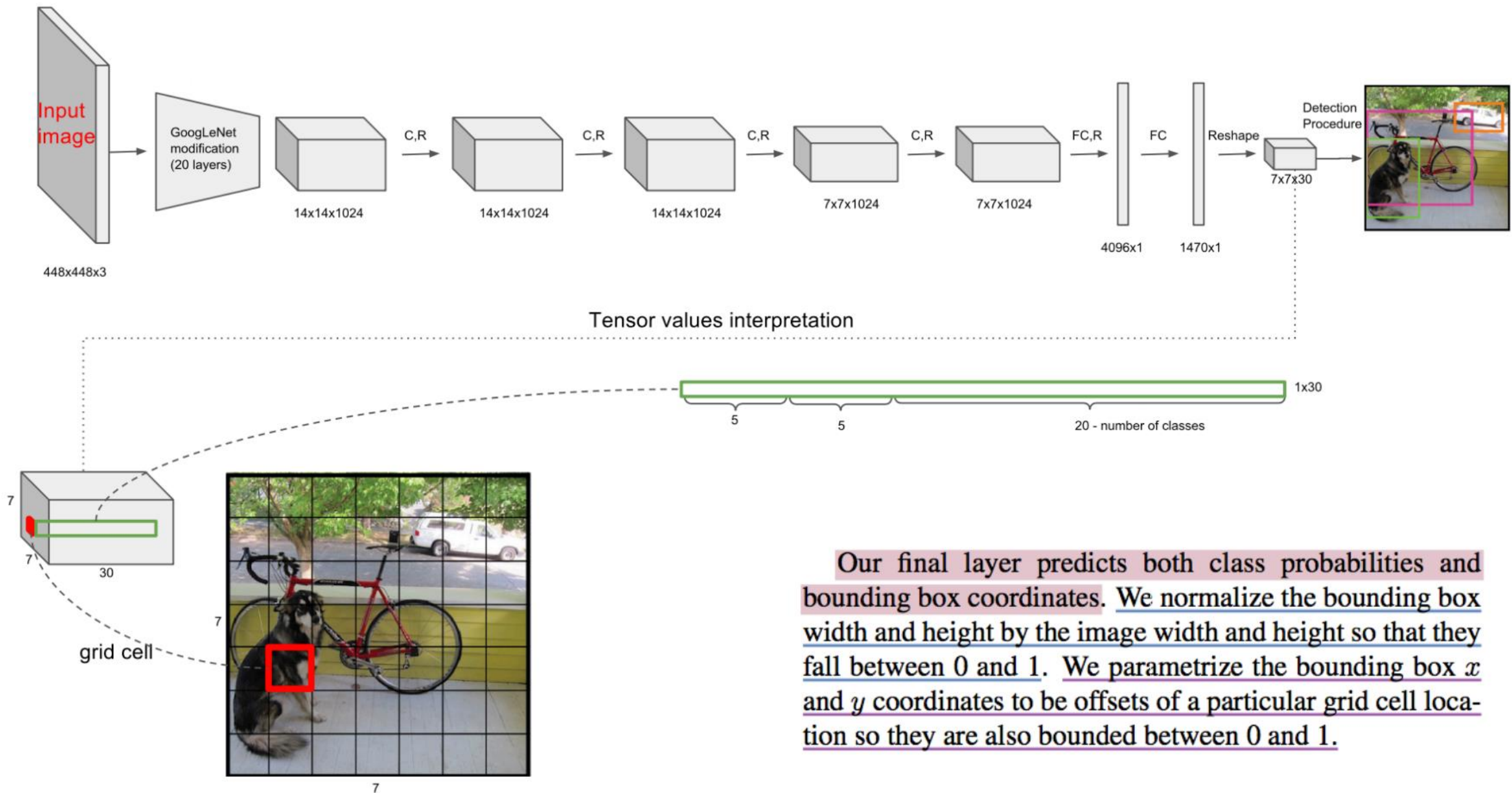
2) “Network on Convolutional Feature Maps”

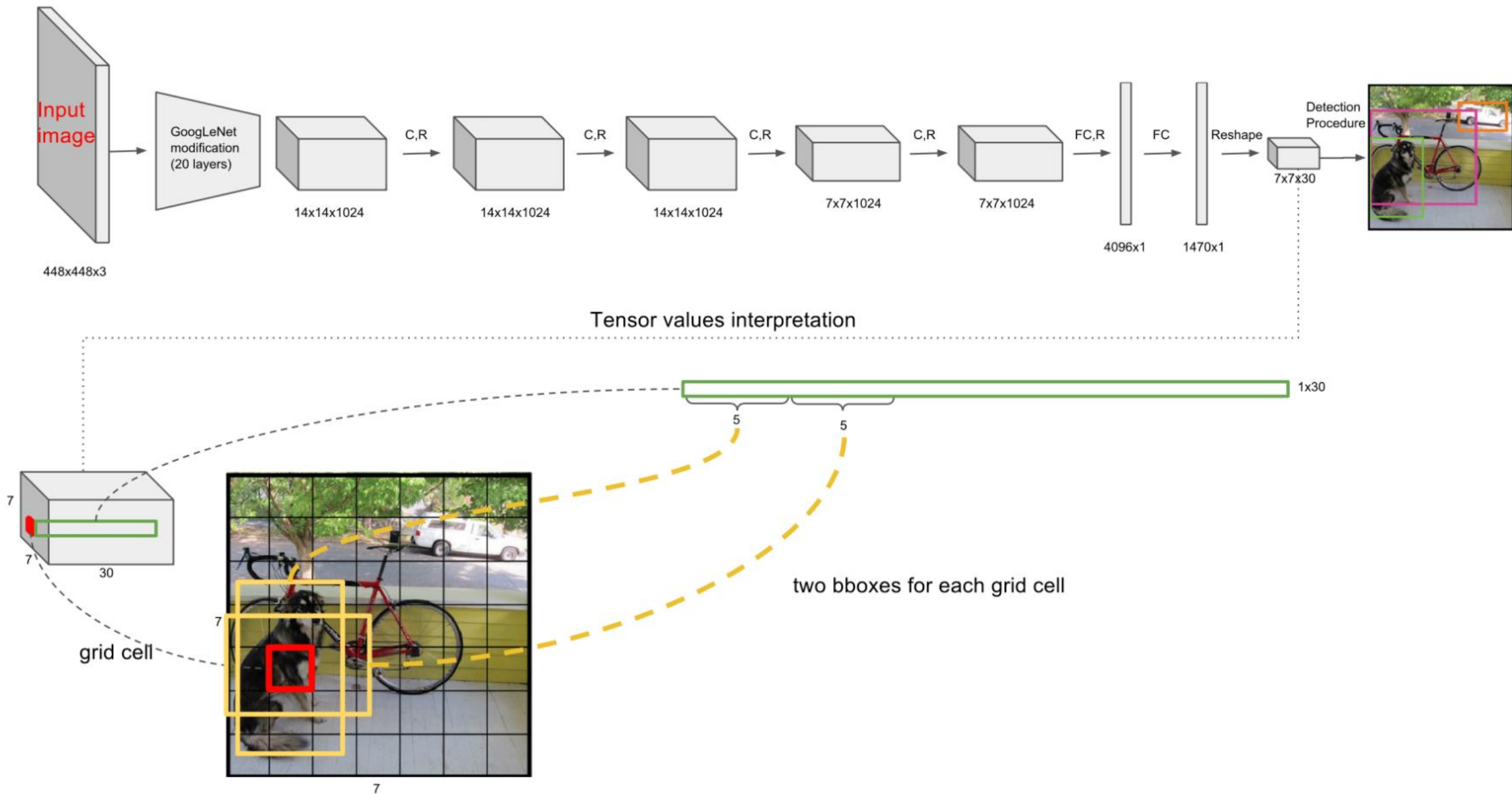
Increased input resolution (224×224)

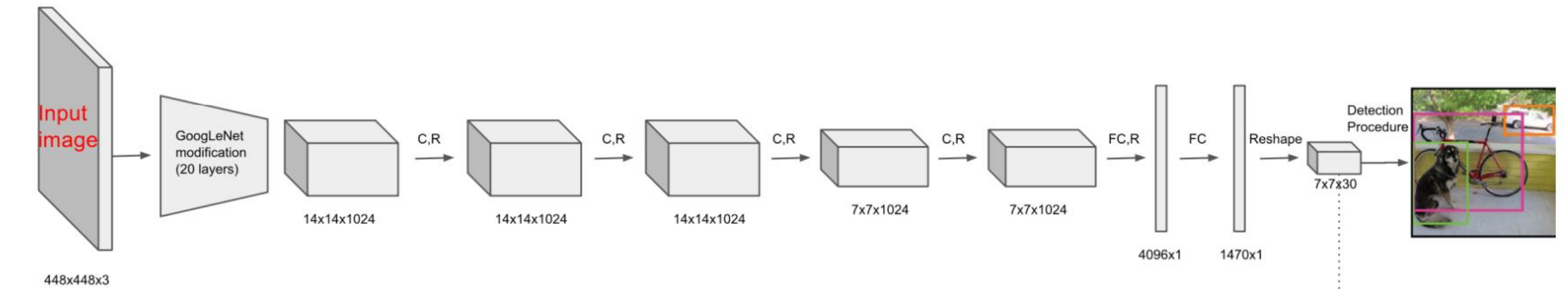


We then convert the model to perform detection. Ren et al. show that adding both convolutional and connected layers to pretrained networks can improve performance [29]. Following their example, we add four convolutional layers and two fully connected layers with randomly initialized weights. Detection often requires fine-grained visual information so we increase the input resolution of the network from 224×224 to 448×448 .

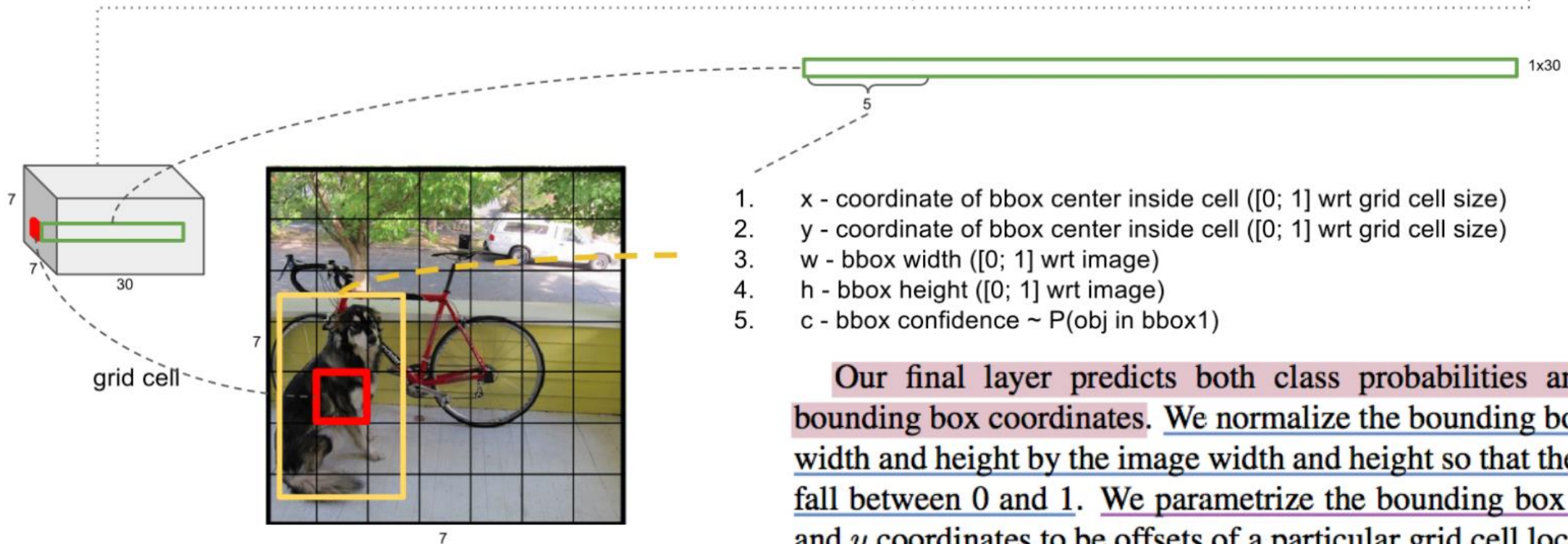






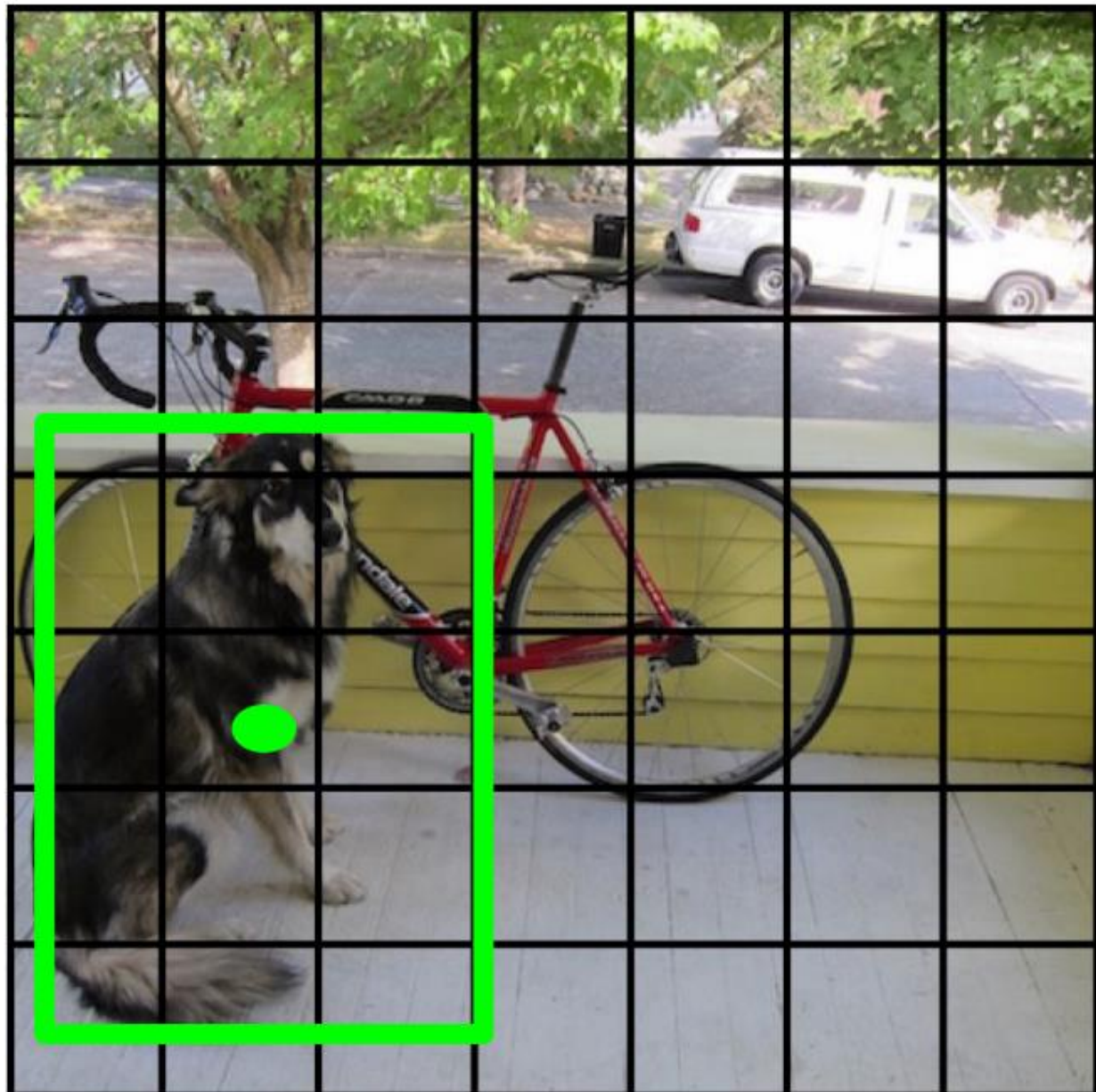


Tensor values interpretation

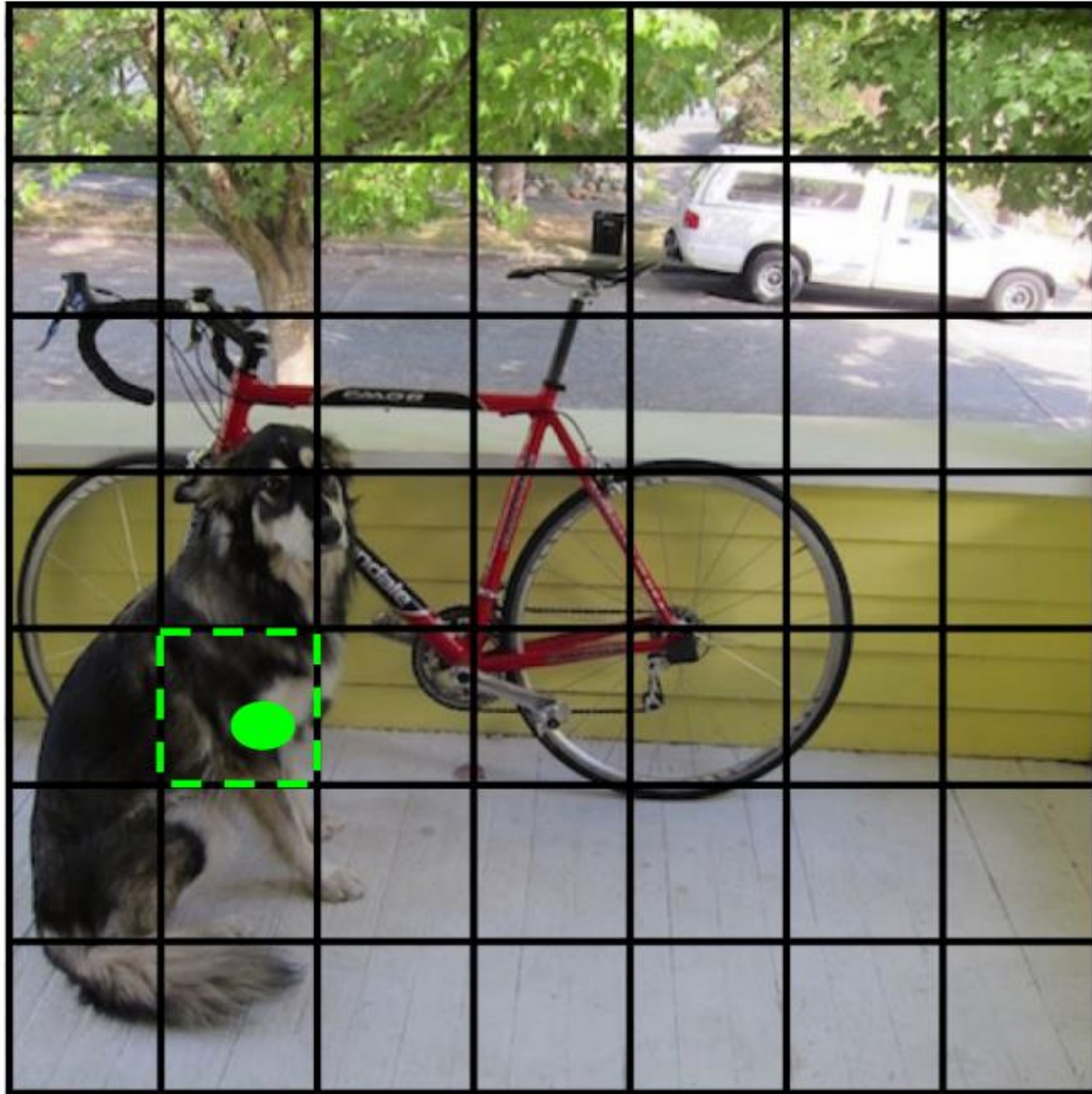


Our final layer predicts both class probabilities and bounding box coordinates. We normalize the bounding box width and height by the image width and height so that they fall between 0 and 1. We parametrize the bounding box x and y coordinates to be offsets of a particular grid cell location so they are also bounded between 0 and 1.

During training, match example to the right cell

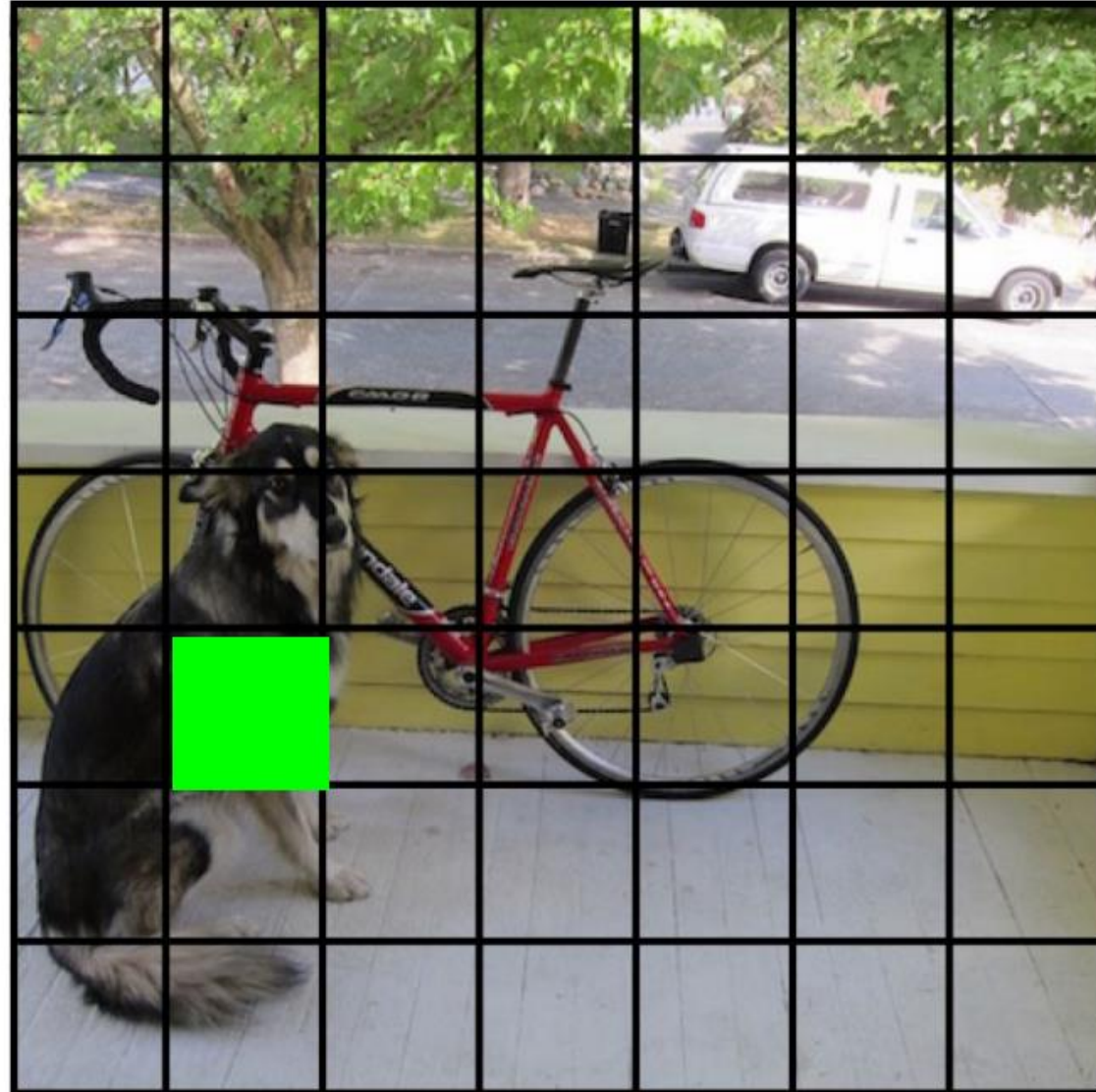


During training, match example to the right cell

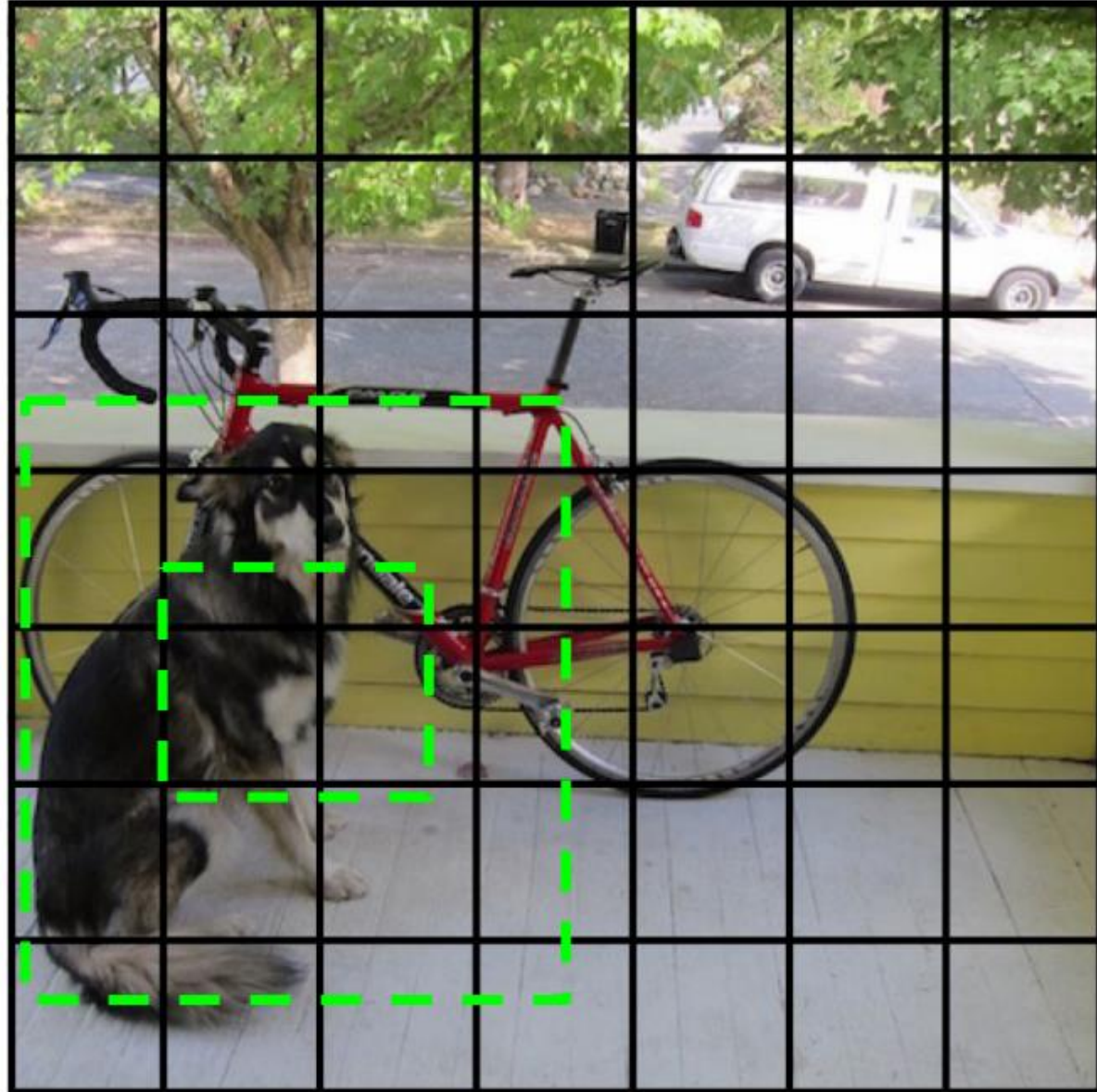


Adjust that cell's class prediction

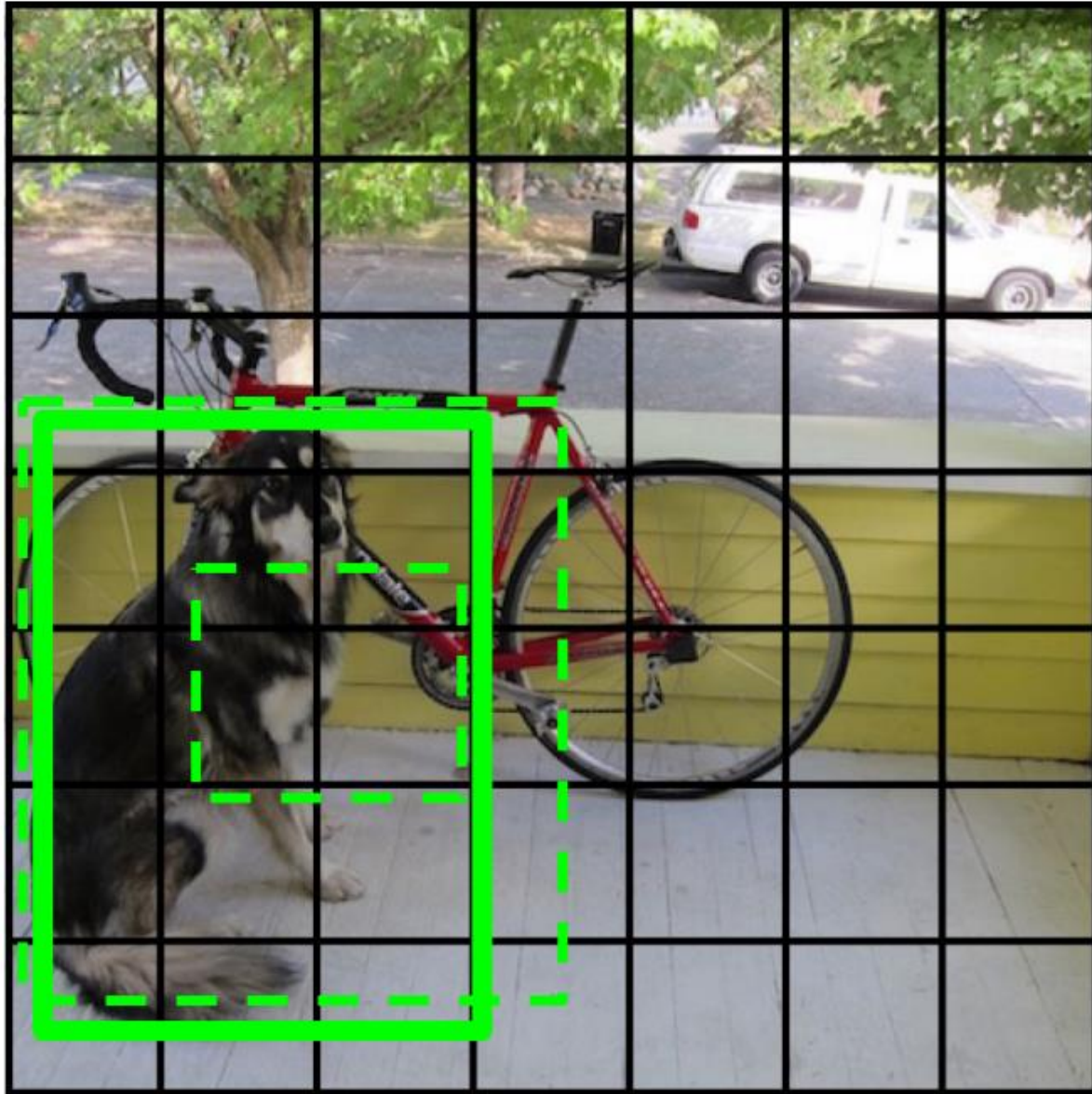
Dog = 1
Cat = 0
Bike = 0
...



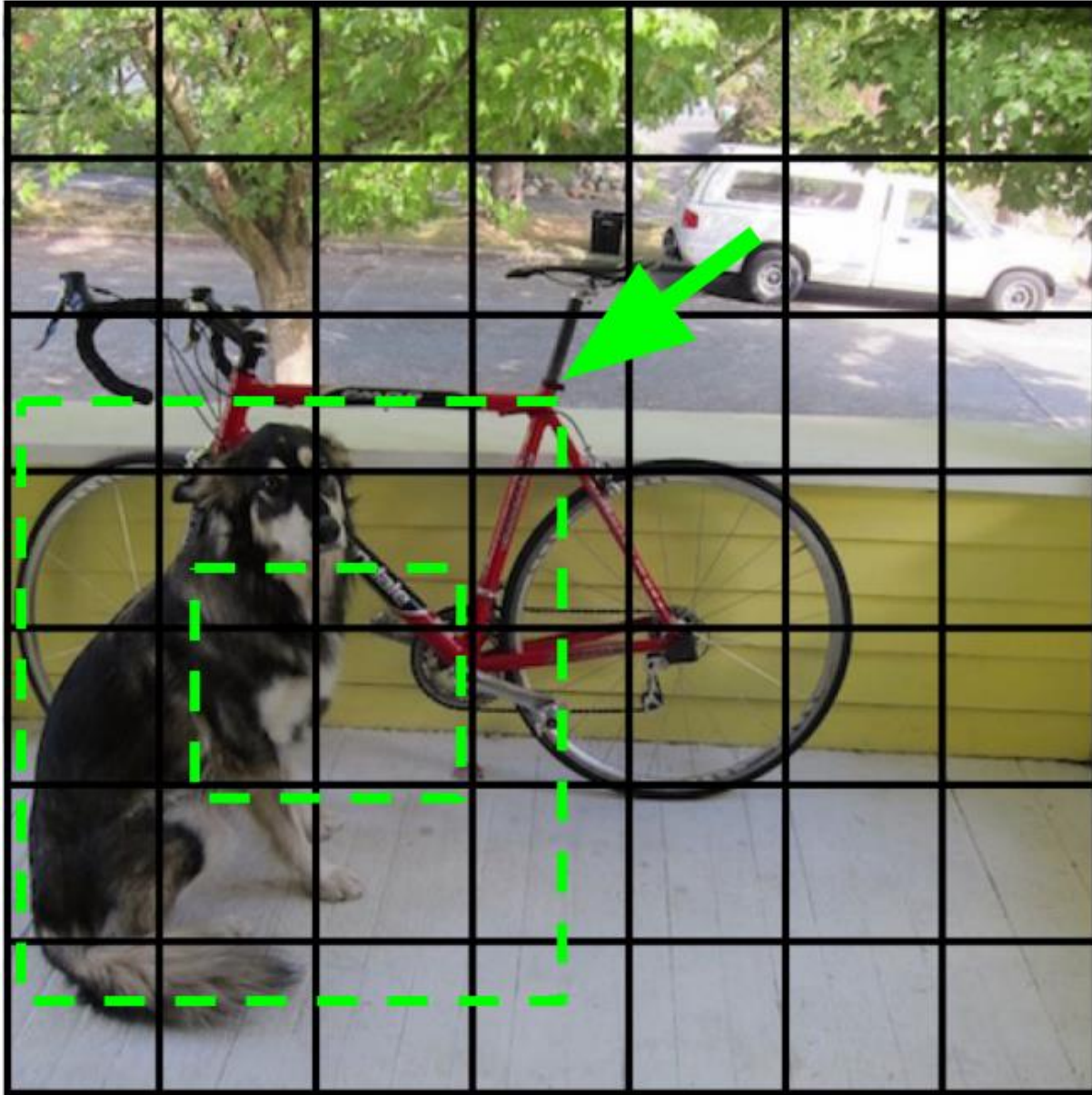
Look at that cell's predicted boxes



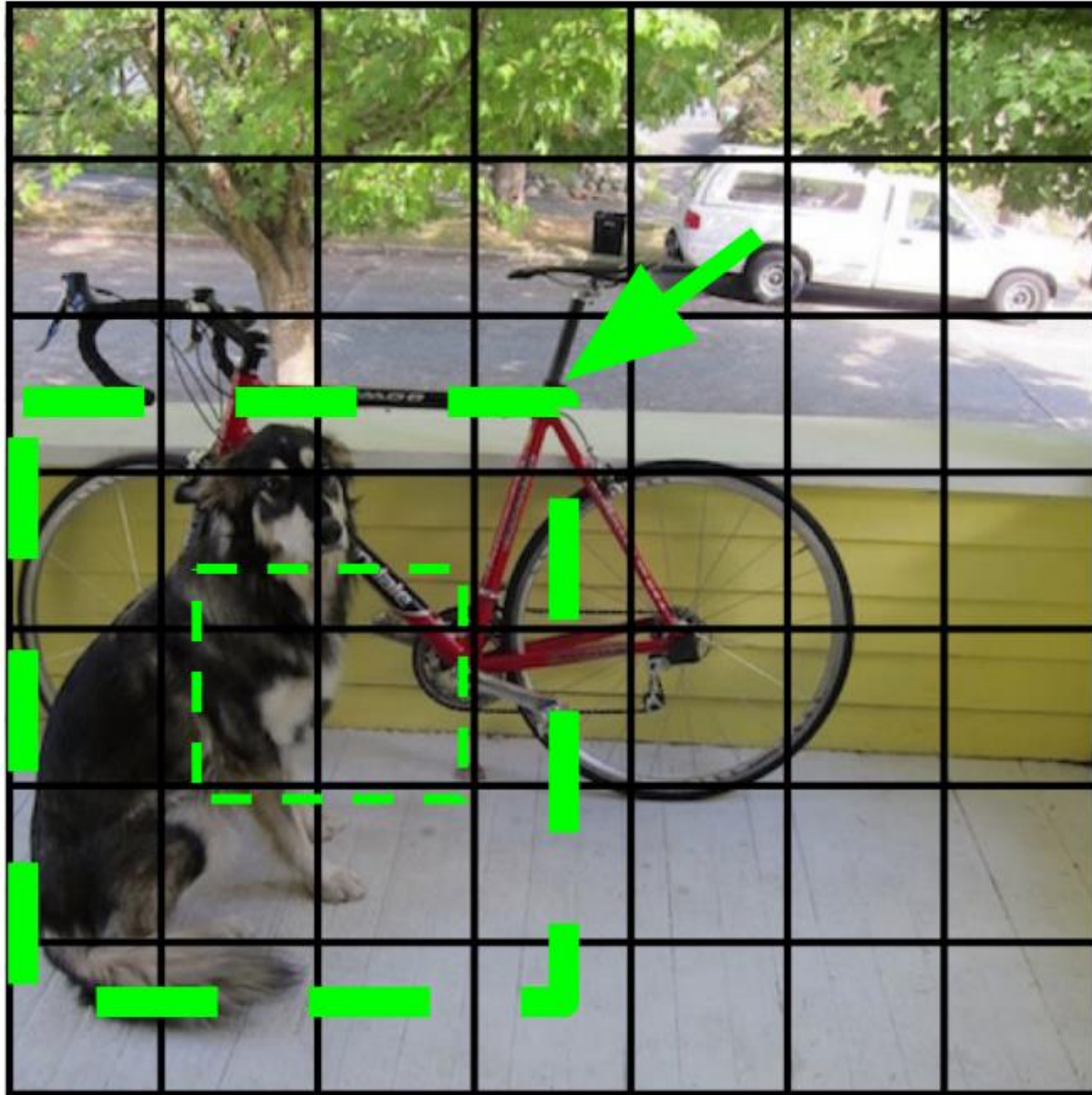
Find the best one, adjust it, increase the confidence



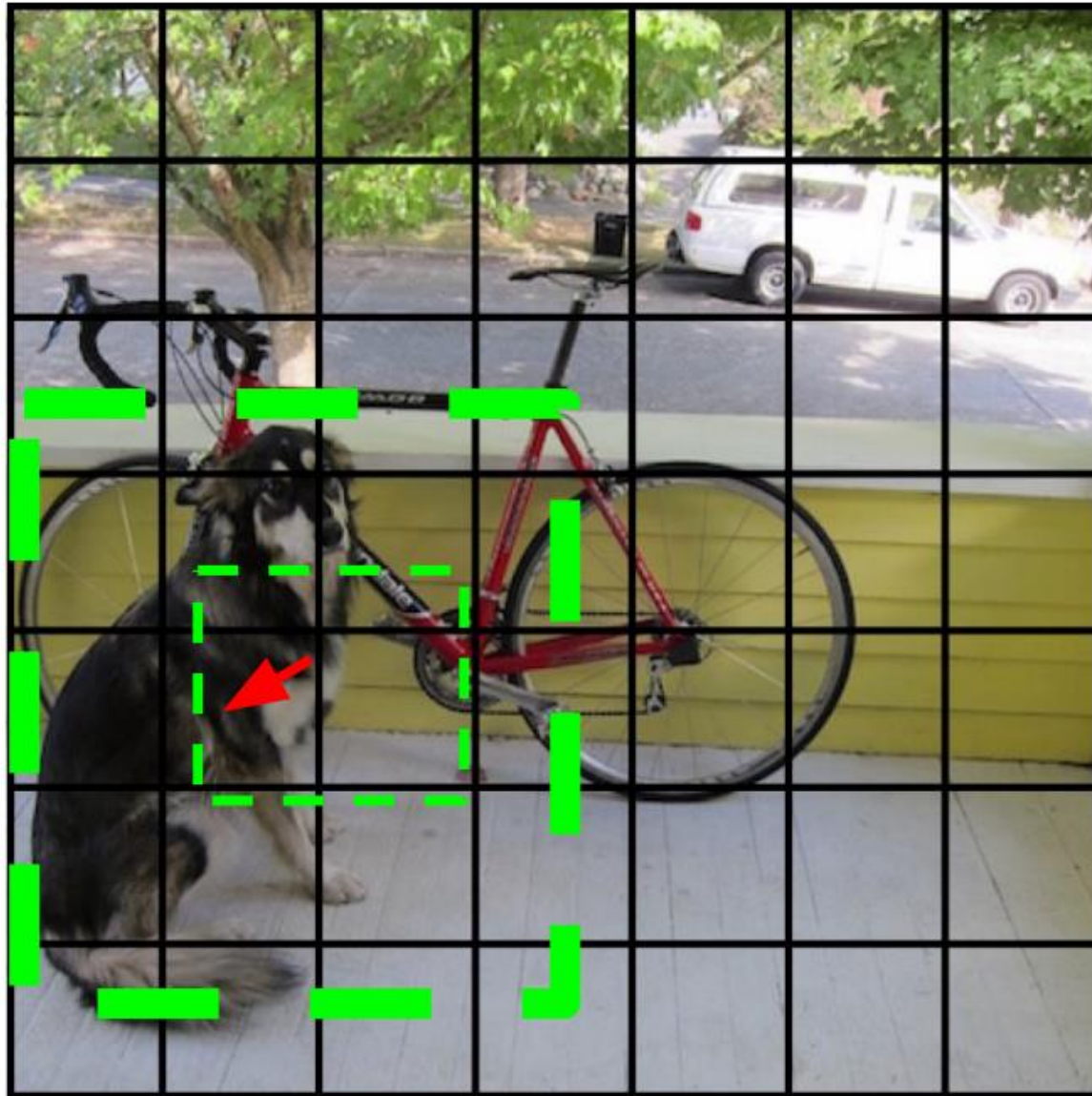
Find the best one, adjust it, increase the confidence



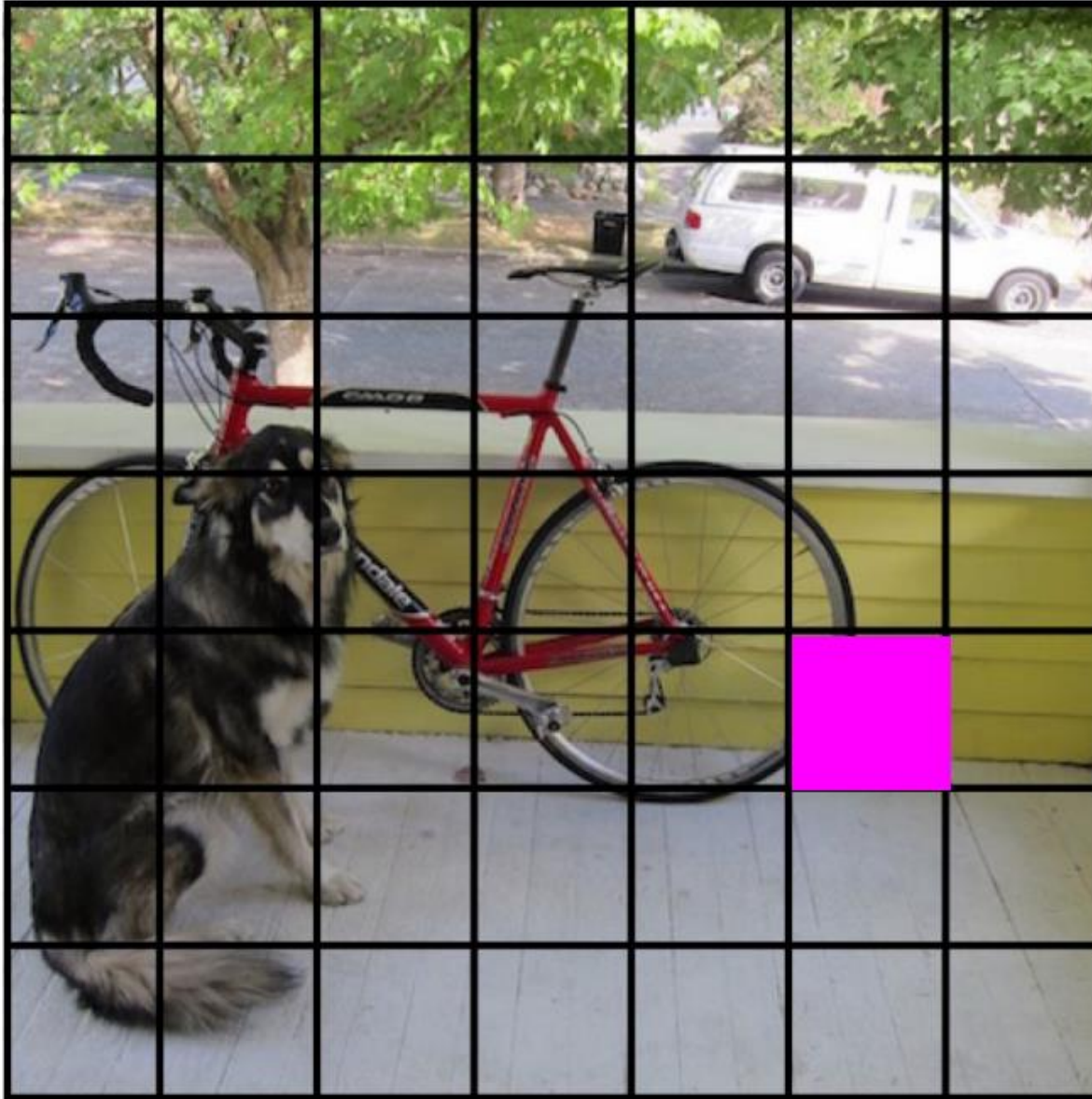
Find the best one, adjust it, increase the confidence



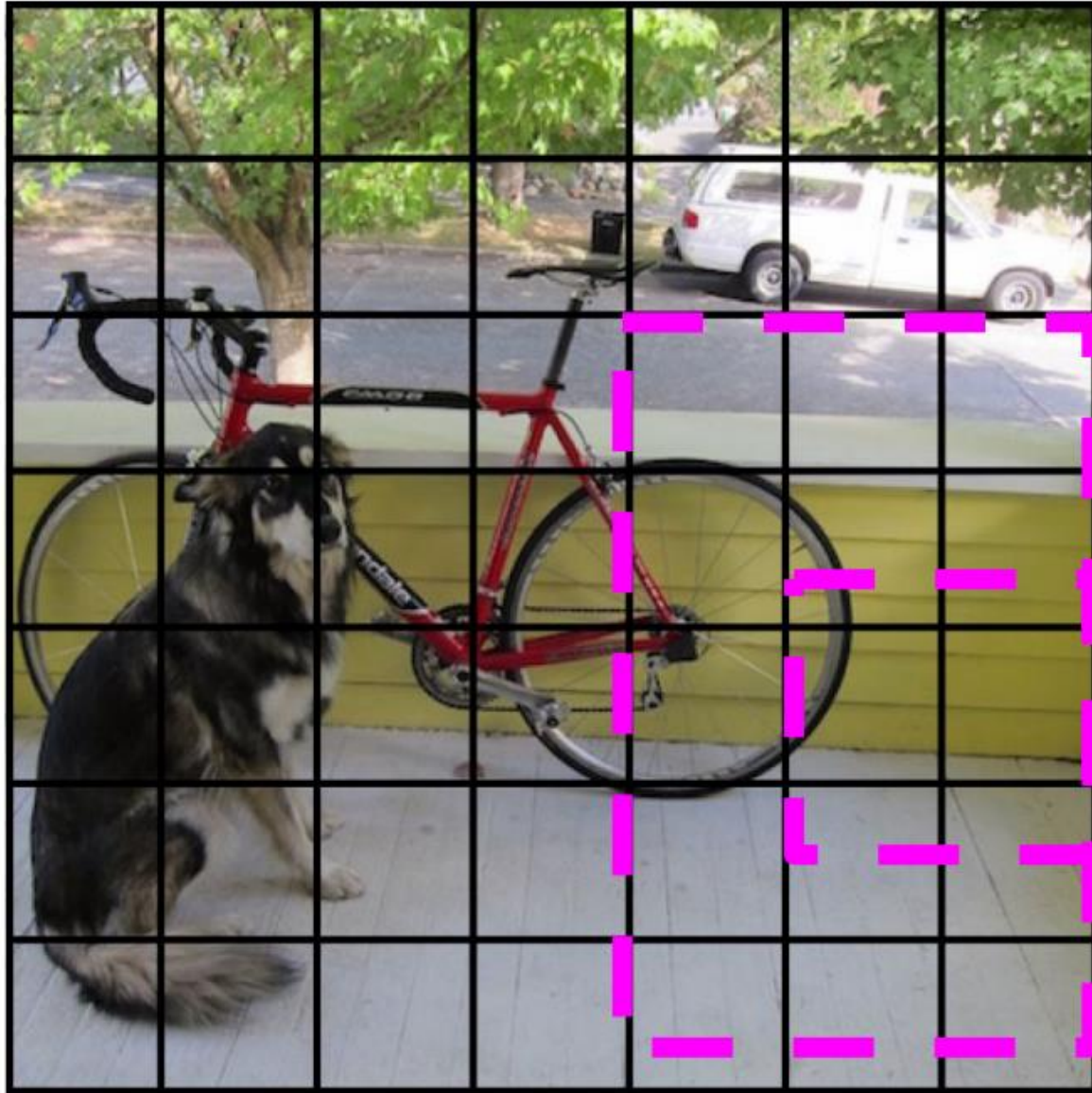
Decrease the confidence of the other box



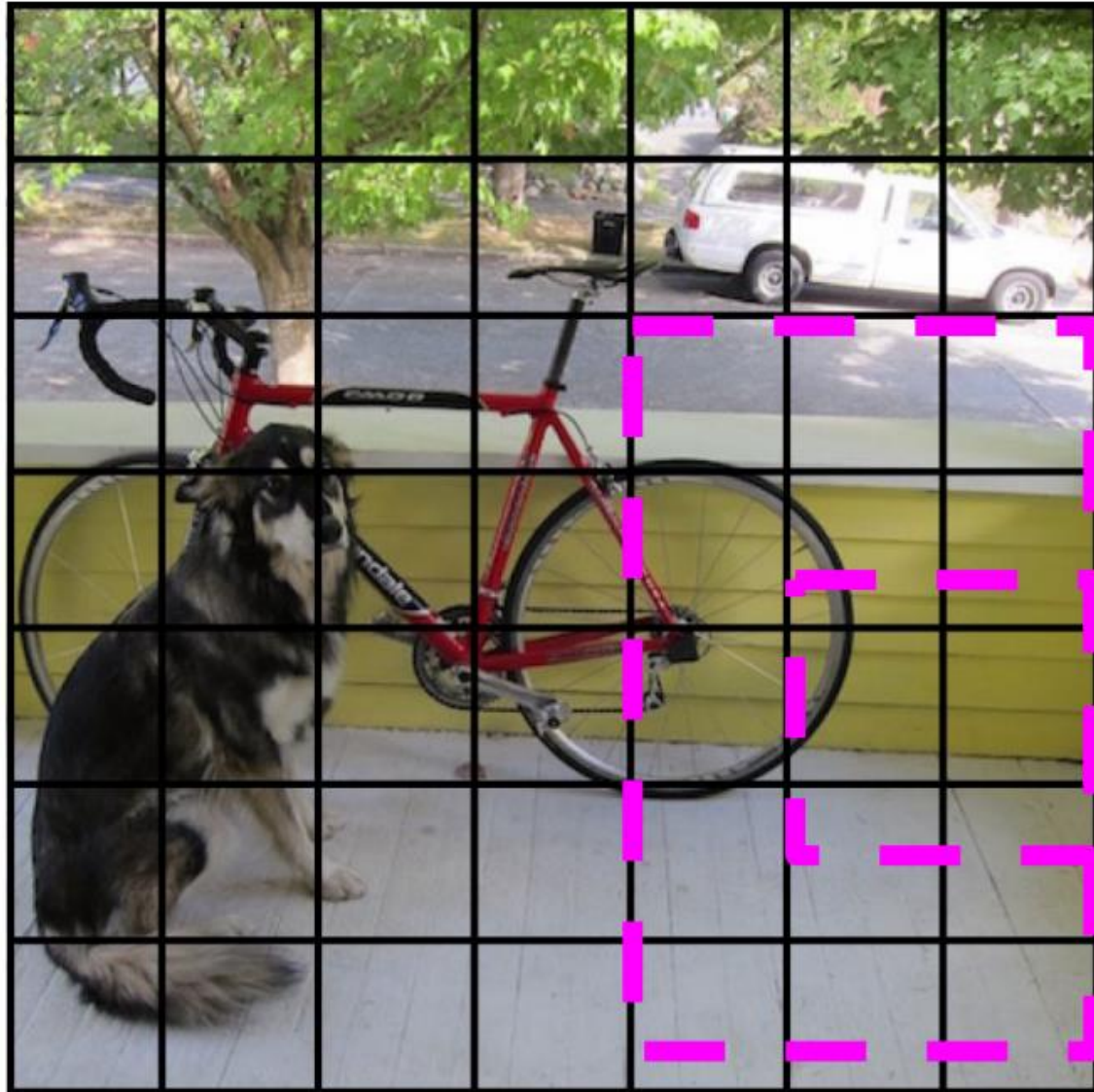
Some cells don't have any ground truth detections!



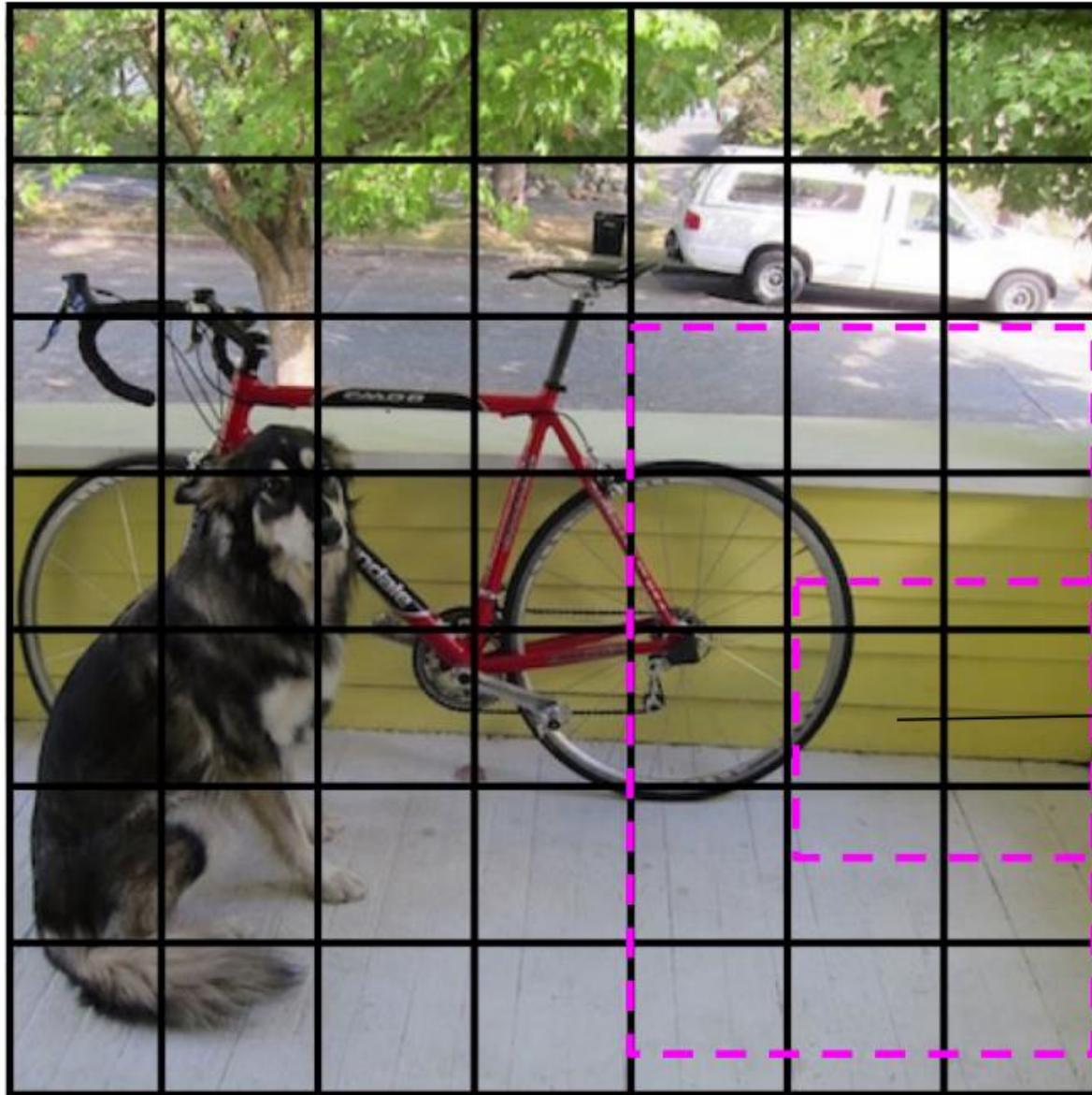
Some cells don't have any ground truth detections!



Decrease the confidence of boxes boxes



Don't adjust the class probabilities or coordinates



Loss Function (sum-squared error)

loss function:

$$\begin{aligned} & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\ & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\ & + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\ & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\ & + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \quad (3) \end{aligned}$$

model. We use sum-squared error because it is easy to optimize, however it does not perfectly align with our goal of maximizing average precision. It weights localization error equally with classification error which may not be ideal. Also, in every image many grid cells do not contain any object. This pushes the “confidence” scores of those cells towards zero, often overpowering the gradient from cells that do contain objects. This can lead to model instability, causing training to diverge early on.

To remedy this, we increase the loss from bounding box coordinate predictions and decrease the loss from confidence predictions for boxes that don’t contain objects. We use two parameters, λ_{coord} and λ_{noobj} to accomplish this. We set $\lambda_{\text{coord}} = 5$ and $\lambda_{\text{noobj}} = .5$.

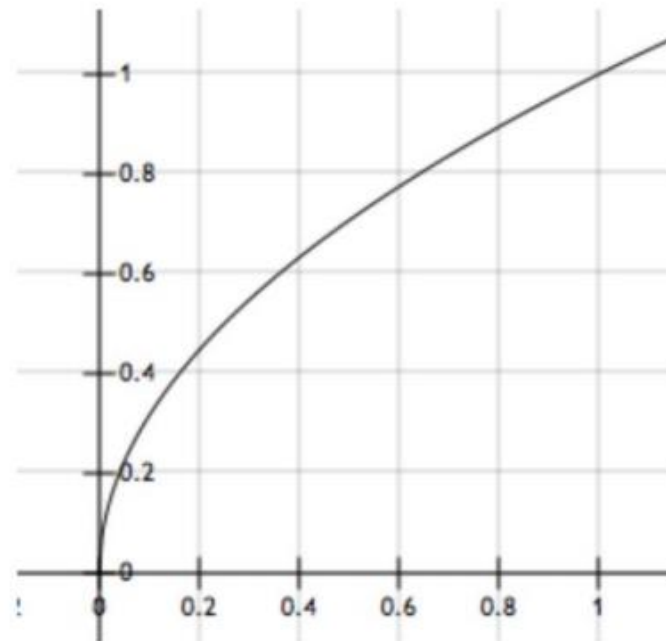
$$\lambda_{\text{coord}}=5, \quad \lambda_{\text{noobj}}=0.5$$

Loss Function (sum-squared error)

loss function:

$$\begin{aligned} & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\ & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\ & + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\ & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\ & + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \quad (3) \end{aligned}$$

Sum-squared error also equally weights errors in large boxes and small boxes. Our error metric should reflect that small deviations in large boxes matter less than in small boxes. To partially address this we predict the square root of the bounding box width and height instead of the width and height directly.



Loss Function (sum-squared error)

loss function:

$$\begin{aligned}
 & \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\
 & + \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right] \\
 & + \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\
 & + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\
 & + \sum_{i=0}^{S^2} \mathbb{1}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \quad (3)
 \end{aligned}$$

$$\mathbb{1}_{ij}^{\text{obj}}$$

The j th bbox predictor in *cell i* is “responsible” for that prediction

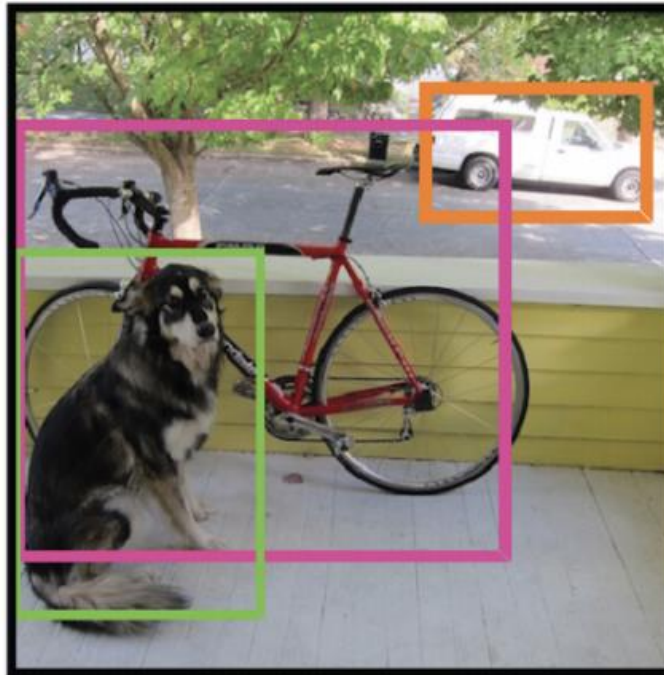
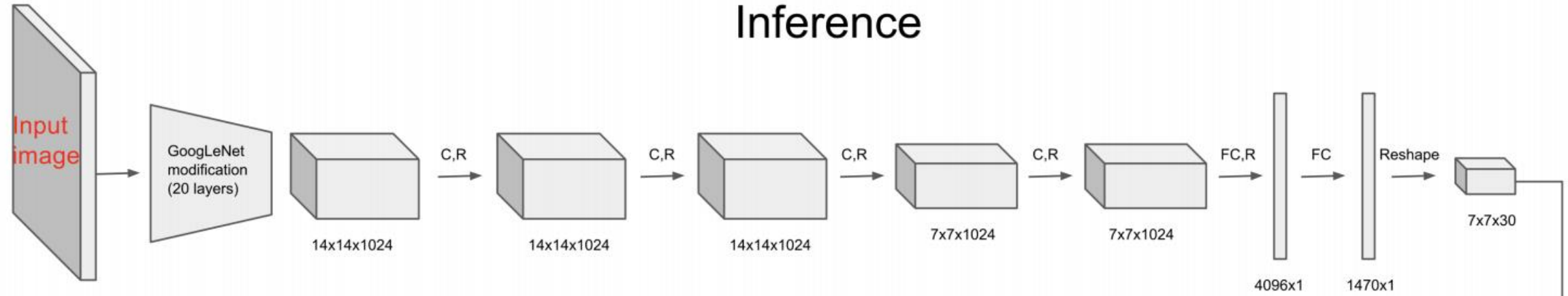
$$\mathbb{1}_{ij}^{\text{noobj}}$$

$$\mathbb{1}_i^{\text{obj}}$$

If object appears in *cell i*

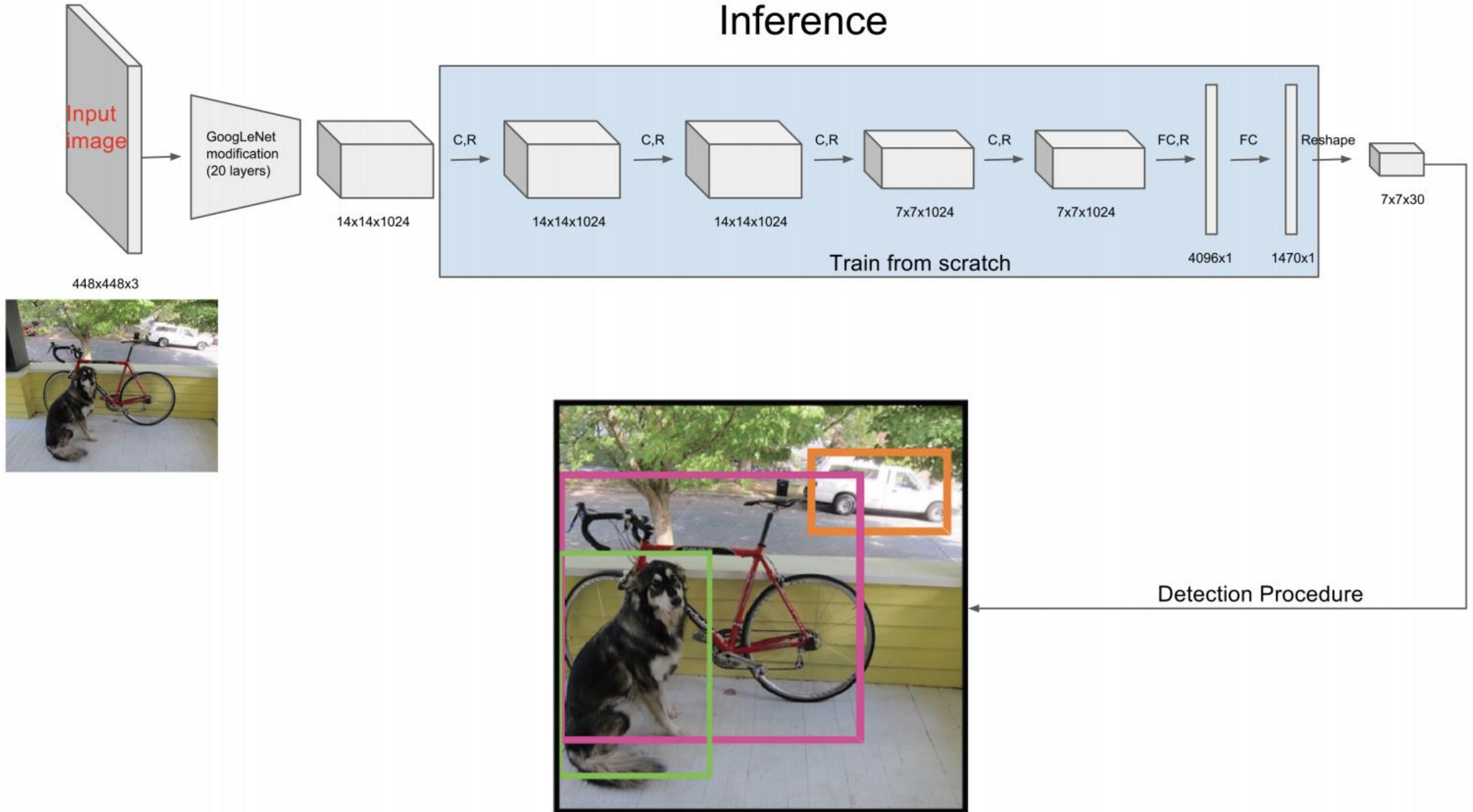
Note that the loss function only penalizes classification error if an object is present in that grid cell (hence the conditional class probability discussed earlier). It also only penalizes bounding box coordinate error if that predictor is “responsible” for the ground truth box (i.e. has the highest IOU of any predictor in that grid cell).

Inference

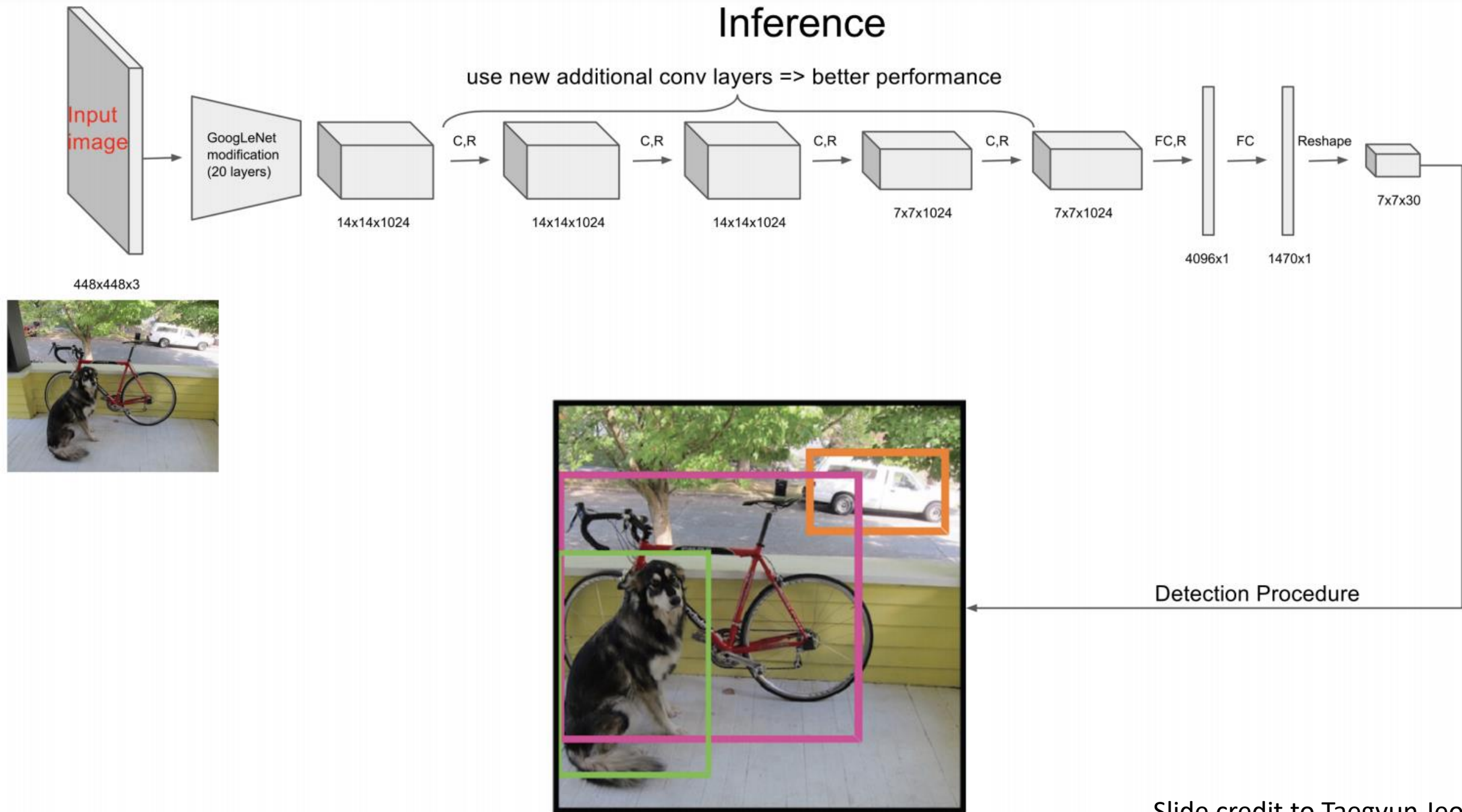


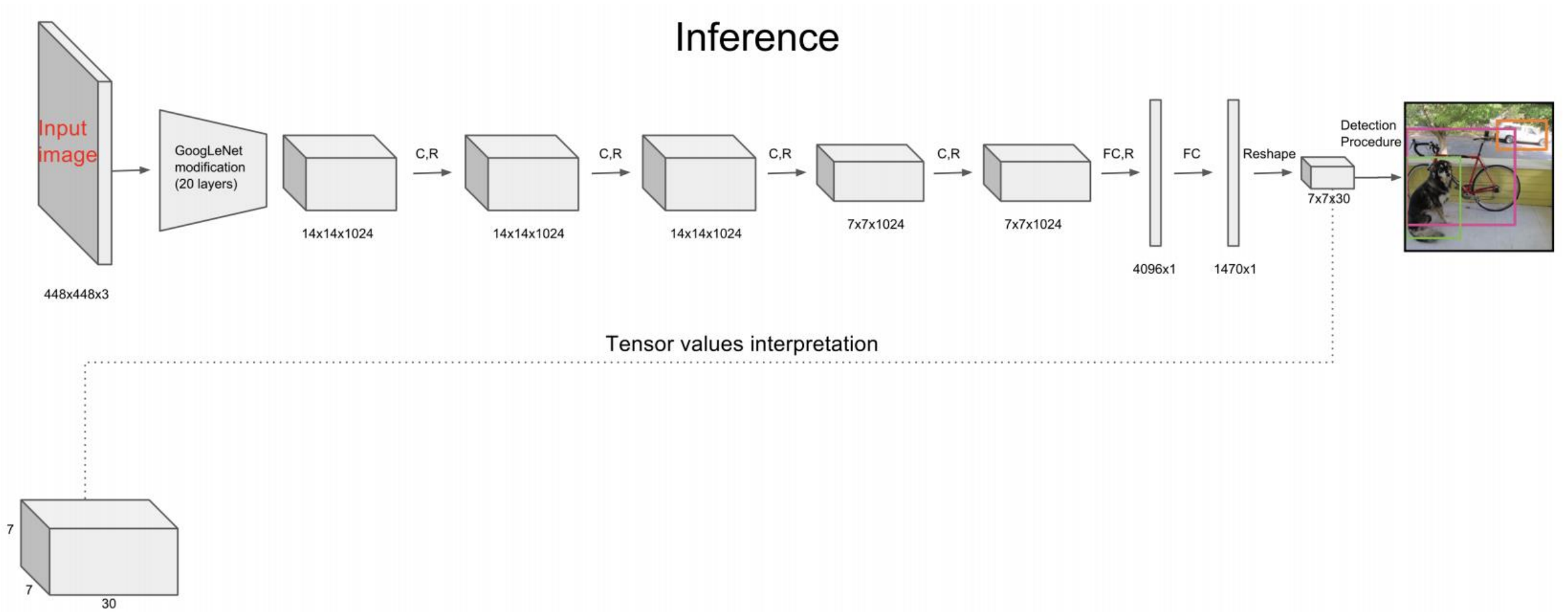
Detection Procedure

Inference

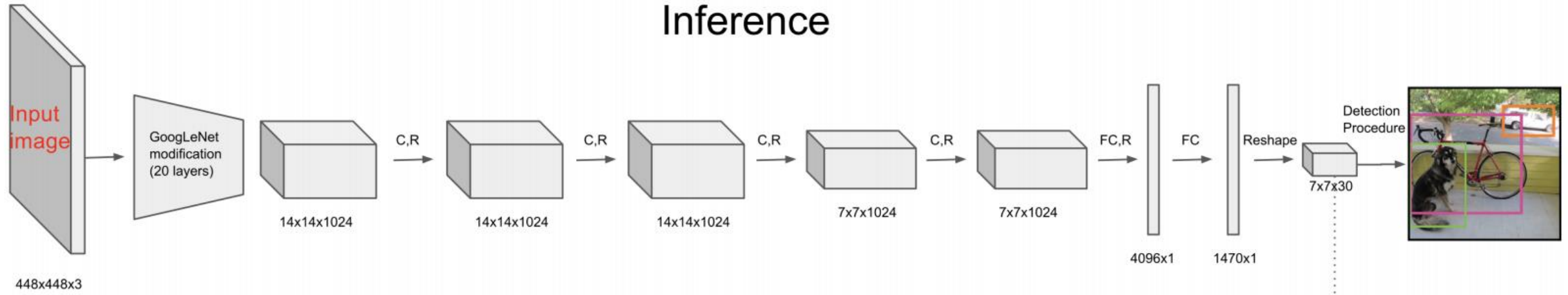


Inference

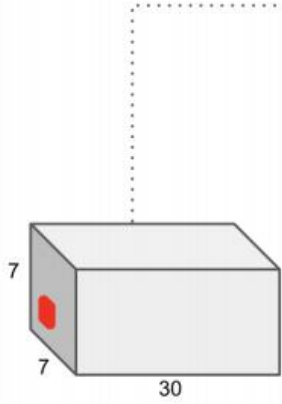




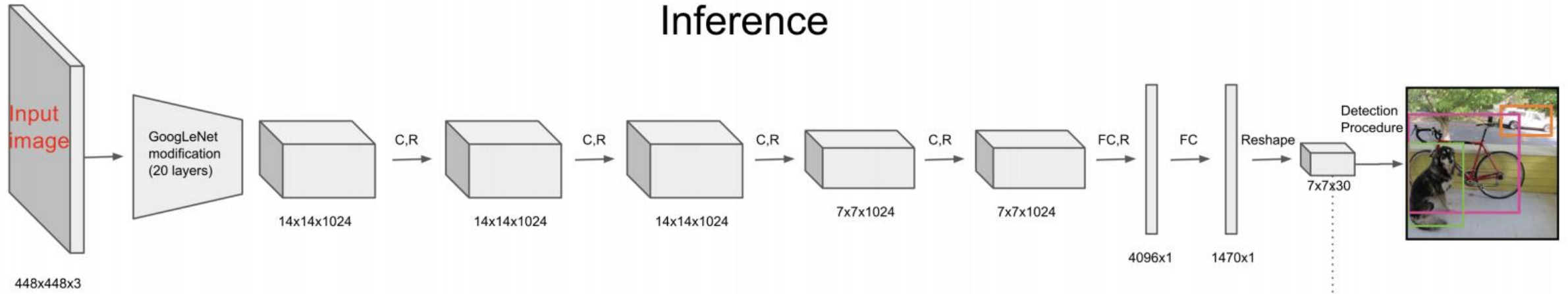
Inference



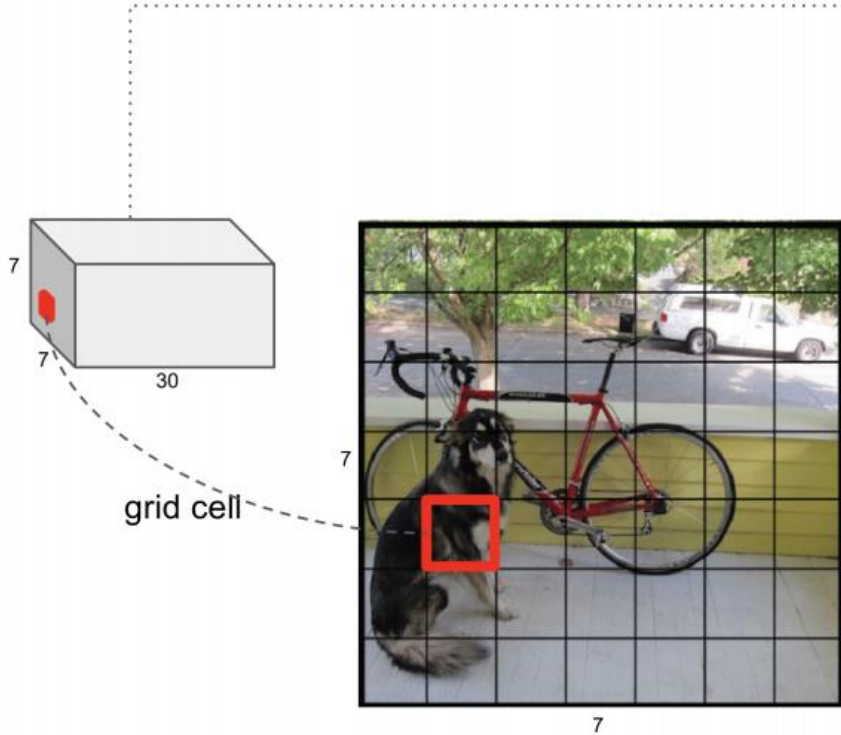
Tensor values interpretation



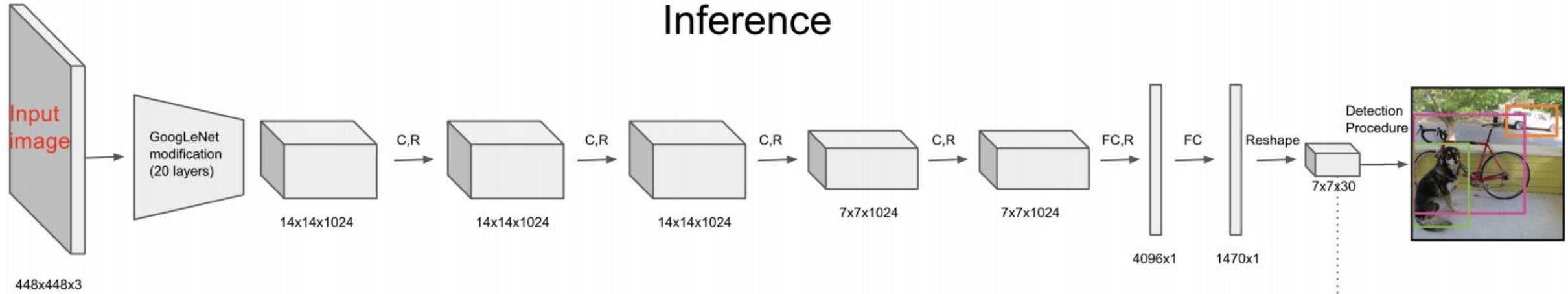
Inference



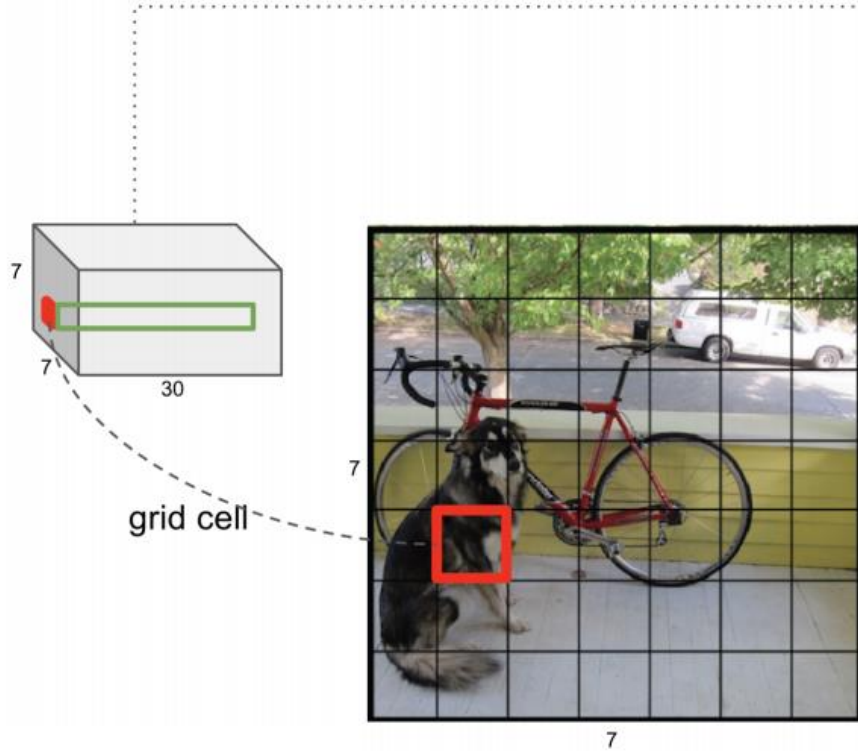
Tensor values interpretation



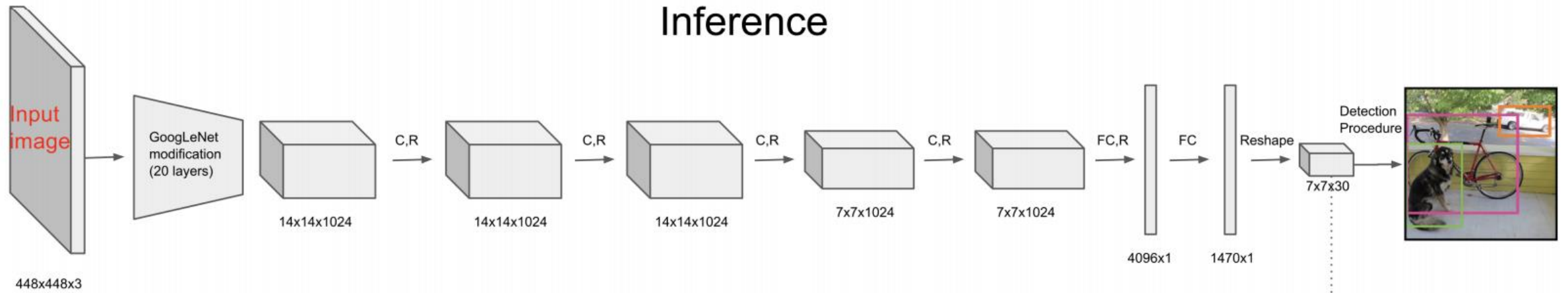
Inference



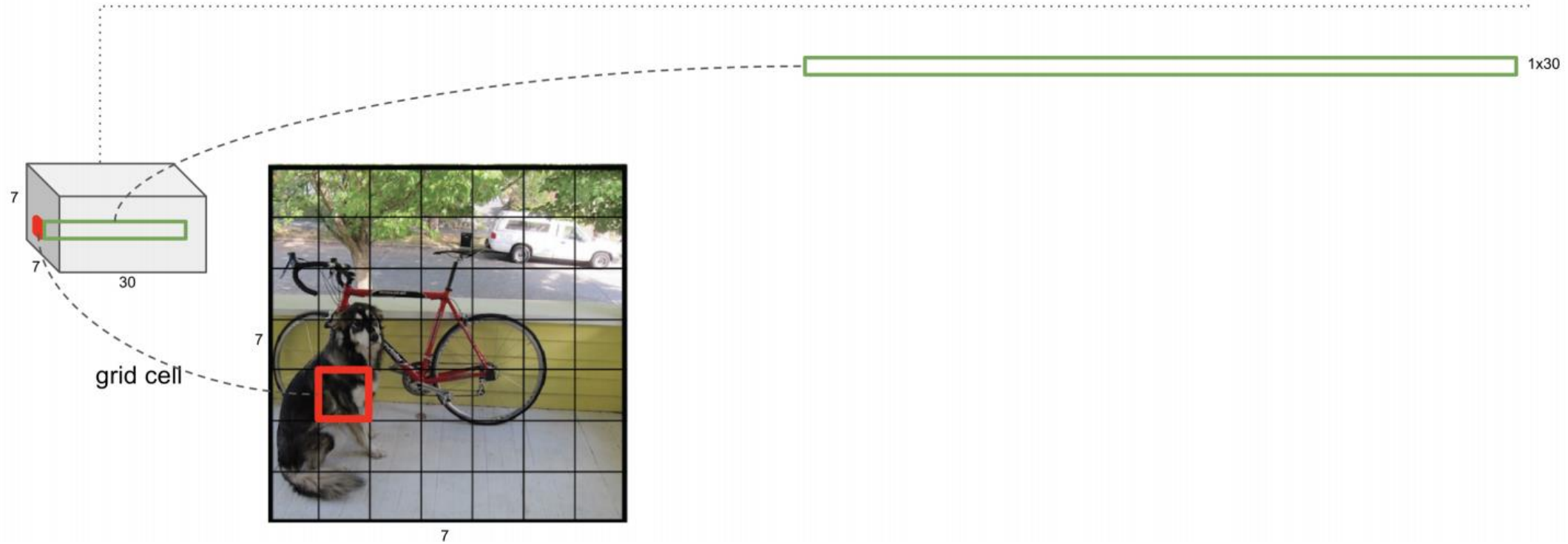
Tensor values interpretation



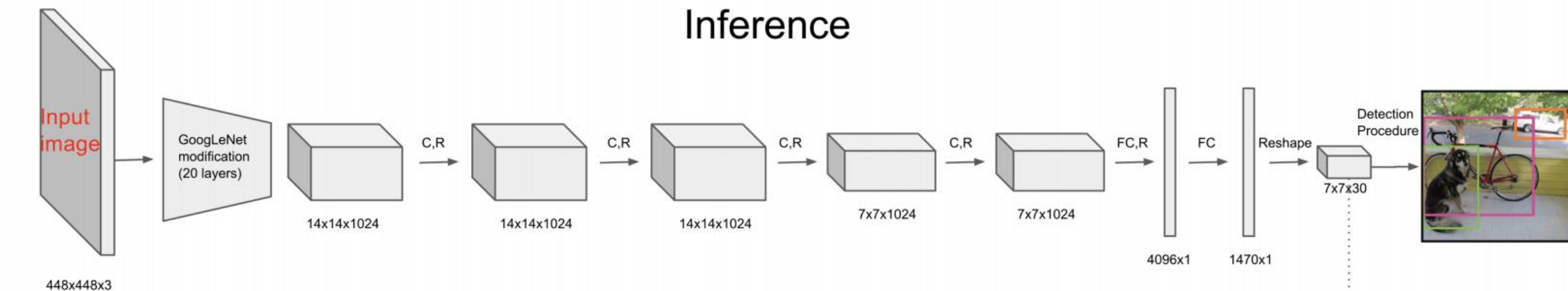
Inference



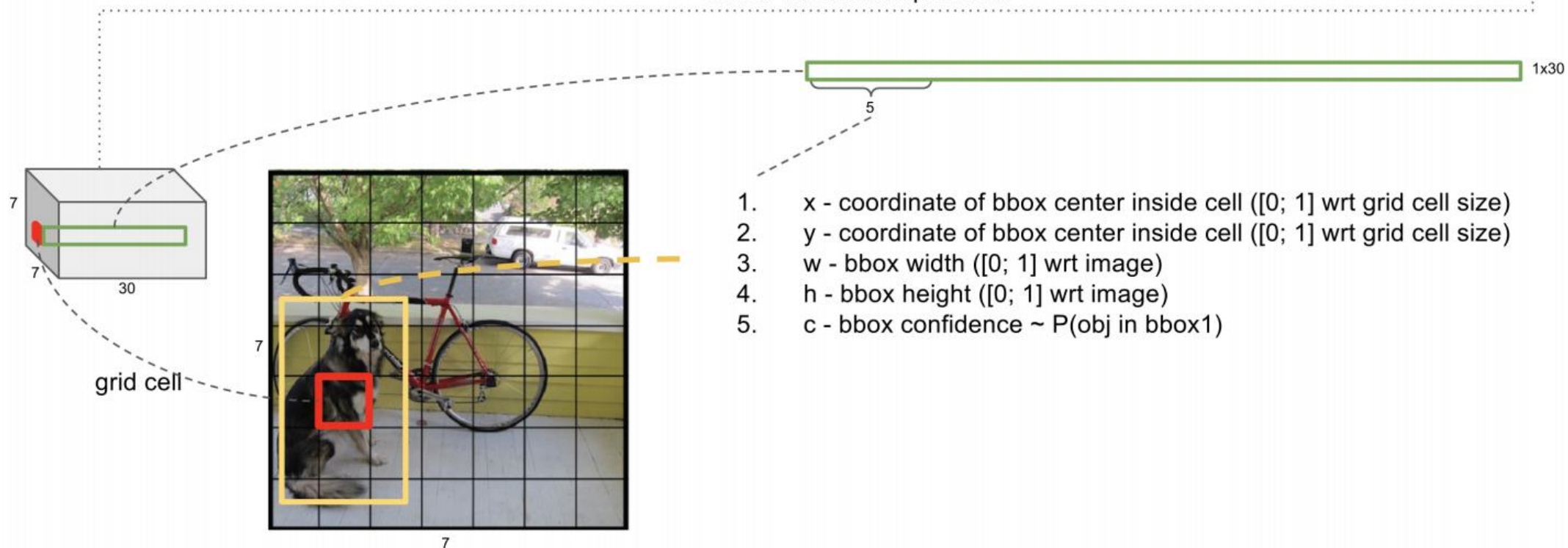
Tensor values interpretation



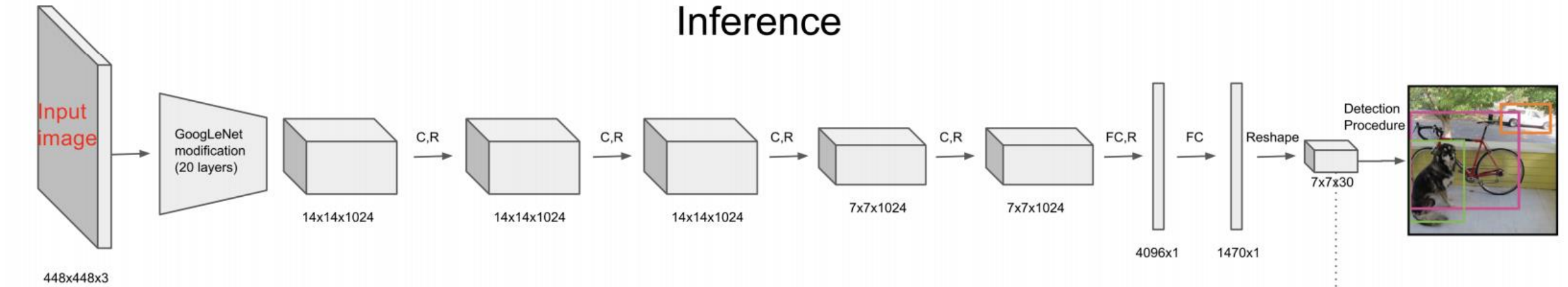
Inference



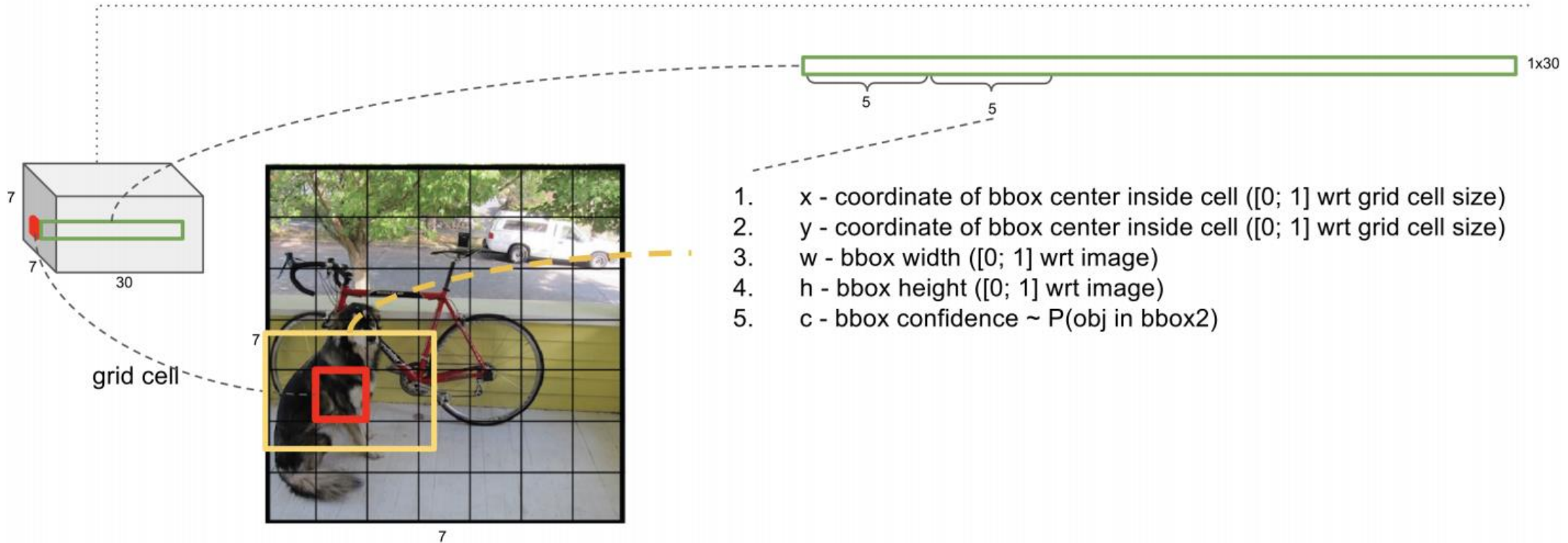
Tensor values interpretation



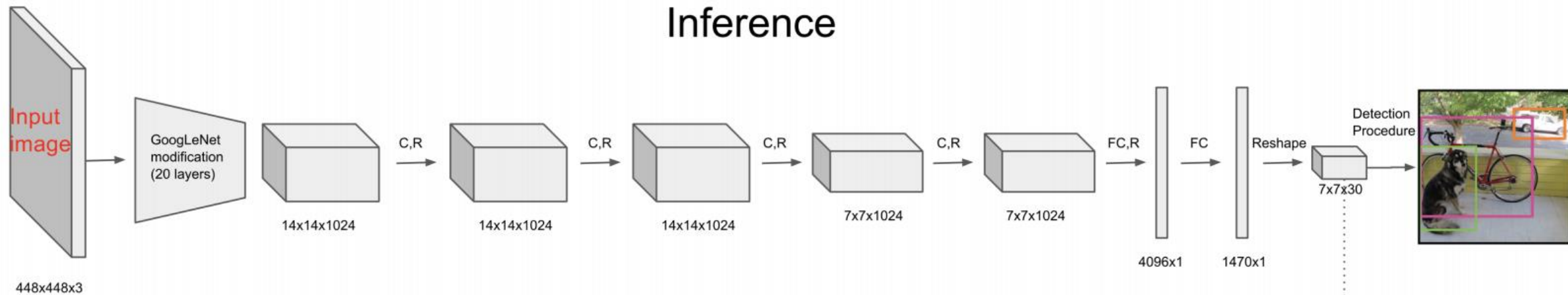
Inference



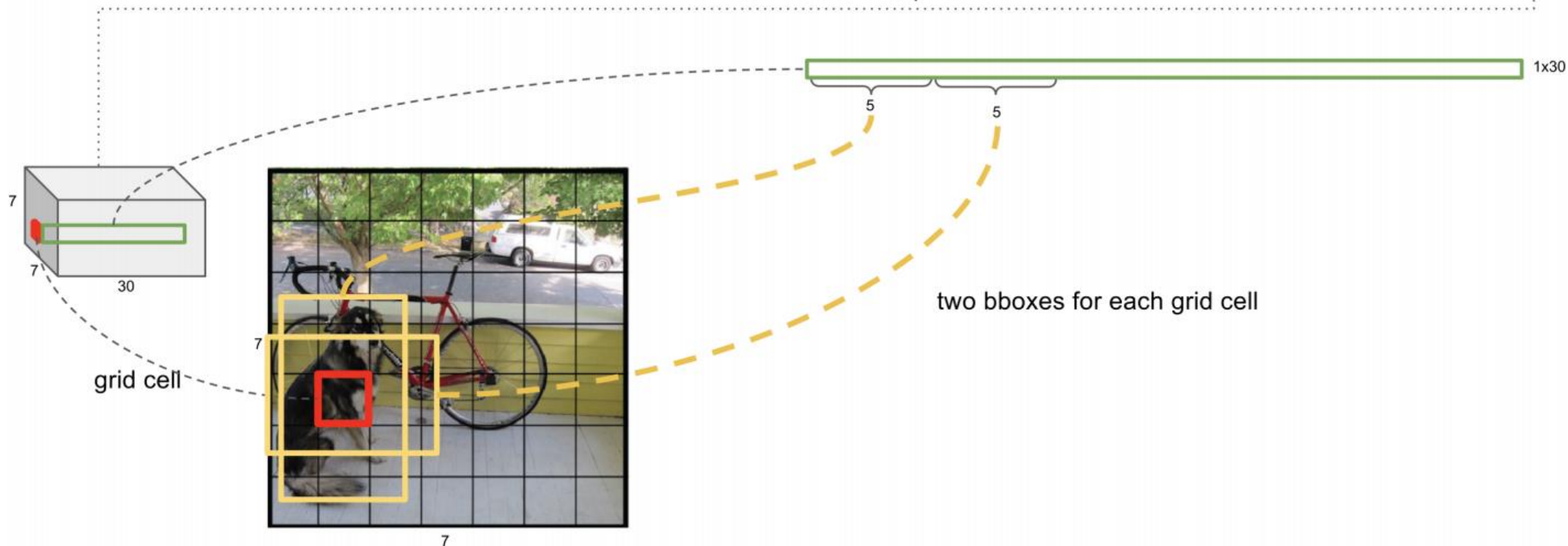
Tensor values interpretation



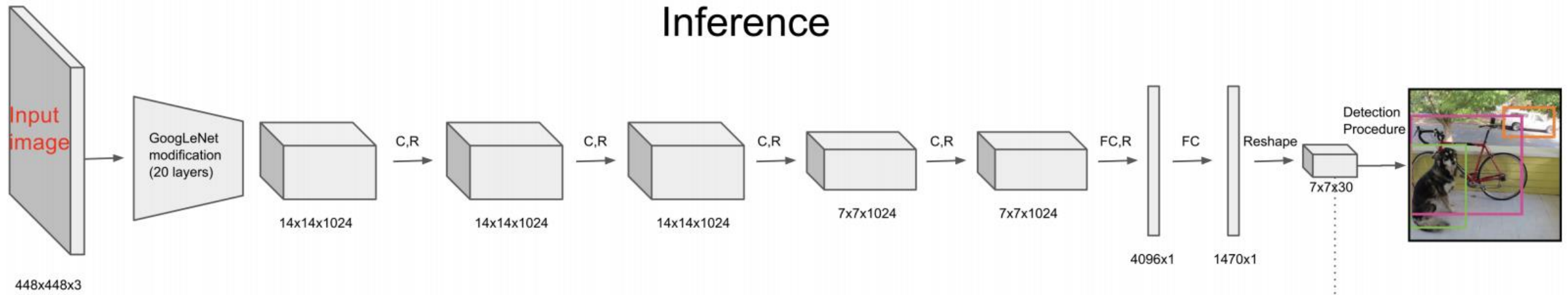
Inference



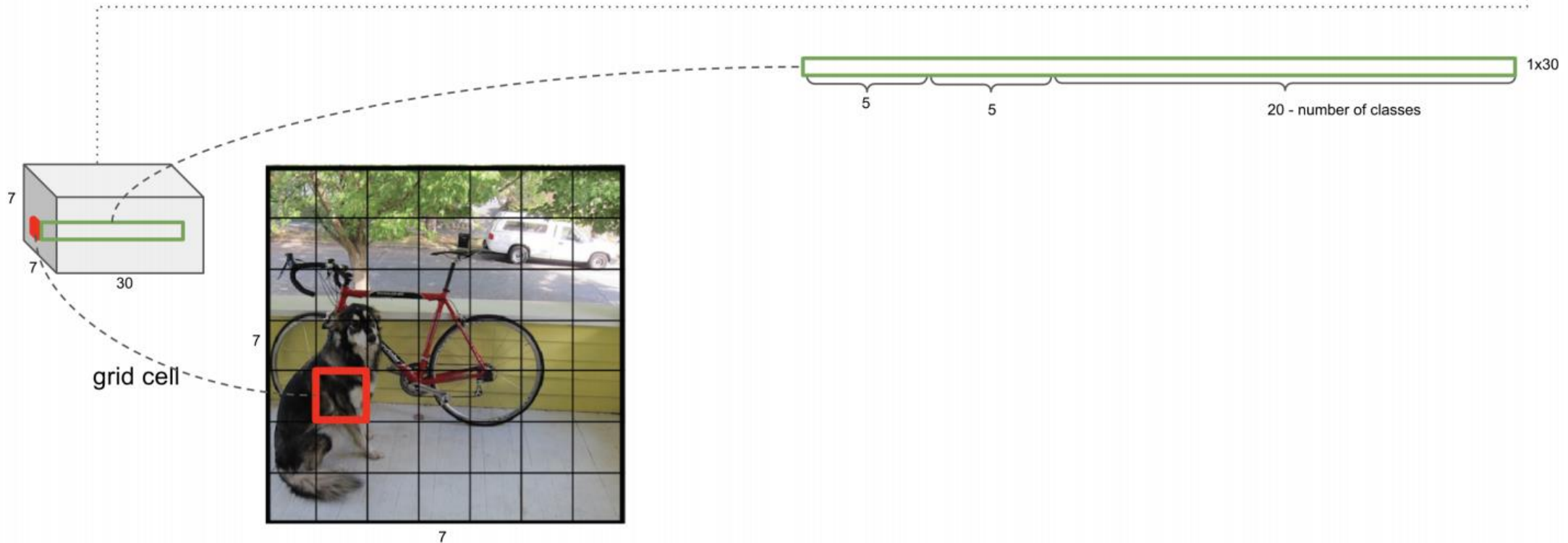
Tensor values interpretation



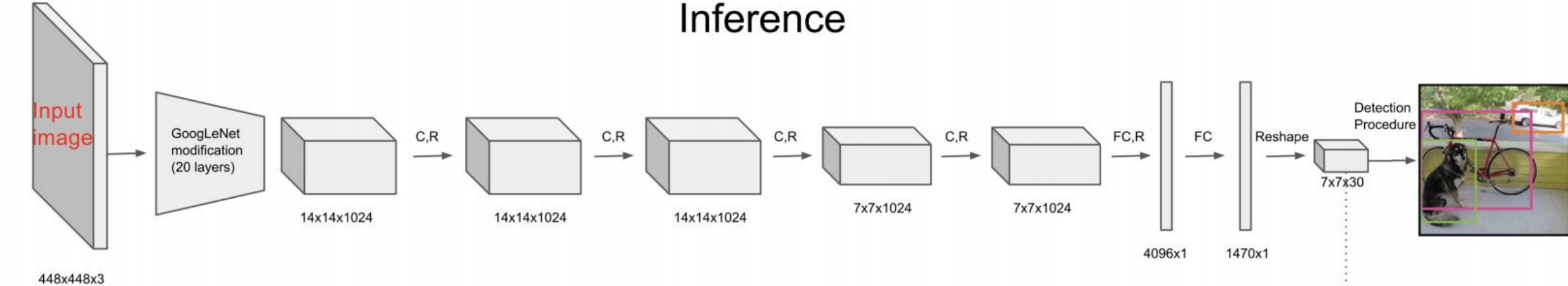
Inference



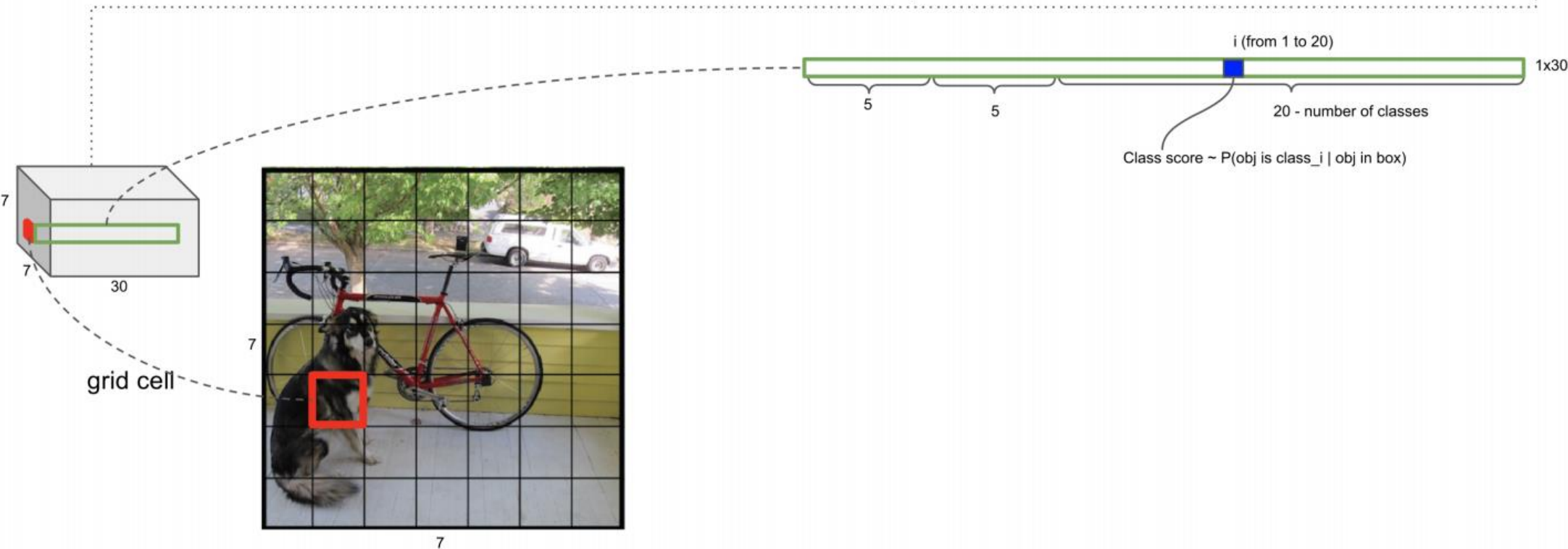
Tensor values interpretation



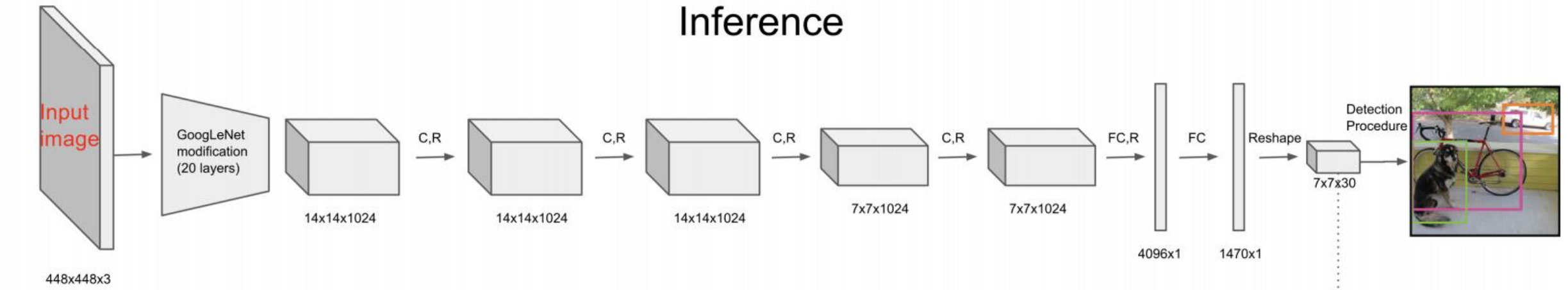
Inference



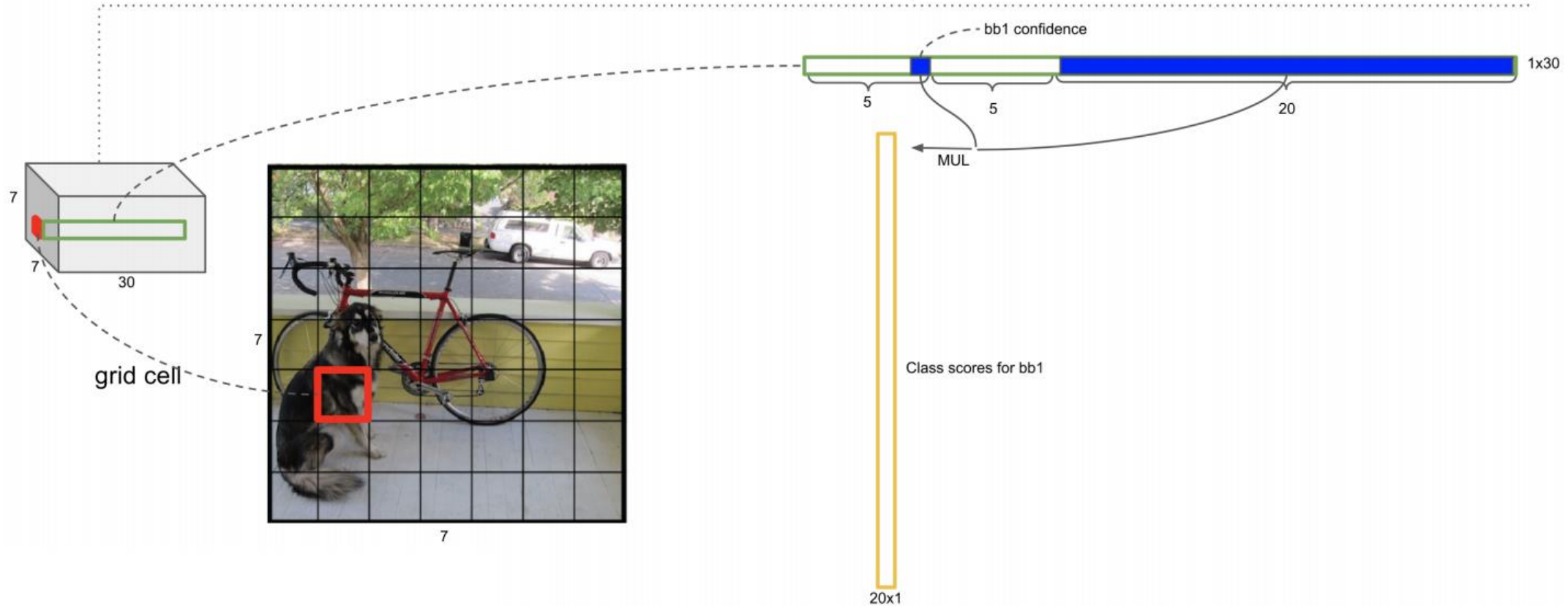
Tensor values interpretation



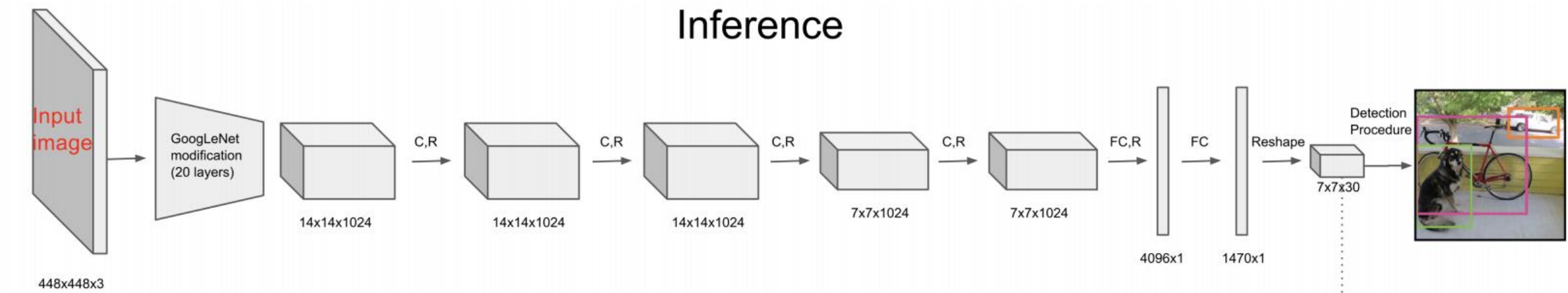
Inference



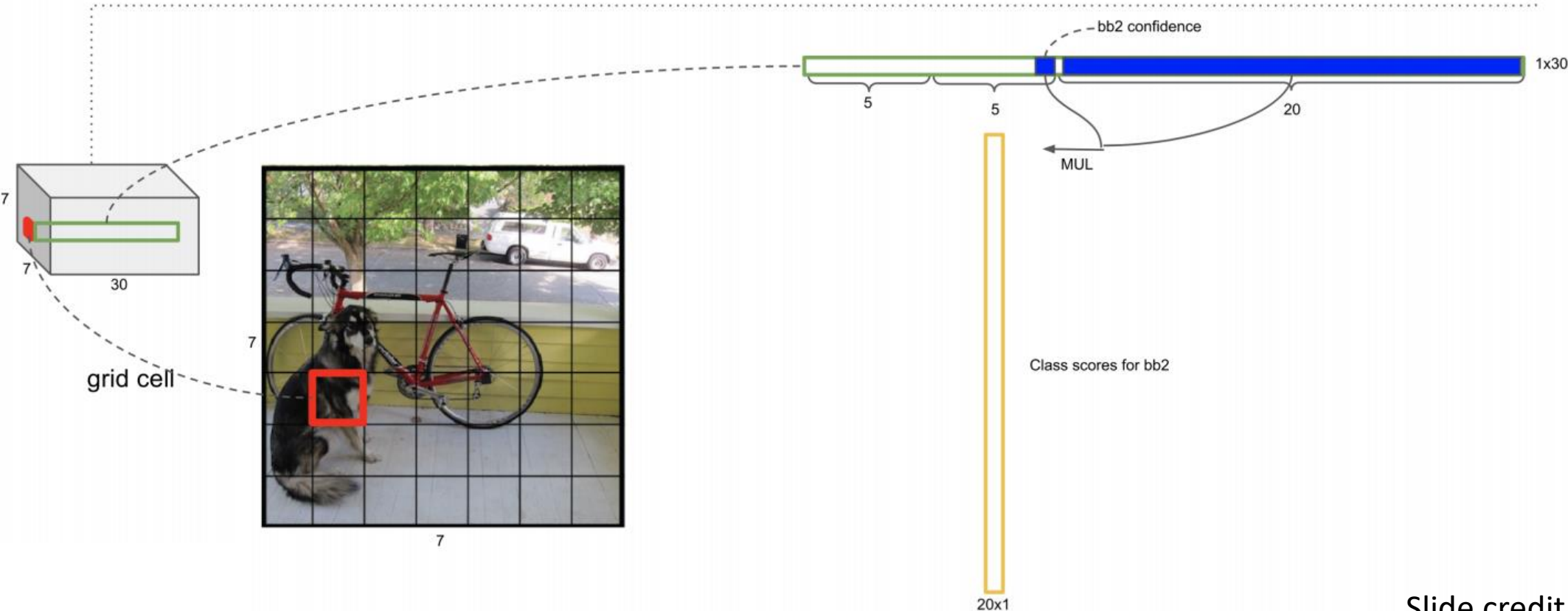
Tensor values interpretation



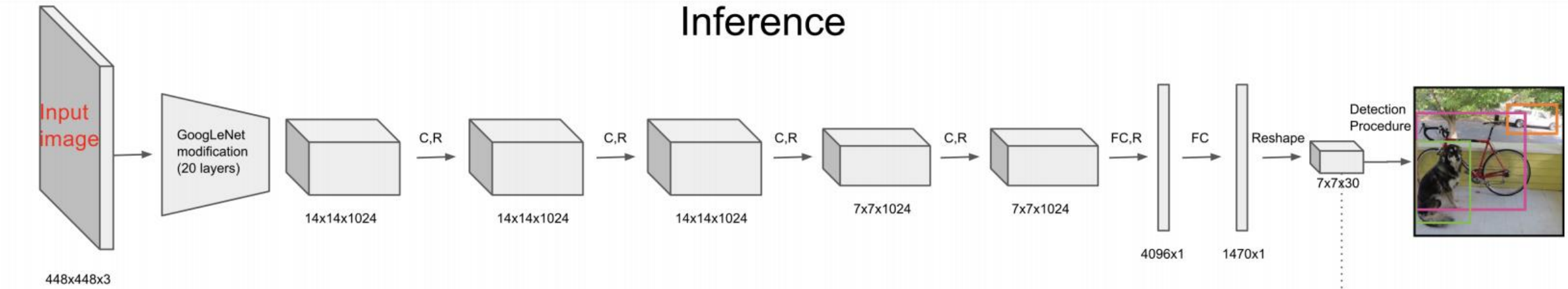
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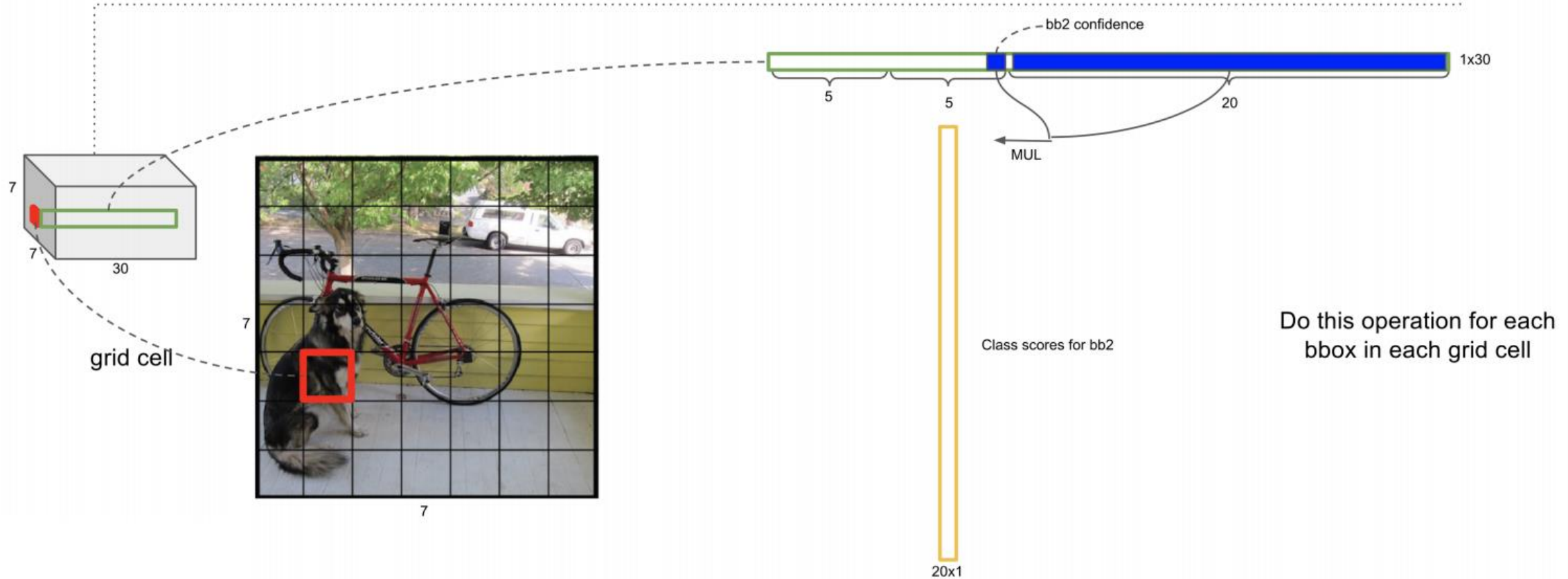
Tensor values interpretation



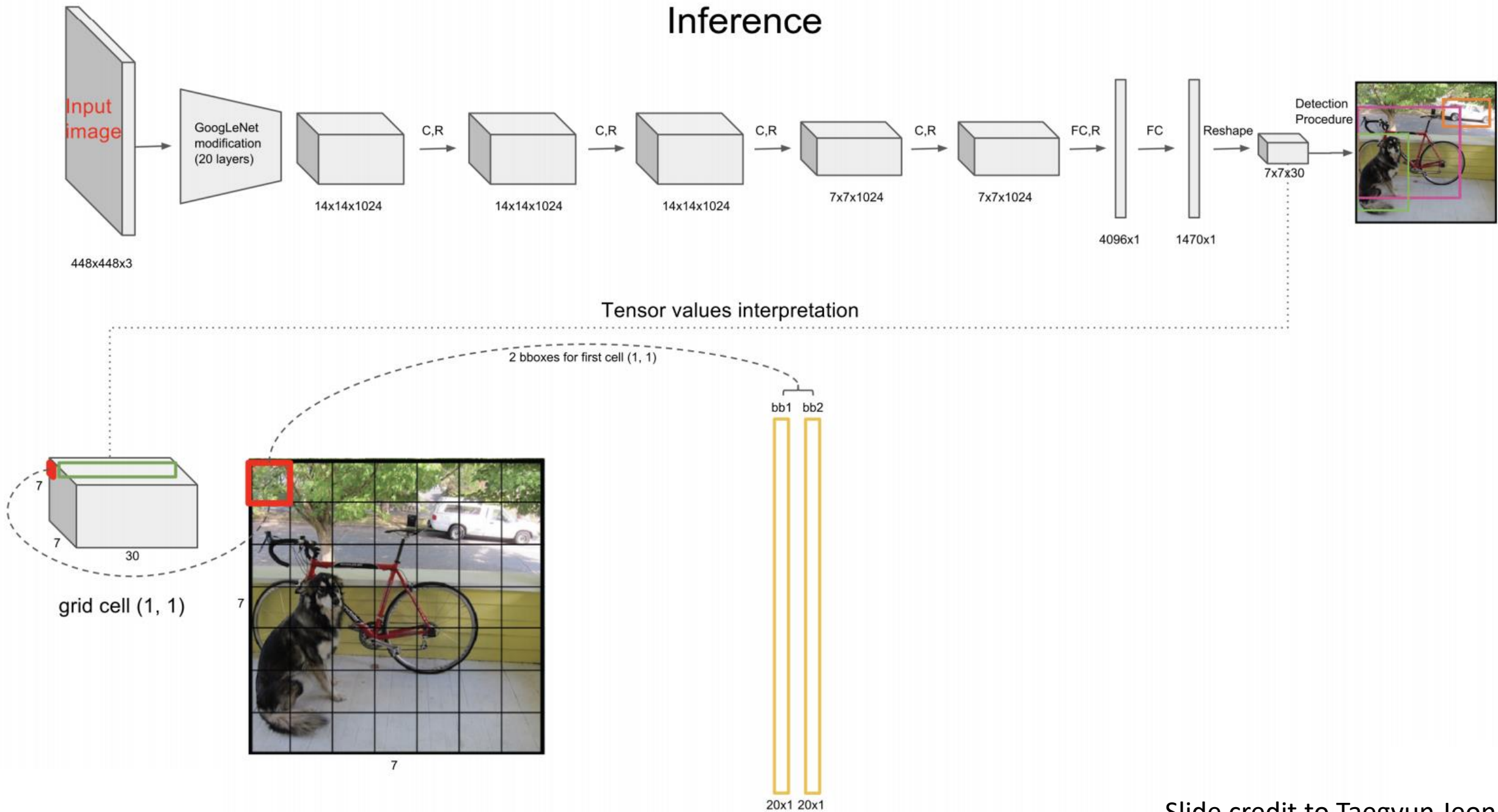
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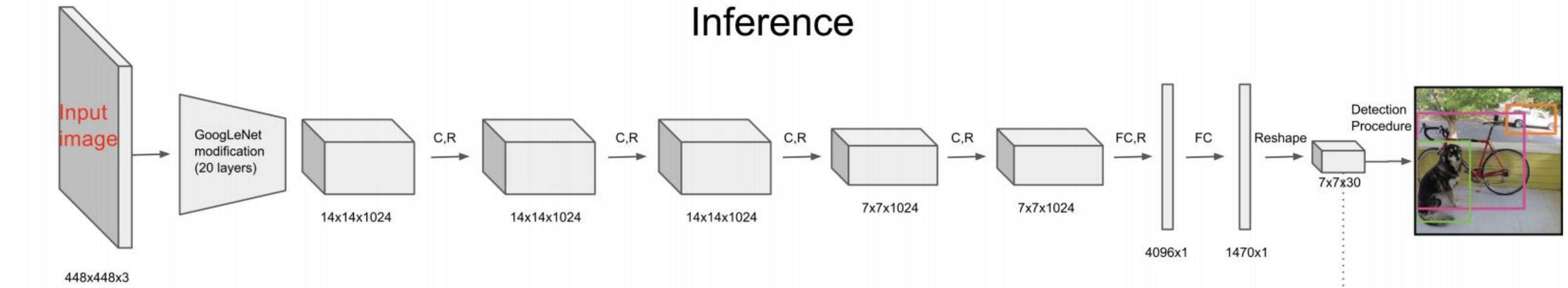
Tensor values interpretation



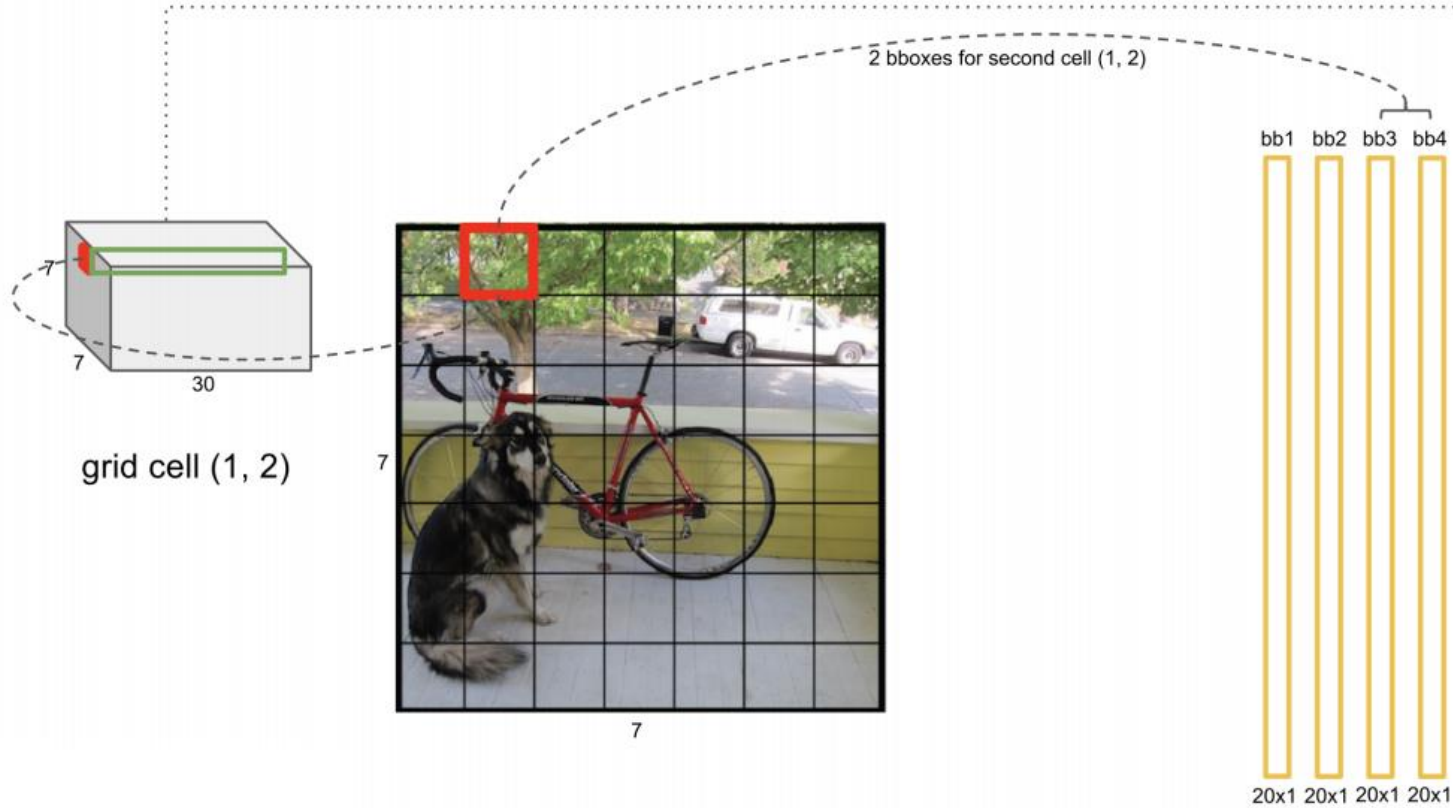
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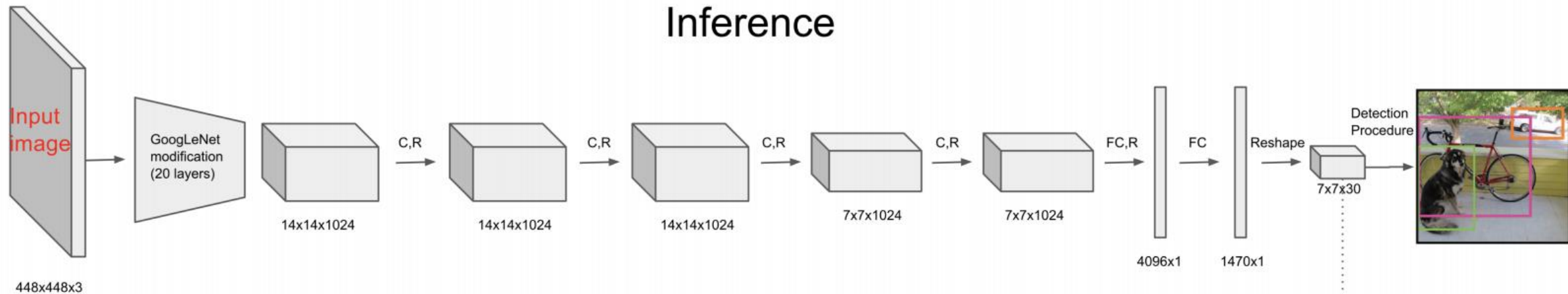
Inference



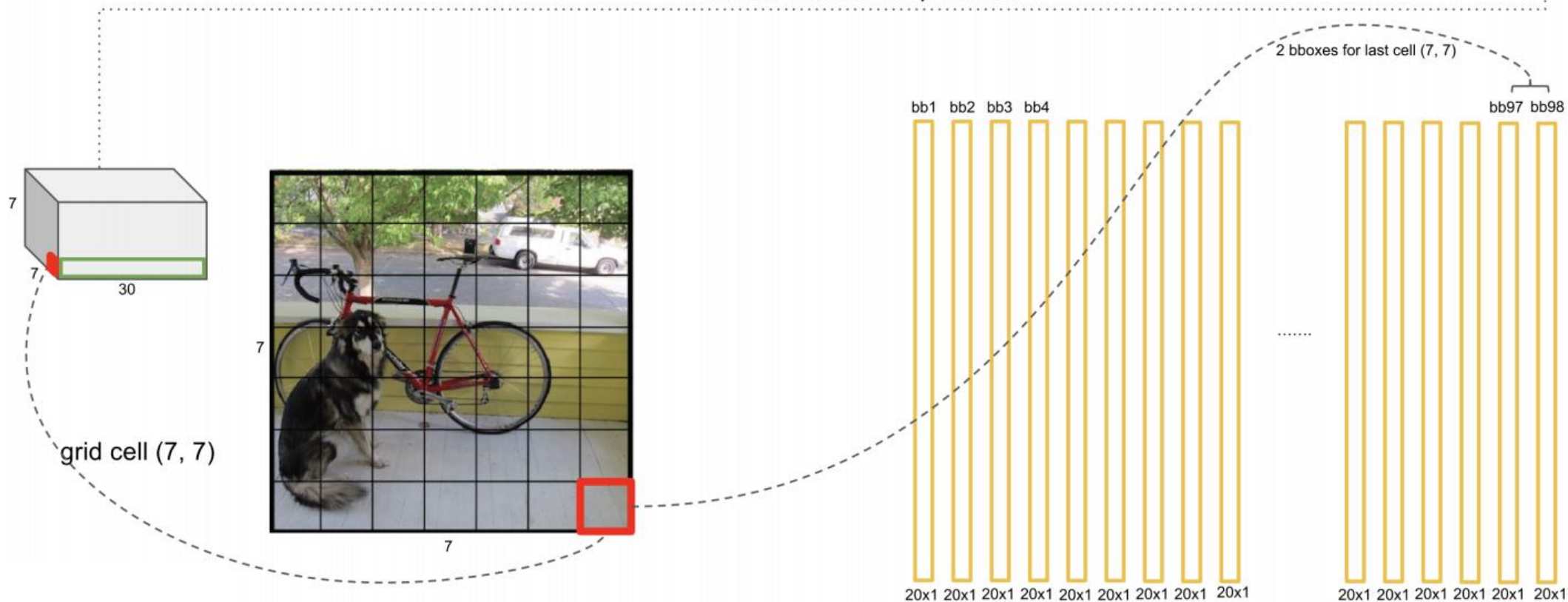
Tensor values interpretation



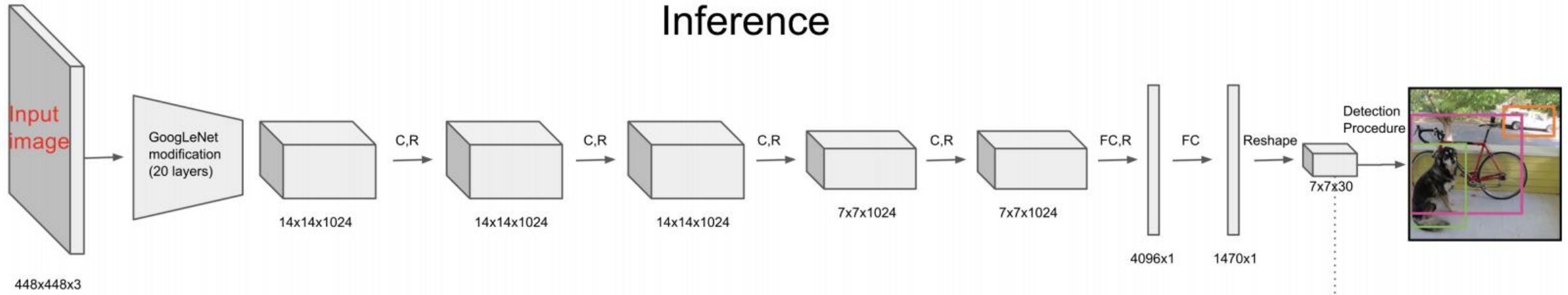
Inference



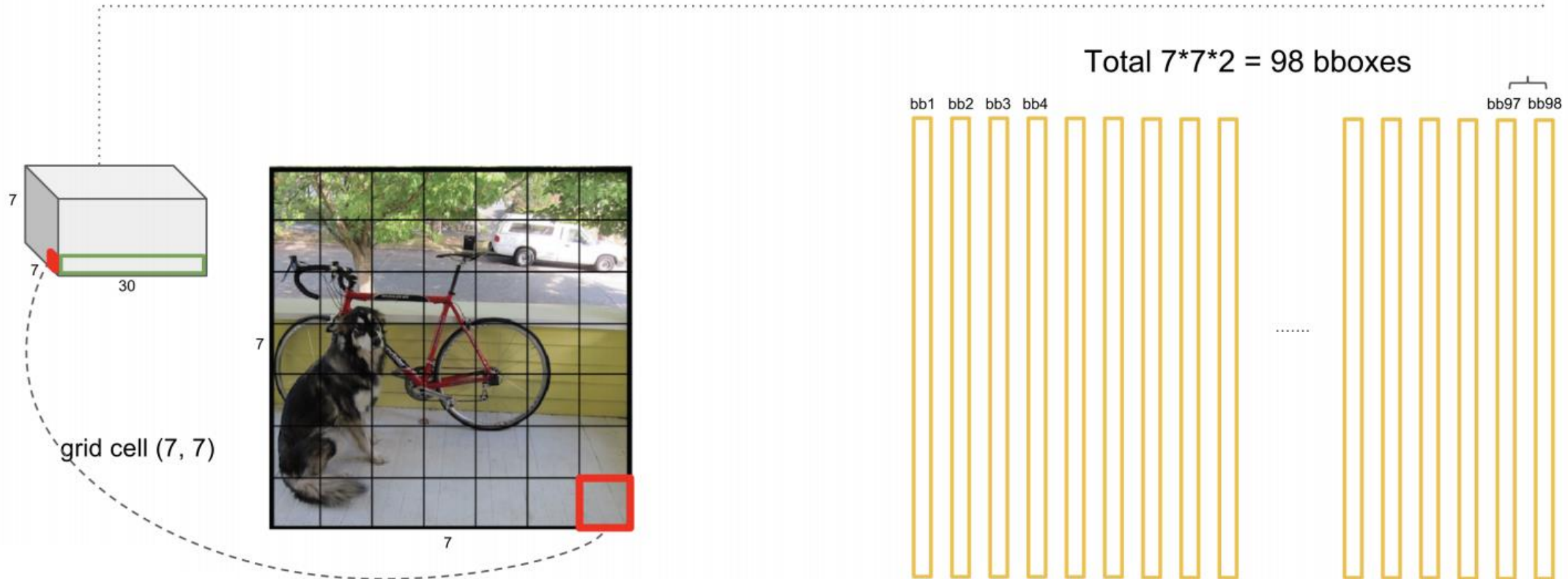
Tensor values interpretation



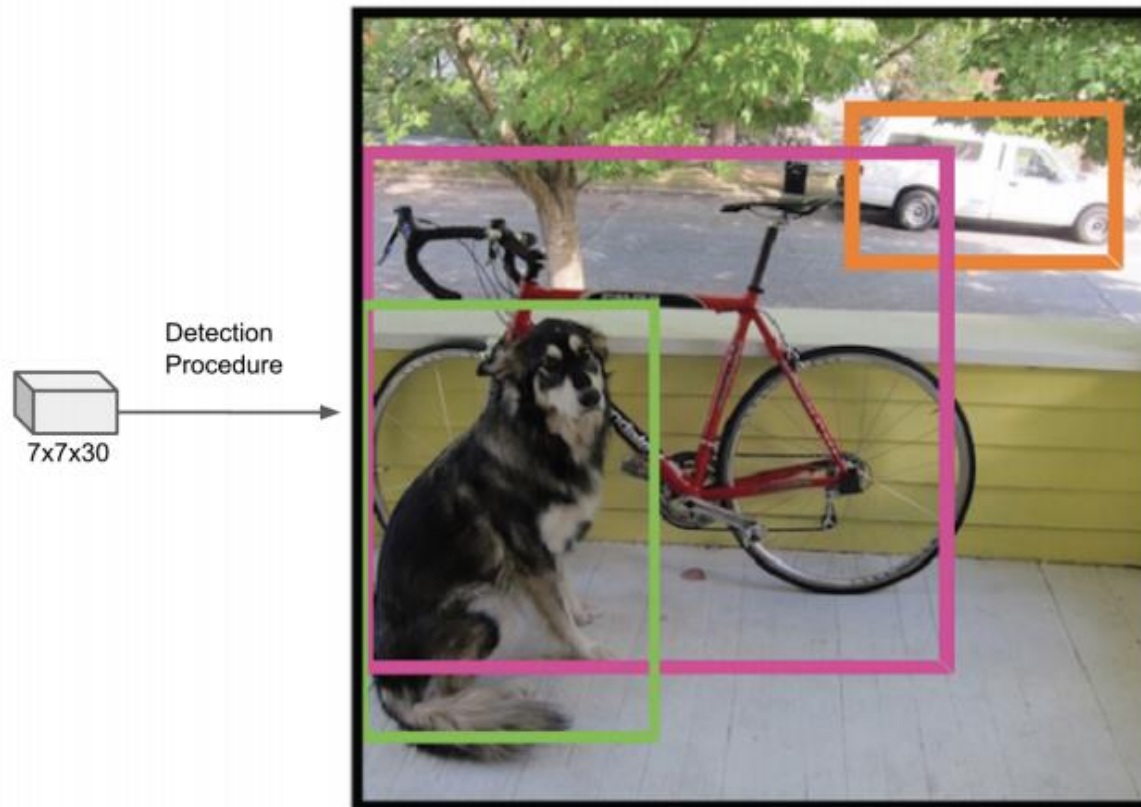
Inference

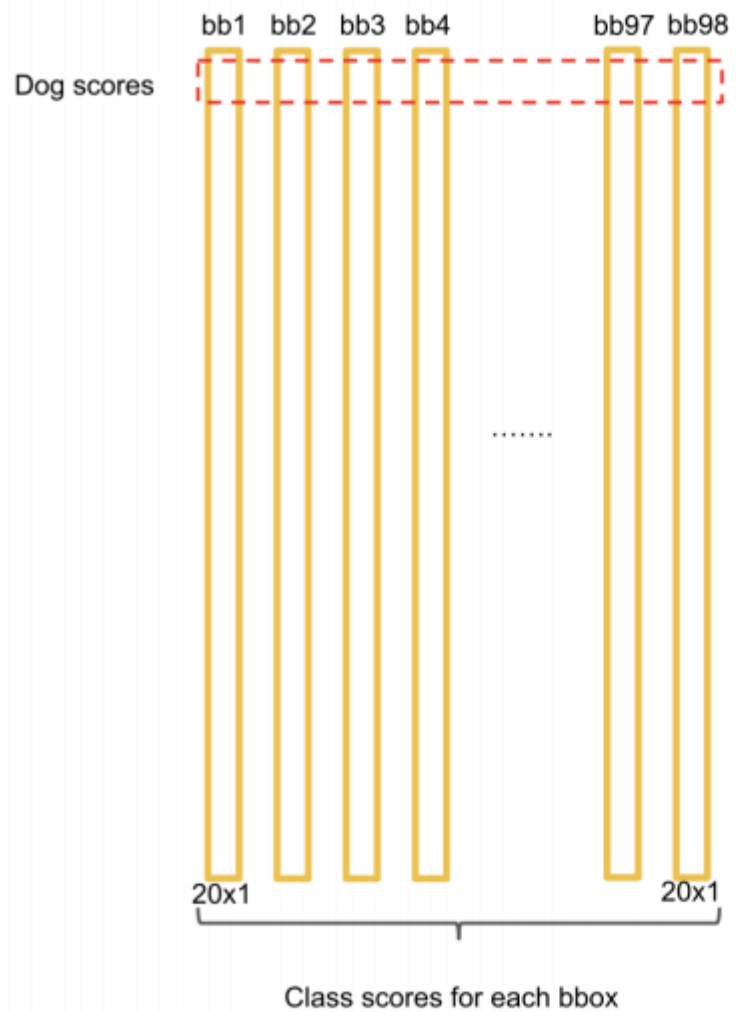


Tensor values interpretation

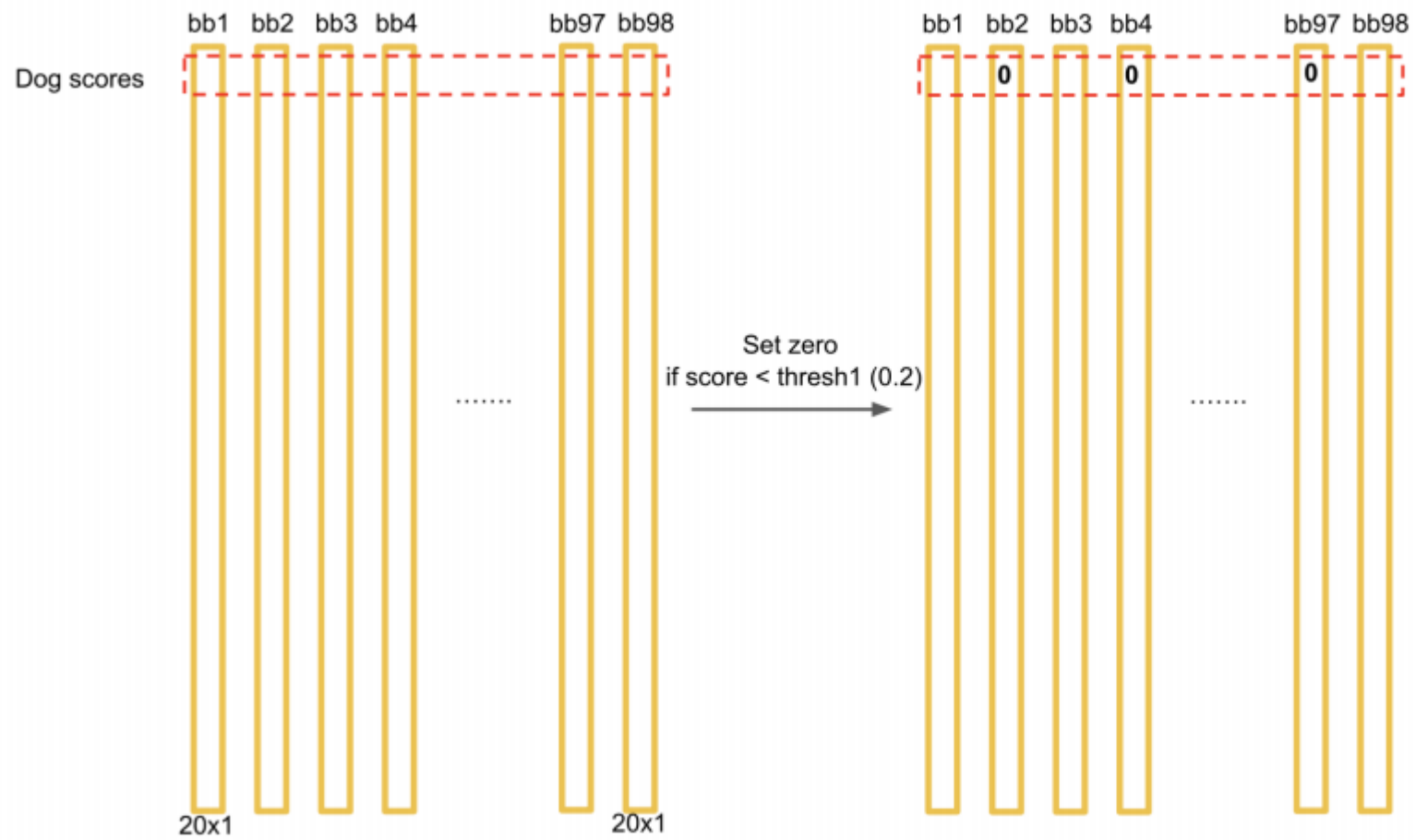


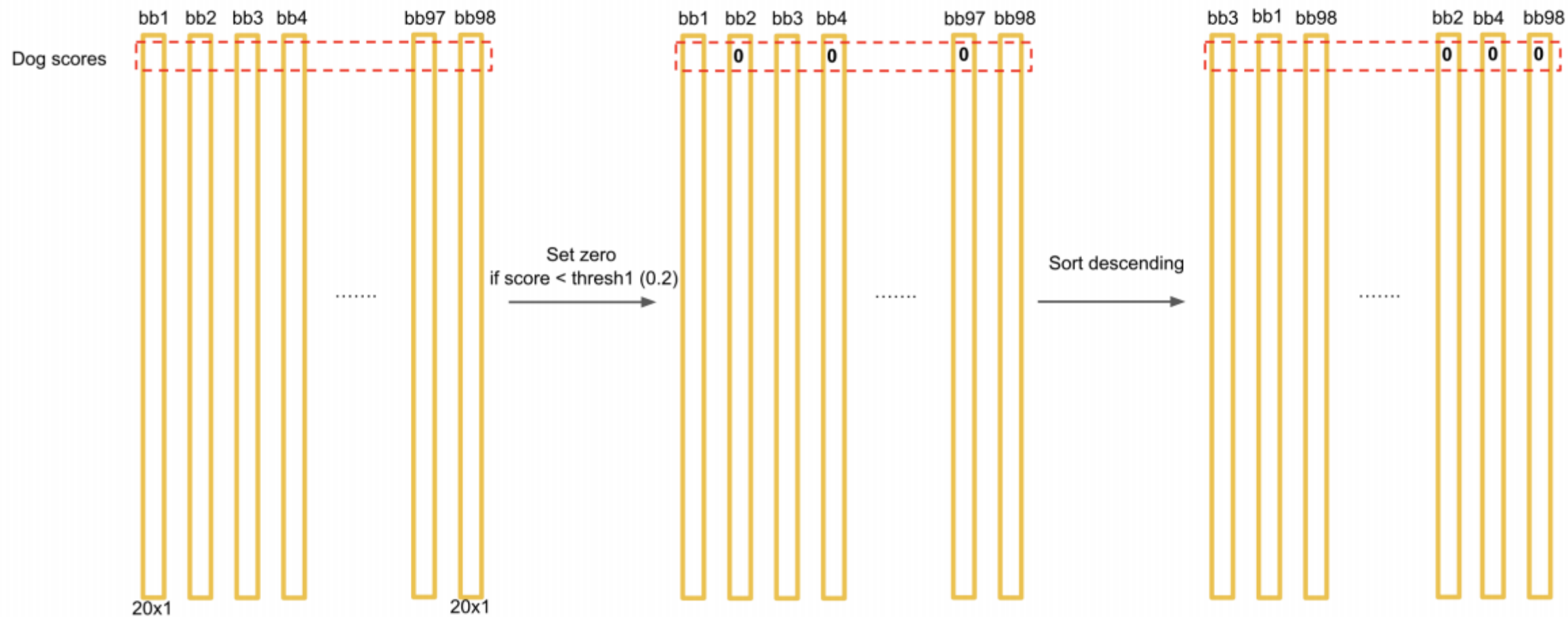
Look at detection procedure

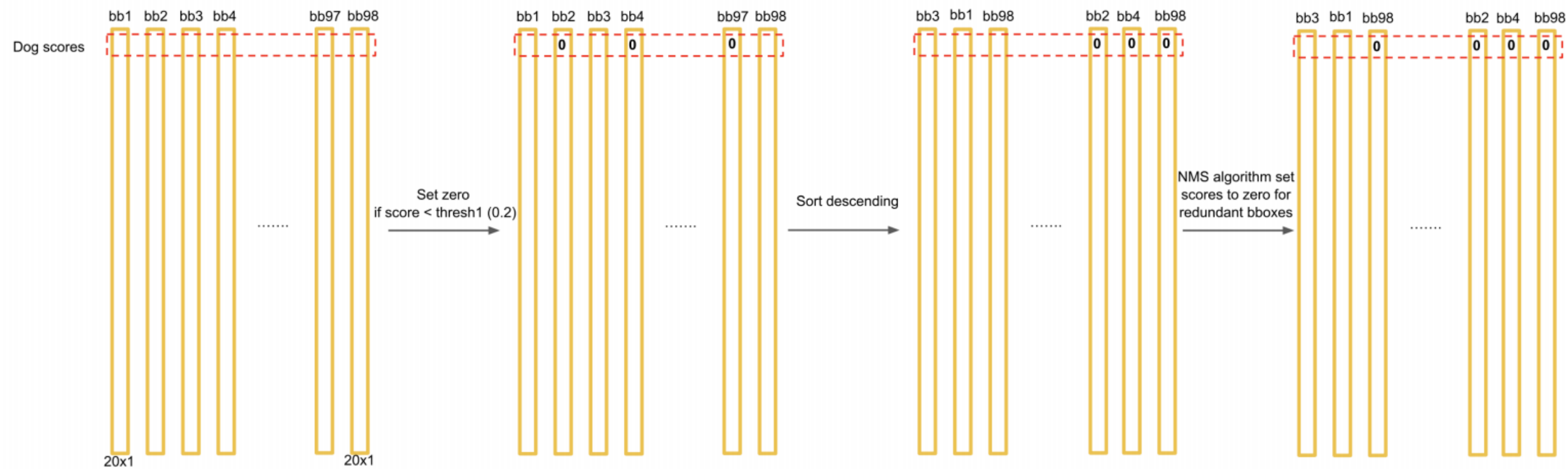


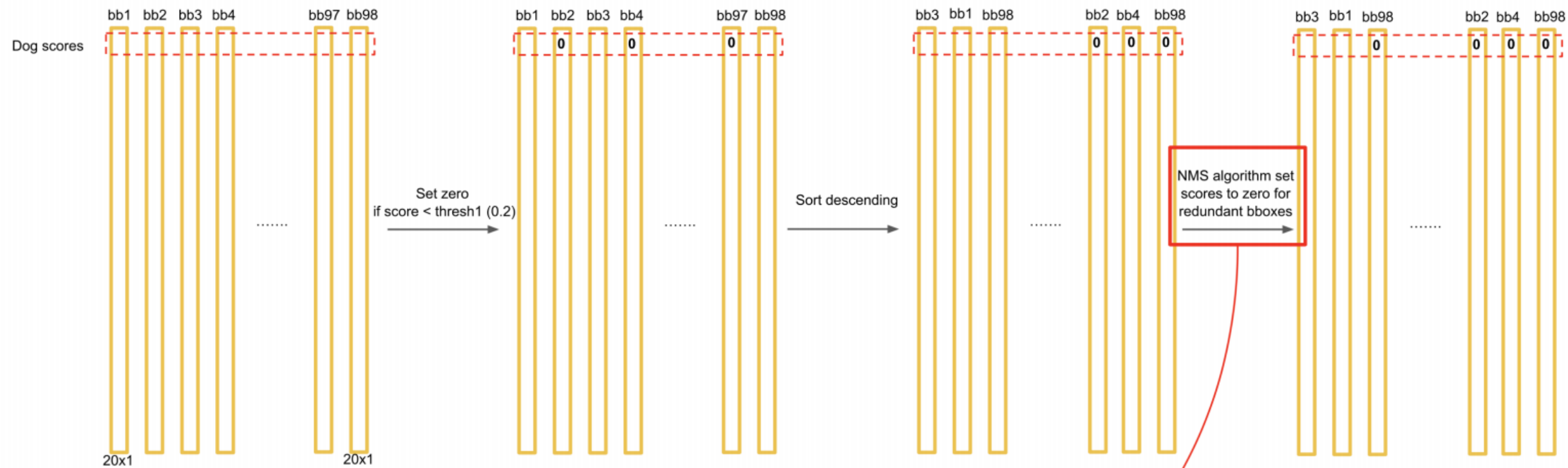


Get first class scores for each bbox







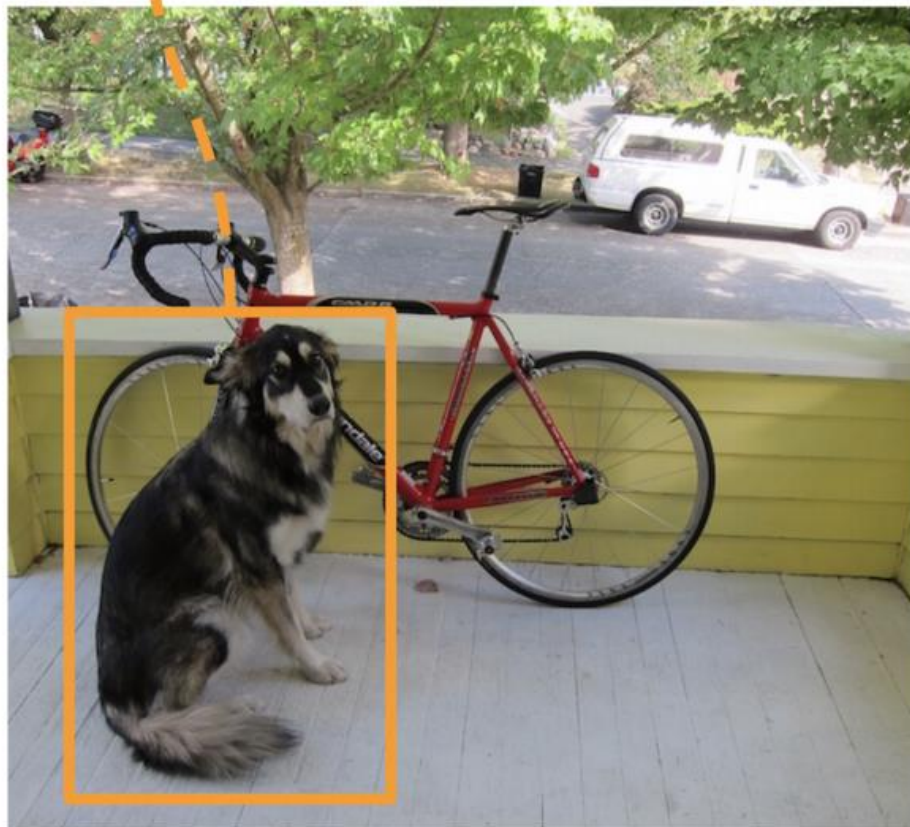


How it works

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

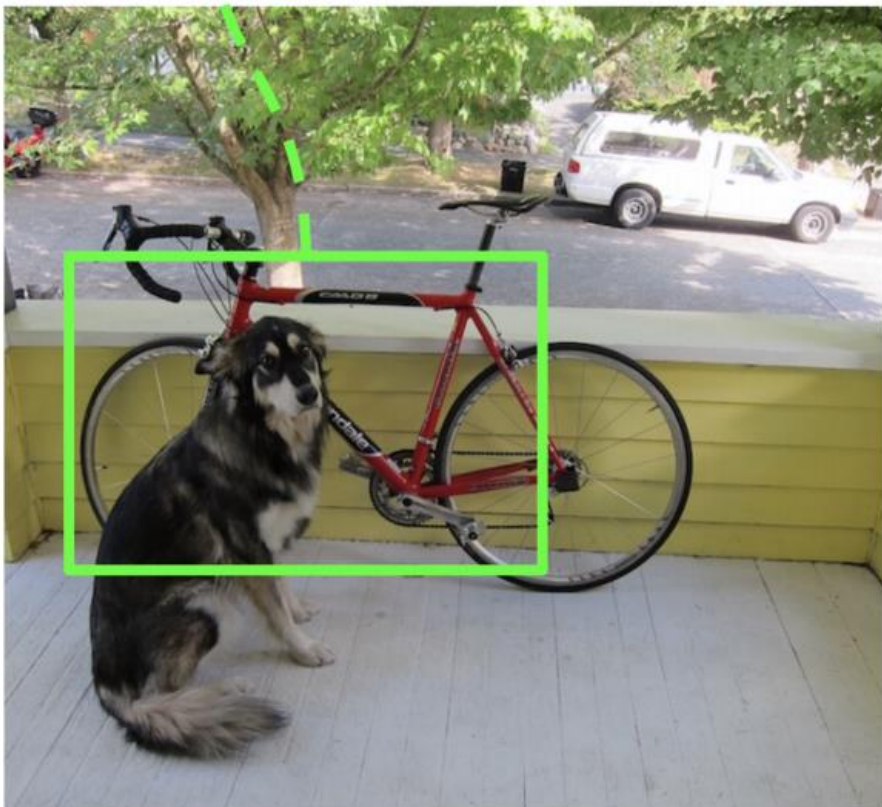
class: dog	bb47	bb20	bb15	bb7									bb1	bb4	bb8	bb98	1x98
	0.5	0.3	0.2	0.1									0	0	0	0	



Non-Maximum Suppression: intuition

class (dog) scores for each bbox

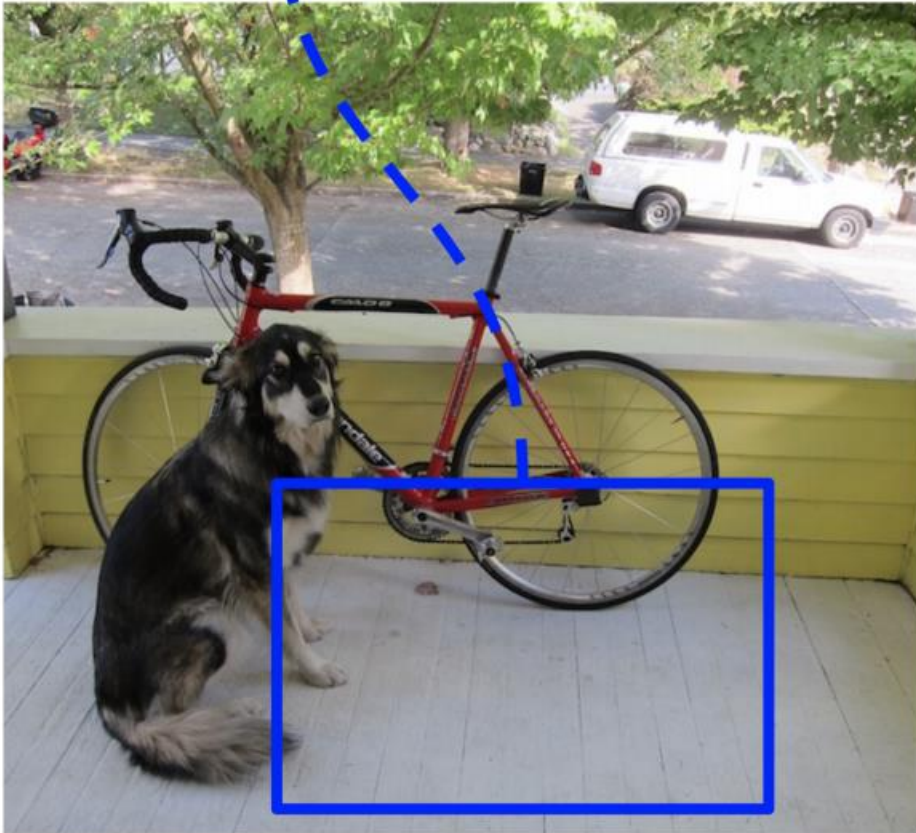
												bb1	bb4	bb8	bb98	1x98
bb47	bb20	bb15	bb7									0	0	0	0	



Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47 bb20 bb15 bb7										bb1 bb4 bb8 bb98				1x98
class: dog	0.5	0.3	0.2	0.1						0	0	0	0	

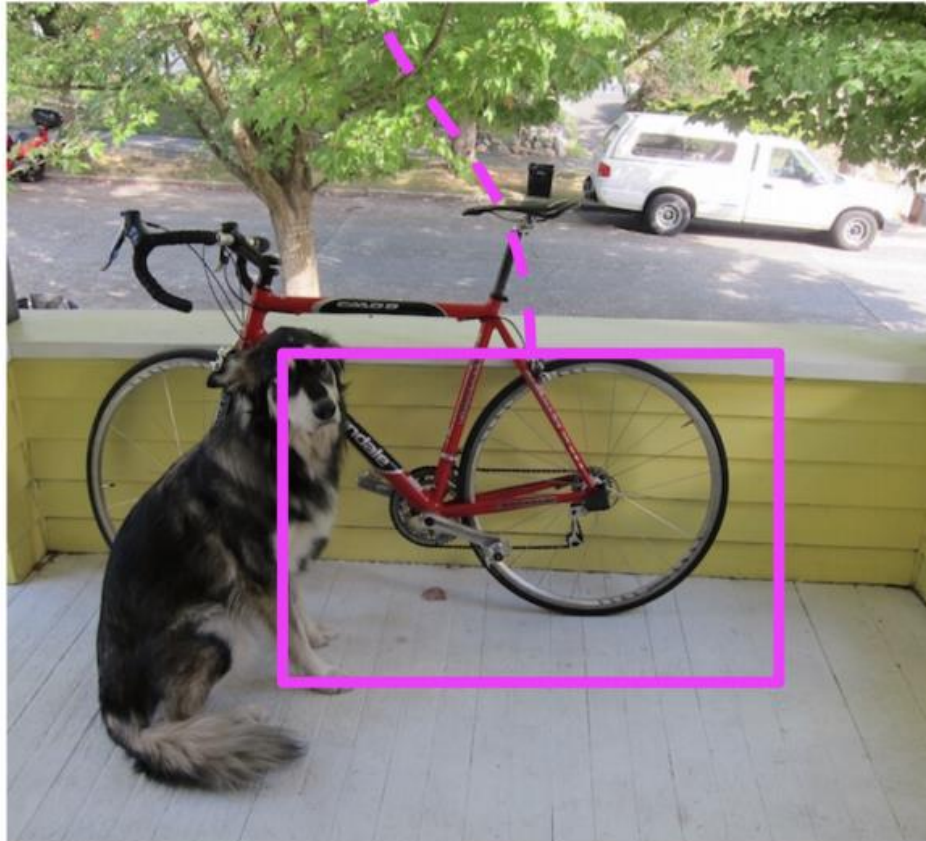


Non-Maximum Suppression: intuition

class (dog) scores for each bbox

	bb47	bb20	bb15	bb7						bb1	bb4	bb8	bb98
class: dog	0.5	0.3	0.2	0.1						0	0	0	0

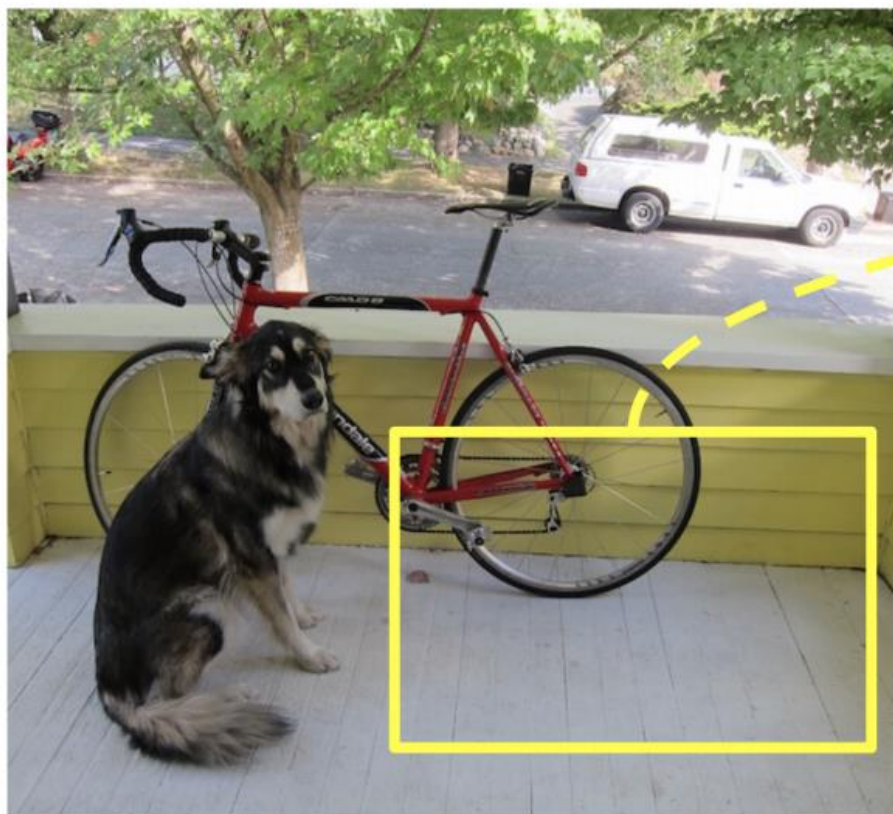
1x98



Non-Maximum Suppression: intuition

class (dog) scores for each bbox

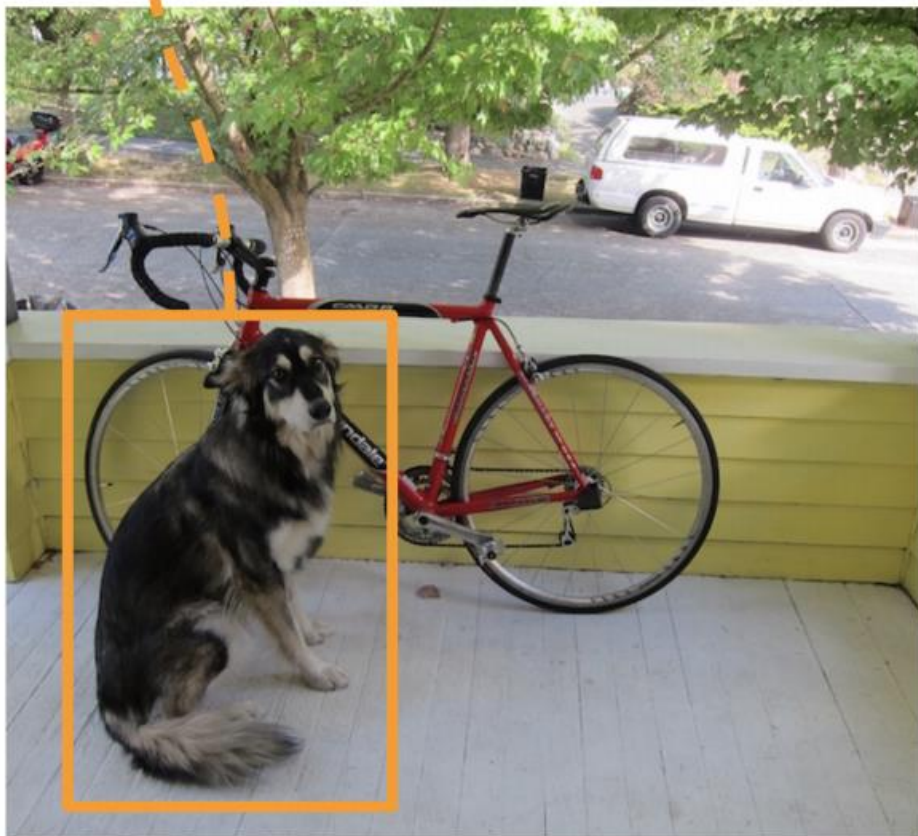
										bb1	bb4	bb8	bb98	
class: dog	bb47	bb20	bb15	bb7										
	0.5	0.3	0.2	0.1										
										0	0	0	0	1x98



Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47 bb20 bb15 bb7										bb1 bb4 bb8 bb98					
class: dog	0.5	0.3	0.2	0.1							0	0	0	0	1x98



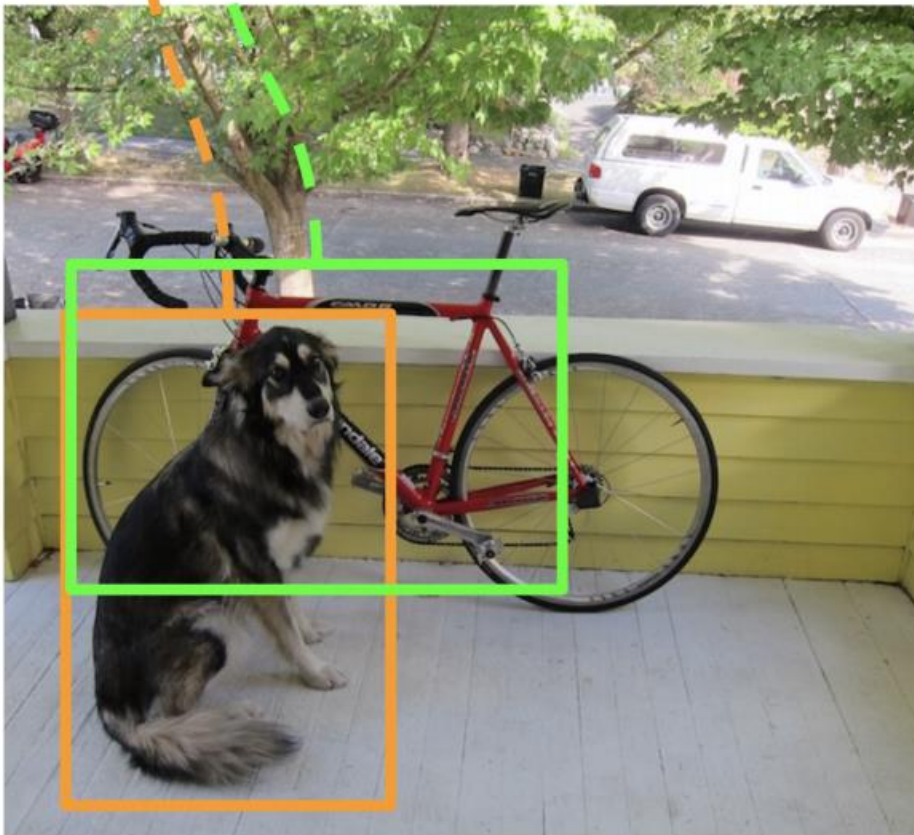
Get bbox with max score. Let's denote it "bbox_max"

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

	bb47	bb20	bb15	bb7																bb1	bb4	bb8	bb98
class: dog	0.5	0.3	0.2	0.1																0	0	0	0

x98

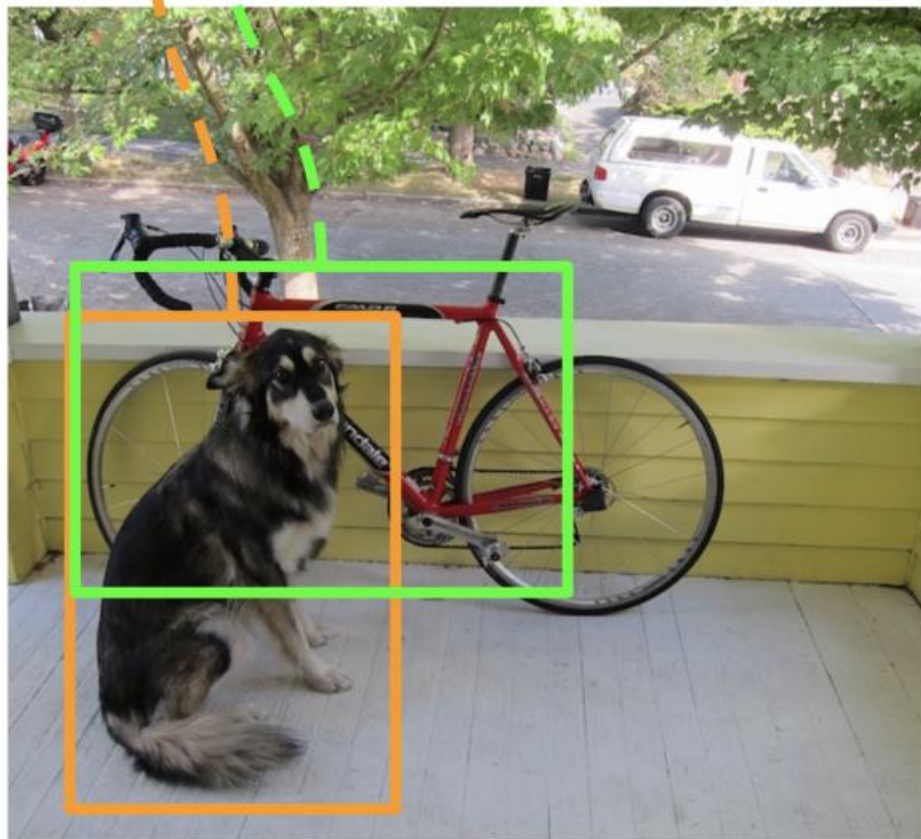


Compare “bbox_max” with others less score (non-zero!) bboxes. Let’s denote it “bbox_cur”

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47 bb20 bb15 bb7										bb1	bb4	bb8	bb98	
0.5	0.3	0.2	0.1							0	0	0	0	1x98

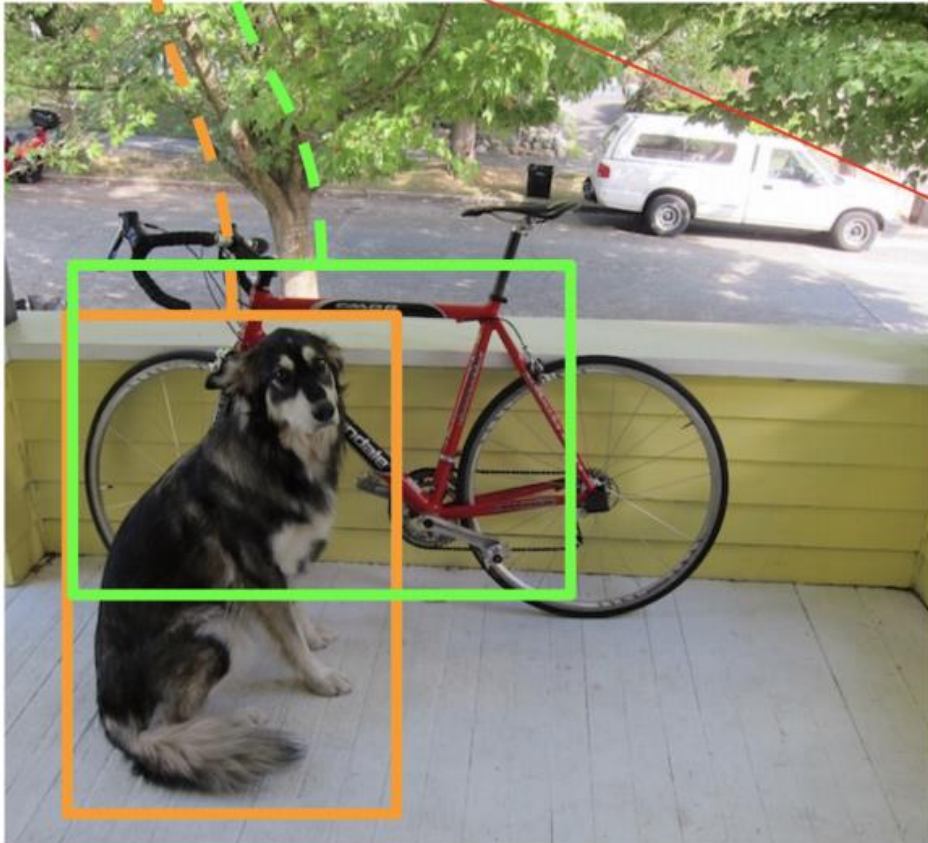


If $\text{IoU}(\text{bbox_max}, \text{bbox_cur}) > 0.5$ then set 0 score to bbox_cur .

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

	bb47	bb20	bb15	bb7						bb1	bb4	bb8	bb98	
class: dog	0.5	0	0.2	0.1						0	0	0	0	1x98



If $\text{IoU}(\text{bbox_max}, \text{bbox_cur}) > 0.5$ then set 0 score to bbox_cur .

In this case: set to 0.

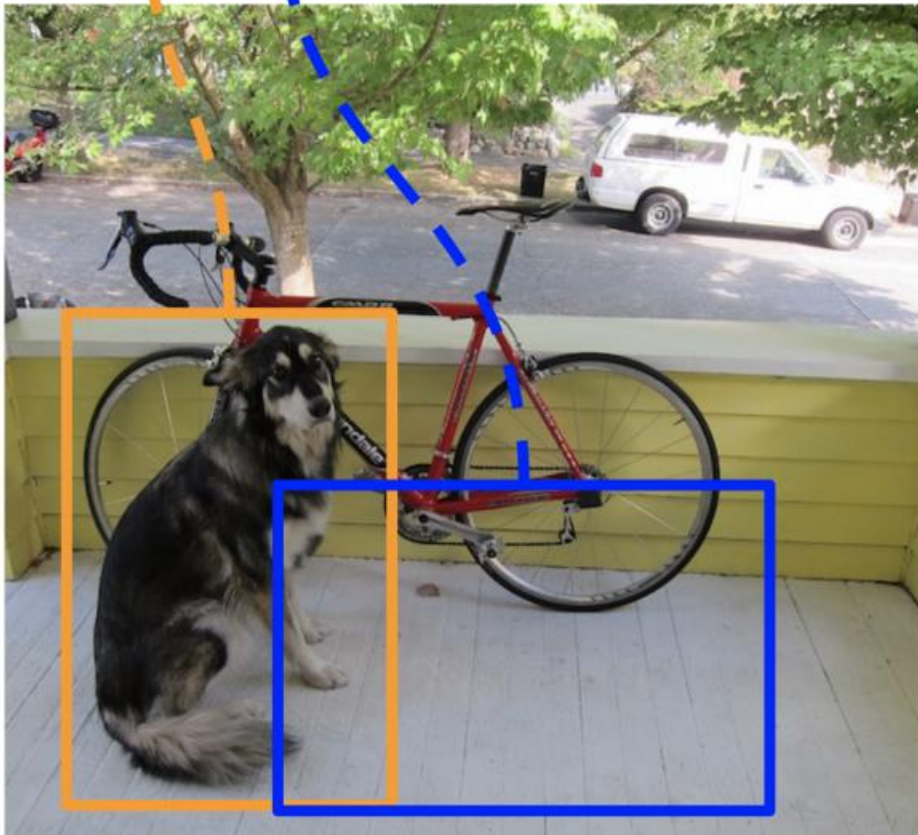
Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47bb20bb15bb7

bb1bb4bb8bb98

0.5	0	0.2	0.1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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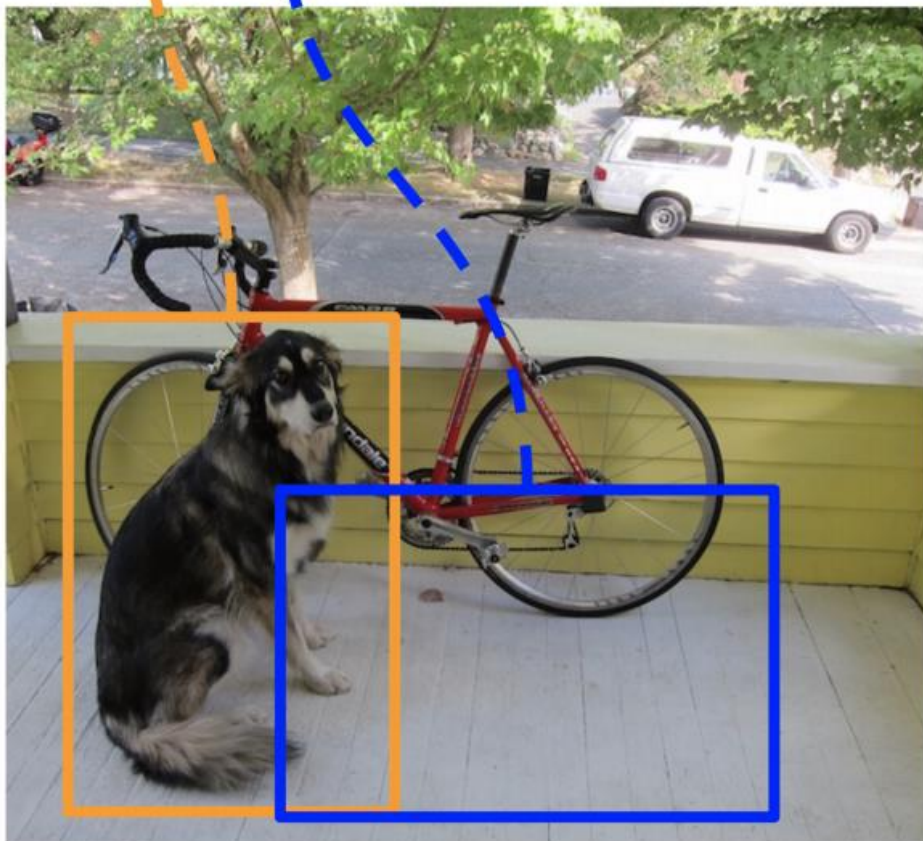
Go to next `bbox_cur`.

If $\text{IoU}(\text{bbox_max}, \text{bbox_cur}) > 0.5$ then set 0 score to `bbox_cur`.

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

	bb47	bb20	bb15	bb7						bb1	bb4	bb8	bb98	
class: dog	0.5	0	0.2	0.1						0	0	0	0	1x98



Go to next `bbox_cur`.

If $\text{IoU}(\text{bbox_max}, \text{bbox_cur}) > 0.5$ then set 0 score to bbox_cur .

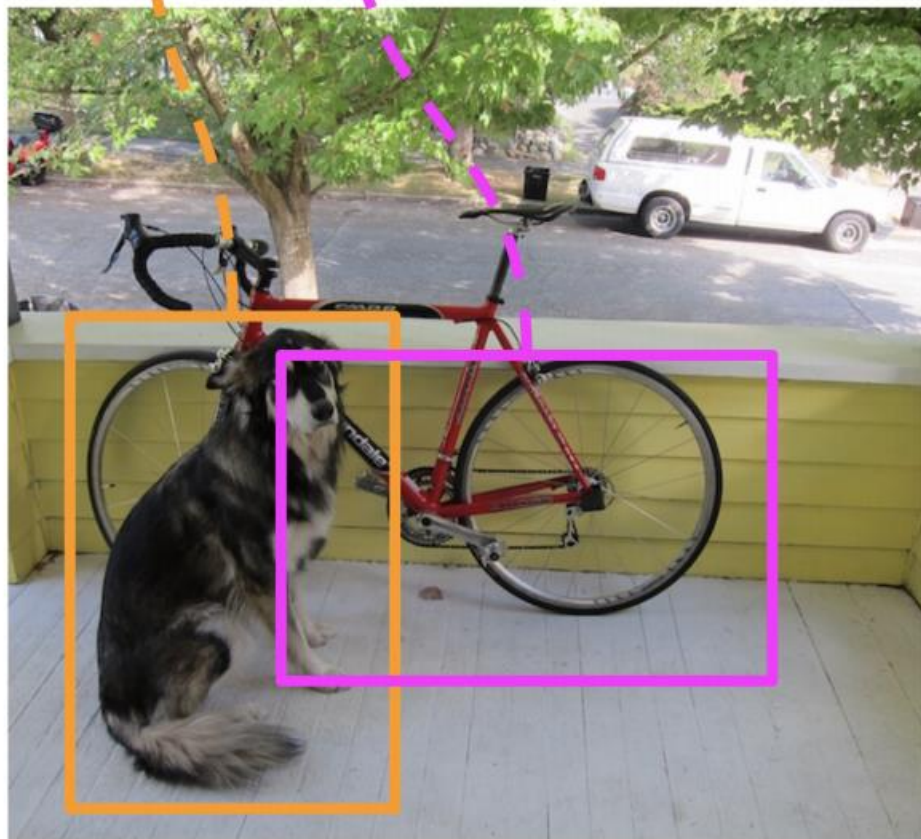
In this case: continue.

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

	bb47	bb20	bb15	bb7																	bb1	bb4	bb8	bb98
class: dog	0.5	0	0.2	0.1																	0	0	0	0

1x98



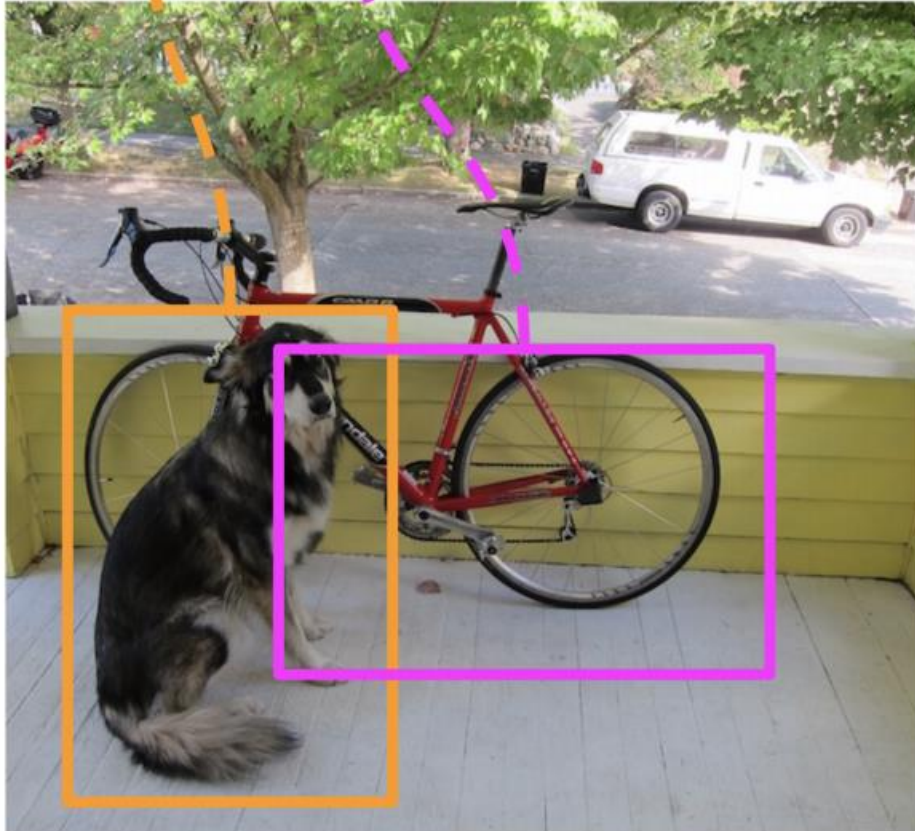
Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47bb20bb15bb7

bb1bb4bb8bb98

class: dog	0.5	0	0.2	0.1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											</
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Go to next **bbox_{cur}**.

If $\text{IoU}(\text{bbox}_{\text{max}}, \text{bbox}_{\text{cur}}) > 0.5$ then set 0 score to **bbox_{cur}**.

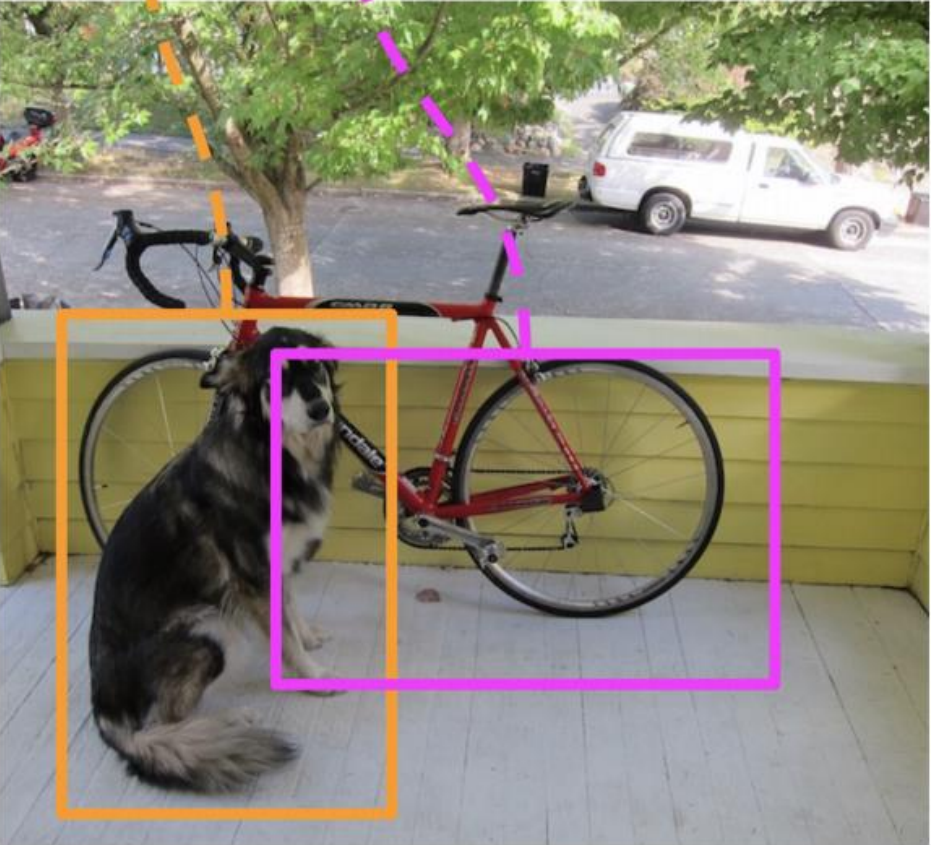
Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47	bb20	bb15	bb7																		bb1	bb4	bb8	bb98
0.5	0	0.2	0.1																		0	0	0	0

class: dog

1x98



Go to next **bbox_{cur}**.

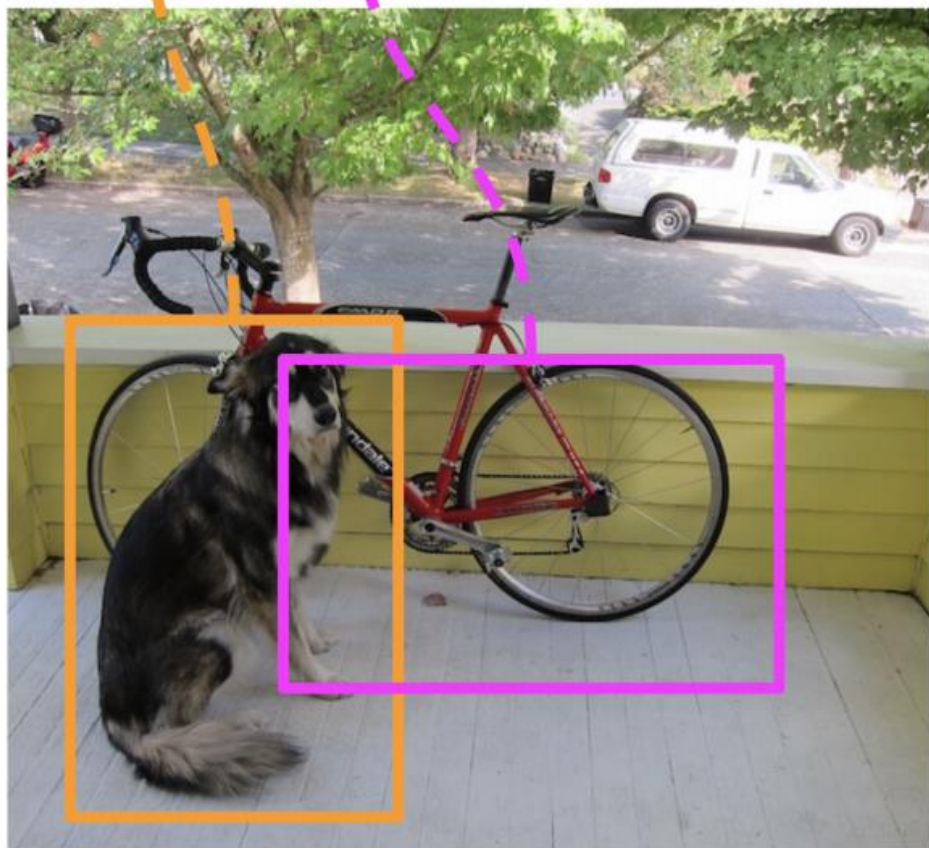
If $\text{IoU}(\text{bbox}_{\text{max}}, \text{bbox}_{\text{cur}}) > 0.5$ then set 0 score to **bbox_{cur}**.

In this case: continue.

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

										bb1	bb4	bb8	bb98	1x98
bb47	bb20	bb15	bb7											
0.5	0	0.2	0.1							0	0	0	0	



Go to next **bbox_cur**.

If $\text{IoU}(\text{bbox_max}, \text{bbox_cur}) > 0.5$ then set 0 score to **bbox_cur**.

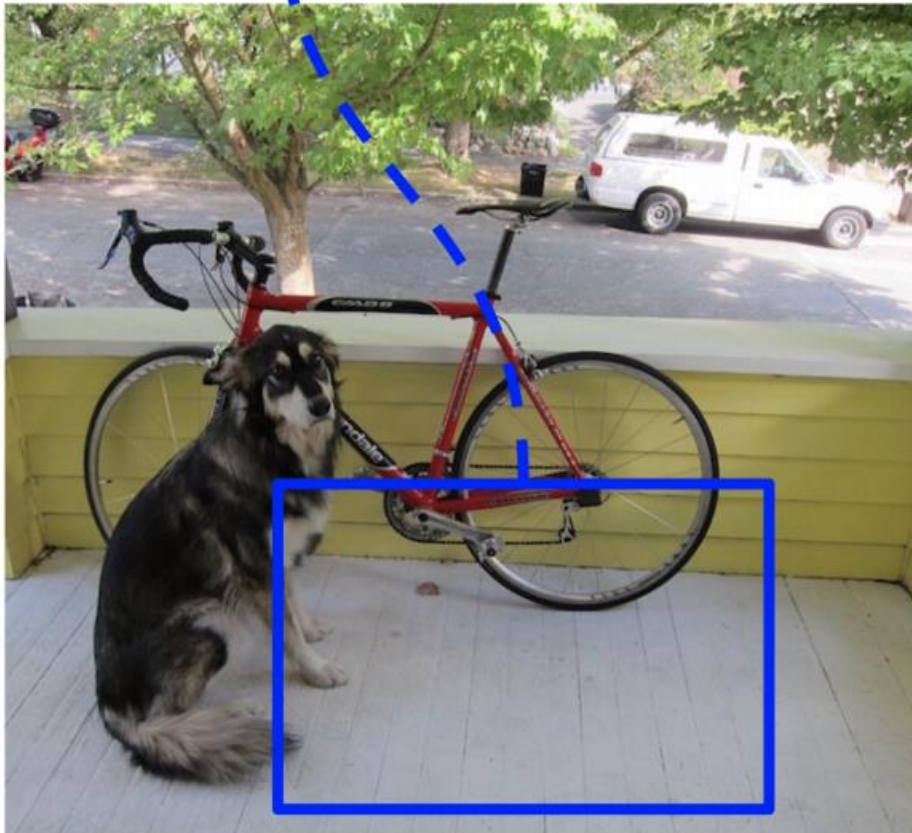
In this case: continue.

Do this procedure for other “bbox_cur”. After that ...

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47 bb20 bb15 bb7												bb1 bb4 bb8 bb98								
0.5	0	0.2	0.1									0	0	0	0					1x98

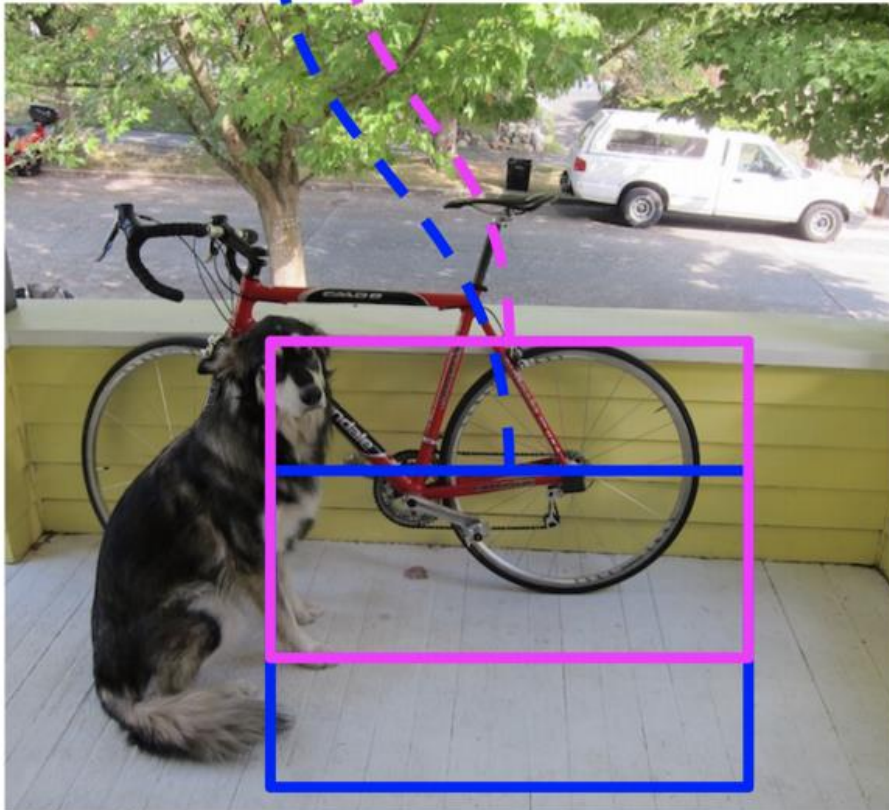


Go to next bbox with big score.
Let's denote it "**bbbox_max**"

Non-Maximum Suppression: intuition

class (dog) scores for each bbox

	bb47	bb20	bb15	bb7						bb1	bb4	bb8	bb98	
class: dog	0.5	0	0.2	0.1						0	0	0	0	1x98



Go to next `bbox_cur`.

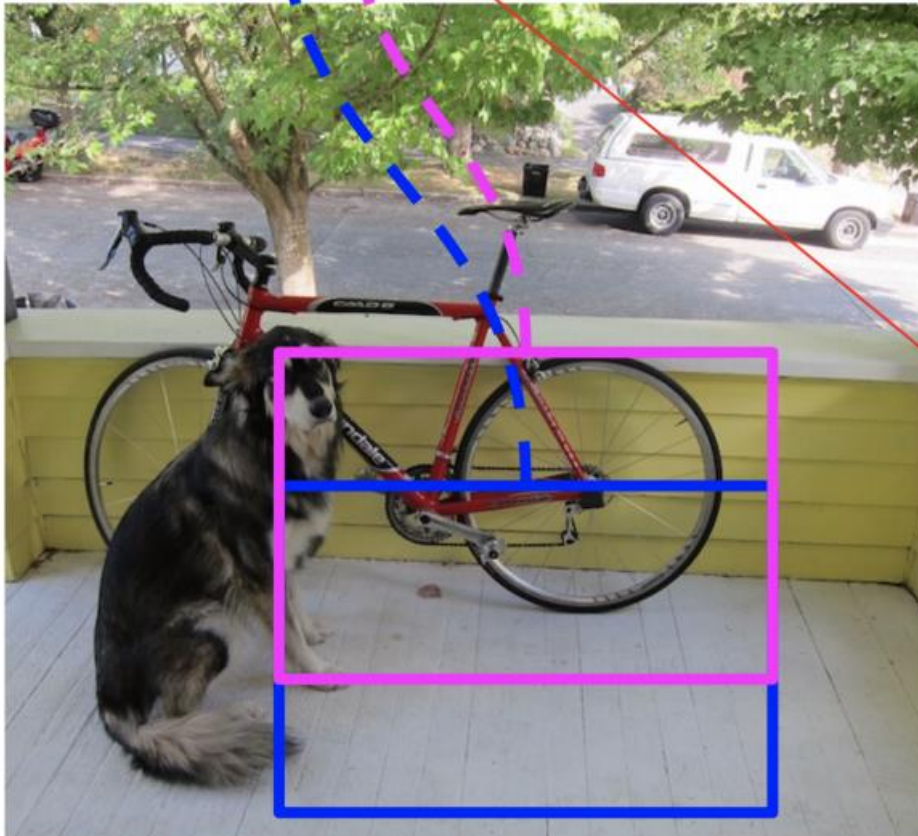
Non-Maximum Suppression: intuition

class (dog) scores for each bbox

bb47	bb20	bb15	bb7																					bb1	bb4	bb8	bb98
0.5	0	0.2	0																					0	0	0	0

class: dog

1x98



Go to next `bbox_cur`.

If $\text{IoU}(\text{bbox_max}, \text{bbox_cur}) > 0.5$ then set 0 score to bbox_cur .

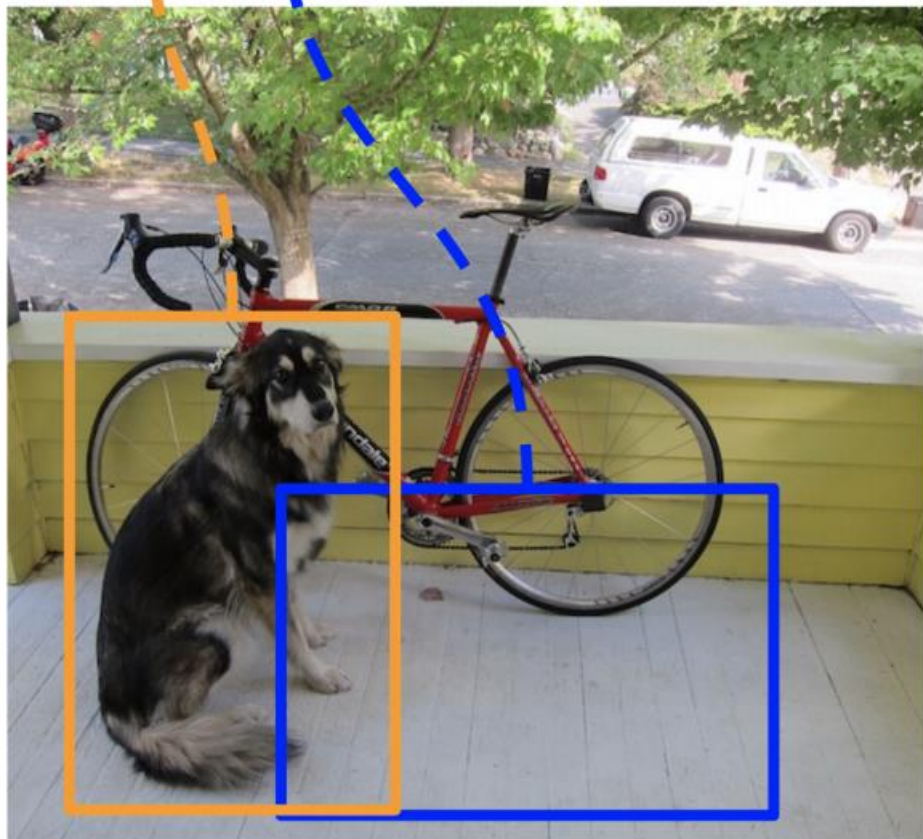
In this case: set to 0.

Do this procedure for other “bbox_max” and for other corresponding “bbox_cur”.

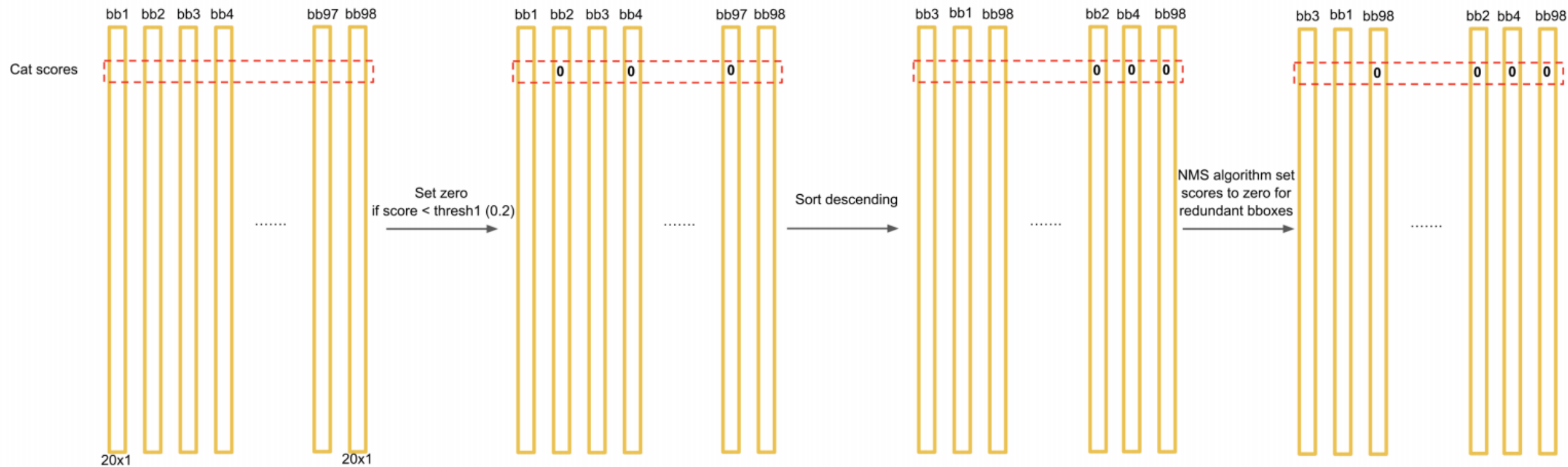
Non-Maximum Suppression: intuition

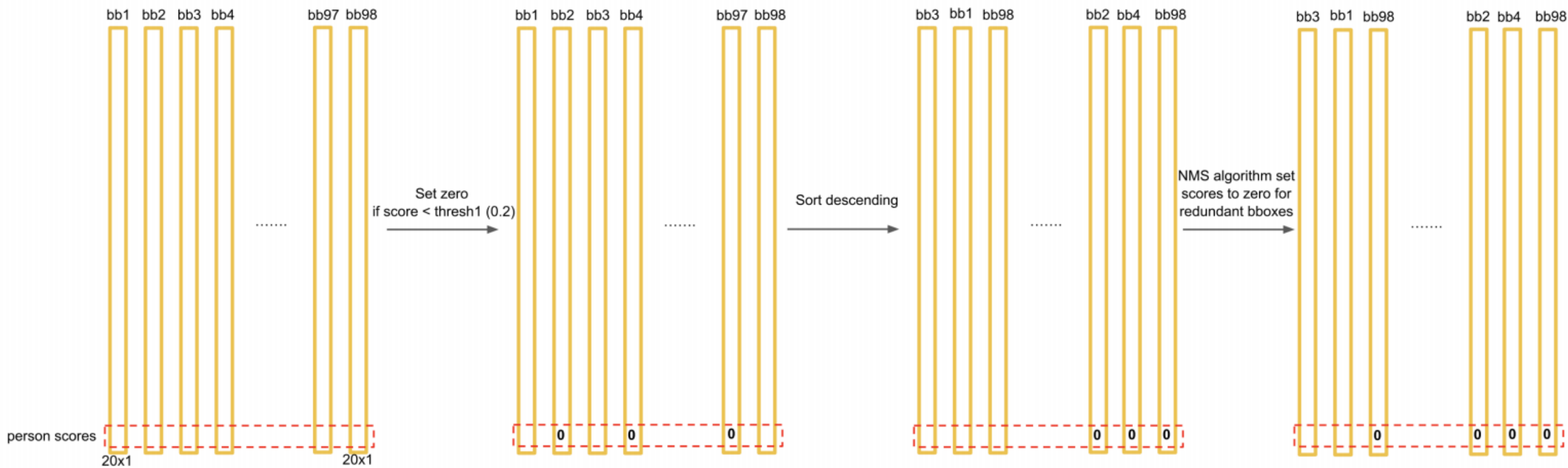
class (dog) scores for each bbox

	bb47	bb20	bb15	bb7						bb1	bb4	bb8	bb98	
class: dog	0.5	0	0.2	0						0	0	0	0	1x98

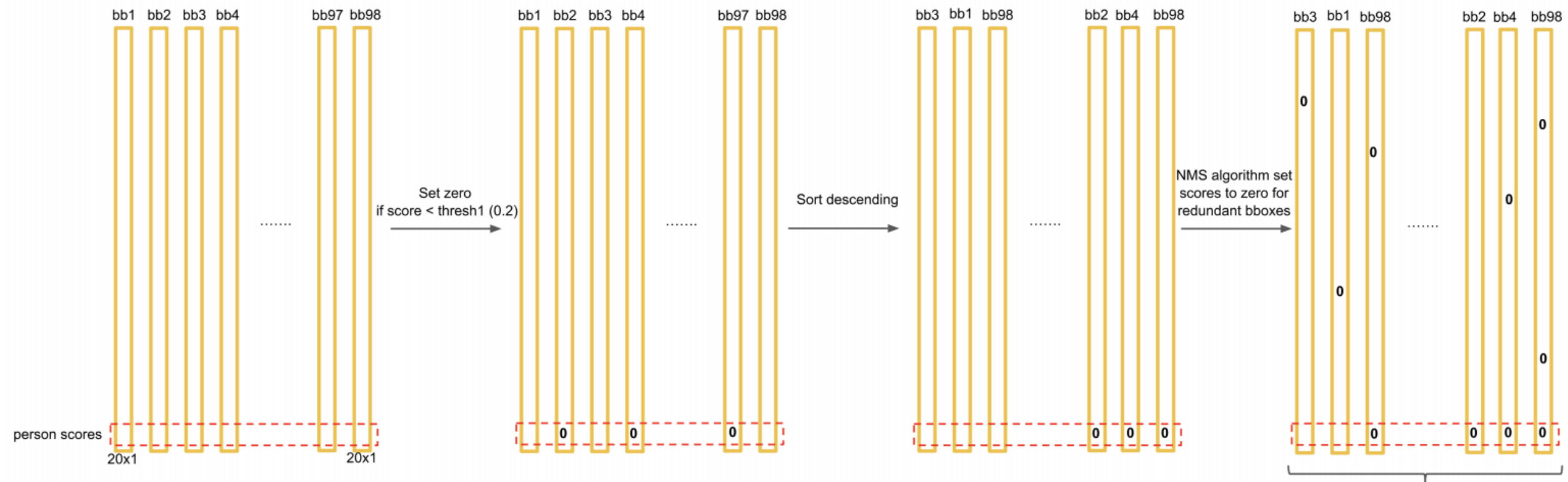


After comparison almost all pairs of bboxes the only two bboxes left with non-zero class score value.

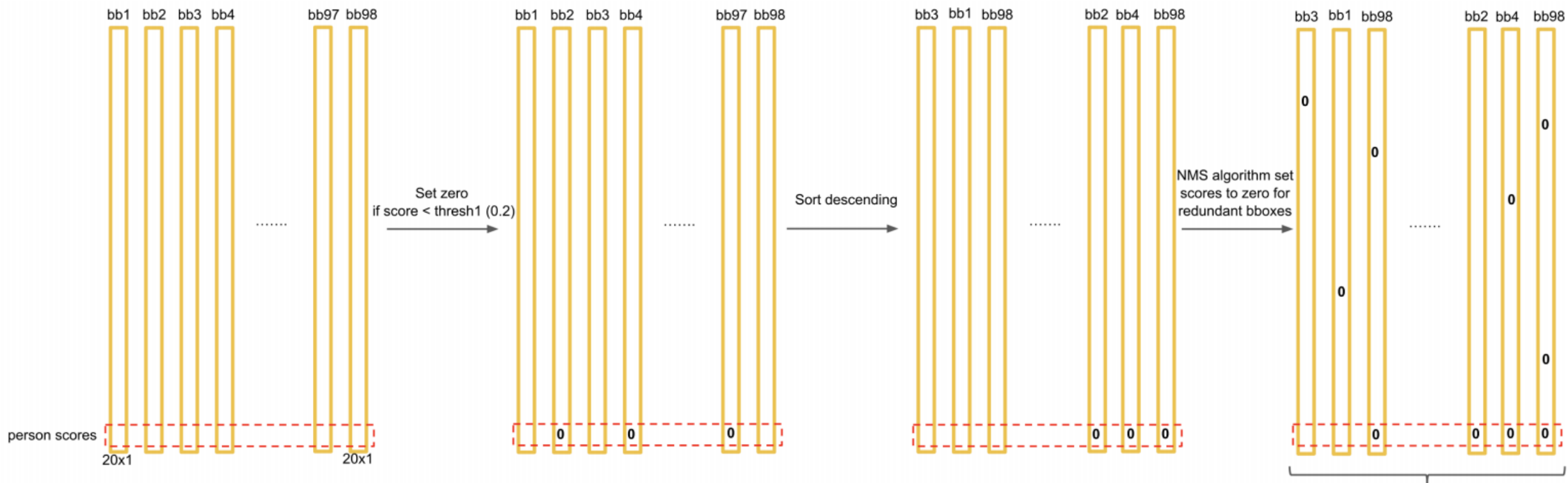




Do this procedure for all classes



After this procedure -
a lot of zeros



Select bboxes to draw by
class score values

